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2025 EVS AND BATTERY SUPPLY CHAINS ISSUES AND IMPACTS

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INTRODUCTION

Anders Hove

This issue of the *Oxford Energy Forum* is dedicated to the topic of global EV and battery supply chains, and specifically how countries are responding to the need to diversify EV supply chains away from China, while also navigating new geopolitical challenges and trade barriers being erected by the Trump administration in the US. The issue builds on the contents of two prior issues. In OEF 142, 'Responding to the China challenge: diversification and de-risking in new energy supply chains', we examined a broad range of technologies and diversification strategies being pursued worldwide. In OEF 143, 'The Geopolitical and Commercial Implications of Green Industrial Policy', we considered the geopolitical and commercial implications of green industrial policy. Many of the themes discussed in those volumes return in this issue.

Among the common themes is industrial policy. Perhaps more so than for wind or solar, many countries not only are eager to capture part of the global EV value chain, but are willing to consider industrial policies to achieve that end. In some cases, the primary consideration is to protect existing industries and enable countries to better compete with the rising tide of low-cost Chinese EVs. Developing-world countries also seek to ensure that the global EV revolution doesn't simply become another import dependency. There are also several instances of countries that seek to use local mineral endowments to move into higher-value aspects of the battery supply chain. Above all, countries see the EV and battery sectors as ways to boost employment, move up the value chain, and upgrade technologically.

As Bassam Fattouh noted in OEF 143, many of these industrial policies are not new at all.[1] Subsidies for EV adoption are common worldwide. Other trade barriers, and tariffs, are becoming more common. These policies can promote the domestic market while favouring or protecting local players. However, this may not be enough. For advanced economies, protection and subsidies can't help close the gap with Chinese players, which are at or near the frontier in battery technology and can therefore offer competitive EVs at attractive price/quality ratios compared with incumbent manufacturers. As for countries in the developing world, technology transfer is even more critical to avoiding the 'value traps' of low-end assembly or polluting mining and extraction industries.

Due to diverse situations and objectives, there are major differences in policy approaches. Initially, under the Biden administration, the US pursued a mixture of domestic subsidies for EV adoption, tariffs on China, manufacturing investment incentives, and international collaboration. The Trump administration seeks to roll back or simply end most of these efforts, while imposing new tariff barriers with its closest economic partners. The impact of tariffs on competitiveness, whether those tariffs are aimed at China or other countries, depends on starting conditions. When the home country already has a technological lead, a skilled workforce, investment capital, and an internal market to achieve scale and reach and maintain a position at or near the technological frontier, tariffs might work to equal out other disadvantages such as higher domestic labour costs. As fewer of these conditions apply, more tariffs could simply isolate its market, raising costs to potential consumers and holding back technological progress at the corporate or societal level.

Licensing, domestic content or joint venture requirements or incentives, and investment deals are another approach. These more closely match China's own development strategy in the past, and hold less potential for retaliation. As with tariffs, the success of these policies hinges on some of the same factors—such as the size of the domestic market, capital availability, and a skilled workforce. However, when one country like China dominates both manufacturing and technology advances, technology cooperation has greater potential to help other countries catch up quickly and reduce costs through technology transfer. That said, for almost all countries, the relatively small size of the domestic market will undoubtedly represent a major barrier—even for those countries that have done the most to boost domestic deployment.

Would China see efforts at technology catch-up as a major risk? That also depends. So far, China appears ready to do business, but with caveats. On one hand, CATL, BYD, Geely, and other companies are actively investing and building abroad. On the other hand, the Chinese government is already pulling the emergency cord on technology transfer in areas such as graphite and lithium-iron-phosphate (LFP) cathodes that could help other countries capture core parts of the battery or EV supply chain. It is unclear whether such restrictions will have a blocking effect, given how quickly battery technology is advancing, and how urgently other countries are seeking to catch up.

[1] Bassam Fattouh, 'Green industrial policy's increasingly complex trade-offs', *in Oxford Energy Forum 143: The Geopolitical and Commercial Implications of Green Industrial Policy*, Oxford Institute for Energy Studies, February 2025, at https://www.oxfordenergy.org/wpcms/wp-content/uploads/2025/02/OEF-143.pdf.

For now, China's lead may be sufficiently large that policymakers perceive limited risk. Further, the major growth engine for China's EV and battery exports is likely to come from markets outside of Europe and the US. Even if vehicle and battery manufacturing ramps up quickly in North America and Europe, these two regions seem unlikely to compete in the lowest-cost markets of Asia, Africa, and Latin America. These markets may be perceived as having lower likelihood of catching up, even with the introduction of some domestic content requirements or local manufacturing.

Contrasting views of the US

The election of Donald Trump in 2024 was already set to upend EV market developments in North America, given his plans to revoke his predecessor's clean energy spending and related policies. Trump's subsequent tariff threats against neighbours and allies have increased the potential for disruption and heightened risks across the EV supply chain. Even if the tariff threats ultimately lead to some grand bargain, it is virtually certain that the prospects of 'friend-shoring' EV and battery production to North America will dim—but the question of how much is open to debate. *Milo McBride* argues that Washington's 'increasingly bombastic' posturing on trade, and efforts to overturn or simply withhold elements of the Inflation Reduction Act (IRA), will hamper US efforts to get back into the battery and EV game. McBride notes that friend-shoring was central to the IRA's promise: its subsidies require that EVs and batteries be produced in North America, and that minerals be produced in free-trade aligned countries. Grants were allocated through the Infrastructure Investment and Jobs Act as well as through the Defense Production Act, including to mines in Canada. Efforts at mining—especially lithium, where the US has made most progress—are at risk. Even more so are projects aimed at processing or battery component manufacturing, where mineral inputs may not be available from the US, or low minerals prices imply the need for the ongoing policy support the IRA provided. Perhaps just as important, negative policy signals and the withdrawal of subsidies and other policy support will slow domestic EV demand, hurting investment in technology and manufacturing alike.

Ray Cai takes a more optimistic tone, albeit agreeing with McBride that Trump's 'policy headwinds' will inevitably affect investment decisions and EV market growth. Cai argues, however, that there exists a growing bipartisan recognition of battery manufacturing as both a national security priority and economic growth driver. Given that EVs will inevitably replace conventional vehicles in the long run, the expansion of US battery production will likely continue. Yet Cai also notes the US faces major challenges—and not just from policy emanating from Washington. For one, the US battery manufacturing value chain needs to more than quadruple its workforce from 2022 levels by 2030. While domestic production capacity is expanding, US firms also have a technology and scale disadvantage compared to the Chinese battery and EV industry leaders. Even if domestic production continues to grow, the US remains far behind and is expected to produce only 10 per cent of global battery capacity for battery-powered EVs in 2027, up modestly from 6.6 per cent in 2021.

The US and Europe have focused on friend-shoring minerals production and increasingly supply, but reducing demand for key battery minerals could mitigate geopolitical and supply risks, and do so more quickly and cost-effectively than efforts focused on mining. *Will Atkinson* argues that battery minerals are not going to become 'the new oil'. In Atkinson's view, efficiency, innovation, and circularity will drive peak demand for mined minerals within a decade, as the world transitions from an extractive model to a circular loop. He presents six solutions to mitigate the need for mineral mining: deploying new battery chemistries, making batteries more energy-dense, recycling their mineral content, extending their lifetime, improving vehicle efficiency, and improving mobility efficiency. Atkinson notes the recent example of lithium-iron-phosphate batteries, which have cut the global nickel and cobalt demand per battery by more than half in just a few years. New chemistries and smaller batteries will help, but only advances in recycling can lead to a true circular economy. Atkinson argues that recycling innovations are progressing, leading to higher quality and lower costs, even with some materials like graphite that are hardest to recover.

Latin America: pursuing different options

The US reversal on trade and industrial policies is set to have an outsized impact on its neighbours, Canada and Mexico. These countries' EV and industrial development strategies may have to go back to the drawing board. In the case of Mexico, however, there is already a plan to draw upon. As *Francisco Parés Olguín, Aditya Ramji, Roland Hwang, and Juan C. Garcia Sanchez* write, the Plan México initiative focuses on boosting the domestic auto industry, and calls for the development of Olinia, a Mexican-made EV tailored to domestic needs. The plan goes together with a Made in Mexico campaign that aims to boost domestic consumption and reduce reliance on exports, particularly as tensions with the US continue to escalate. The author notes that EVs are central to Mexico's strategy: despite internal combustion engine vehicles continuing to constitute the bulk of Mexico's vehicle production, most of its industry's profit growth depends on its ability to supply projected EV demand. The author writes that several developments will be required for Mexico to succeed in the field. These include encouragement of Chinese investment and technology transfer, tightened vehicle standards and better charging infrastructure to encourage domestic EV demand, and reduced reliance on the US market through focus on other trade partners in Latin America.

While countries are a handy unit of analysis, subnational actors also play important roles—and Mexico is no exception. In his contribution, *Carlos Guadarrama* examines the case of Sonora, one of the poorer Mexican states, which already hosts vehicle assembly capacity as well as large reserves of lithium. Under the northern state's own Plan Sonora, policymakers hope to make the state an EV manufacturing hub. The challenges are immense, however. While Mexico has attracted US carmakers to invest—including Ford in Sonora starting in 1986—high-value-added activities such as research and development, design, and advanced manufacturing have remained concentrated in the US. For this reason, Sonora is turning to its upstream advantages: the state has Mexico's largest lithium reserves, which had attracted the Sonora Lithium Project, operated by Bacanora Lithium and partially owned by China's Ganfeng Lithium. However, for Plan Sonora to succeed, Guadarrama writes, greater coordination is needed at the national and local levels, to clarify laws around mineral ownership as well as to promote technology transfer and innovation rather than simple assembly.

Like Mexico, Brazil already has an automotive manufacturing sector, but there are at least three key differences. First, as *José Carlos Díaz Silva* notes, Brazil's car manufacturing is focused on the low-cost Latin American market, including a large domestic market, both of which are potentially more susceptible to a transition to EVs as Chinese offerings reach the market. Second, Brazil has long promoted biofuel and flexible fuels, meaning EVs are certain to coexist with conventional technologies promoted by policy. Third, Brazil is already home to some substantial Chinese EV investments, which are beginning to take market share from other foreign competitors. Díaz Silva notes that government policies favour public investment and private investment in the EV space, and expects that the arrival of Chinese companies, especially BYD, is resulting in not only assembly but also establishment of a complete production chain. Increasingly, this is reorienting Brazil's trade links towards Asia and not with the West, meaning Brazil is partnering with the technological leader in the EV industry. The large Latin American market and the receptivity of government policy to Chinese investment will incentivize Chinese players to relocate supply chains and upgrade Brazil's manufacturing sector, Díaz Silva writes, potentially avoiding some of the pitfalls to the EV transition in other countries.

Also writing on Brazil's EV transition, *Alexis Harmon* largely agrees that Brazil has some of the ingredients to become a major regional player for manufacturing low-cost EVs, in partnership with Chinese firms. Harmon notes that tariffs and industrial policy are already drawing investment: although the initial flood of Chinese EV imports was a response to China's domestic overcapacity, Chinese firms are now investing billions in local production and mines. Since some of these investments will require a decade to come online, this represents a sizeable long-term commitment. Chinese firms are also leaders in plug-in hybrids, which Harmon notes are likely a better match for Brazil's long-term promotion of biofuels. However, promotion of plug-in hybrids and biofuels carries risk: although Brazil's large domestic market allows it to diverge from global trends, delaying full EV adoption for too long could leave its manufacturers at a competitive disadvantage. Further, Harmon notes that while Brazil is working to leverage its lithium deposits and silicon and graphite production into higher-value areas of the battery supply chain, it is unlikely to become a major global EV manufacturing hub. Instead, given its relatively high costs, geographic distance, and uncompetitive trade policies, Brazil is likely to remain a regional leader.

Africa and Southeast Asia: seeking to avoid new import dependencies

Unlike the countries discussed above, the vast continent of Africa plays a minor role in global vehicle manufacturing, and is at risk of once again becoming only a zone of exploitation for the valuable minerals that EVs need. However, a few countries do have some potential to take advantage of existing supply chains to move up the value chain and upgrade manufacturing for the future EV era. In their article, *Godwin Kafui Ayetor, Joseph Mashele, and Issaka Lawerh Tetteh* discuss the particular cases of Morocco, South Africa, and Ghana, each of which is pursuing different strategies, based on their unique situations. Morocco is perhaps the most emblematic example of the production potential of EVs in Africa. The country is working to transition to fully decarbonized automotive supply chains and exploit the growing global EV market, and it is home to 70 per cent of the world's phosphate reserves and is the tenth largest producer of cobalt globally. However, Morocco and other African countries face major challenges, given their current industries' focus on low-value assembly, and they lack human resources to compete in high-tech fields. Ayetor and colleagues highlight the need for Africa to process its minerals before export, enhance labour skills, and establish intra-African component manufacturing and supply value chain clusters through infrastructural and industrial policies.

Writing on Southeast Asian EV supply chains, *Nancy Wei* also notes that countries are pursuing vastly different strategies. Relatively speaking, the countries in this region have more developed supply chains, but they still face some of the same disadvantages. As a result of their different situations, Thailand is focused on attracting manufacturing facilities, while Vietnam is promoting a national champion, Vinfast, and Indonesia is seeking to leverage its minerals wealth to move into processing and other higher-value activities. Major challenges include the small size of each country's domestic market, a lack of charging infrastructure across far-flung rural areas, and the perceived low quality of low-cost EVs currently available in the market. Collaboration with Chinese EV manufacturers offers the potential for accelerated growth and infrastructure development, yet it

also brings accompanying challenges, such as dependency on imported technology and the need for political diplomacy amidst global geopolitical tensions. Wei notes that though the EV transition may yield substantial domestic and regional economic benefits, it will need to be managed with foresight to ensure it contributes to sustainable growth, technological upgrading, and long-term economic development goals.

Europe and China: the co-opetition model

Though the US IRA was perhaps a more thoroughgoing industrial policy than those adopted in Europe, immense uncertainty now hangs over those policies, and over US supply chains in general. Europe is already a far larger EV market than the US, and the continent has been playing catch-up in the field of industrial policy as well. First and foremost, as *Julia Poliscanova* writes, the EU has been attempting to build a local battery value chain, notably cell manufacturing, since the creation of the European Battery Alliance in 2017. Since then, more than 1,000 GWh of battery manufacturing capacity has been announced across the member states by 2030, more than enough to meet the local demand. However, Poliscanova notes, European players are mainly start-ups at higher risk of delays and facing scaling difficulties. The now well-known story of Northvolt's struggles is not a unique case: at least 100 GWh of the planned EU-led capacity was cancelled or delayed in autumn 2024 alone. Meanwhile, Europe faces a dilemma over cooperation versus competition with China. Chinese and South Korean companies' shares of Europe's announced 2030 battery gigafactory plans are 27 per cent and 13 per cent, respectively. Reliance on Chinese players is even more pronounced on the battery component side (anodes and cathodes), underlining the growing role of Chinese companies within EU's battery ecosystem. Currently, Chinese investments in Europe are largely focused on securing affordable battery supplies for EU automakers' operations, while comprehensive local knowledge or technology transfer provisions are missing. Poliscanova argues Europe needs to address policy gaps to ensure capacity build-out goes hand-in-hand with technology catch-up, including through domestic content requirements, simpler and streamlined policies on subsidies and state aid, and more focus on trade and industrial policy versus simple promotion of R&D.

The case of Hungary is almost a microcosm of the challenges Europe faces, both in catching up and in figuring out how to work with China. *Ágnes Szunomár* writes that despite trade tensions, Chinese EV companies are gaining a foothold in European markets, in part with batteries made in Hungary, host to most of the Chinese EV battery projects in Europe. Hungary already ranks fourth globally in EV battery production, and batteries made in Hungary can be employed locally by European carmakers with a presence such as Audi, BMW, and Mercedes. However, Hungary's battery policies have clear tensions with EU policy. The EU's priority is to develop European batteries and create an indigenous European value chain, whereas the Hungarian strategy is based mainly on attracting investment from Asian companies, rather than upgrading the country's own manufacturing capability to move up the technology value chain. In the short and medium term, Chinese batteries will certainly be needed to meet European climate targets. In the longer term, however, as European innovation advances and local content policies grow more strict, Chinese-owned or -invested factories in Europe may become a liability.

Meanwhile, technology continues to race forward

Though most of the articles in this issue deal with strategies for catching up to China's lead in EV technology, the two concluding articles turn to the topic of China itself, which is hardly standing still in EV technology. *Julian Schwabe* looks into the topic of autonomous vehicles, which have immense potential to transform the industry of personal transportation. According to one estimate, 66 per cent of total passenger-kilometres driven in road transport in 2040 will be by autonomous vehicles. While electrification and self-driving functions can be adopted independently of each other, the current robotaxi fleets and vehicles with advanced autonomous driving capability are almost all EVs. Not surprisingly, China is an early leader, due to many of the same factors that propelled its clean energy and EV sectors. Cities are driving deployment, startups are rushing in and competing to drive down prices and improve quality, and government is rolling out supportive measures for investment and deployment while also introducing restrictions on data and mapping that are, in effect, domestic content requirements. As with other clean technologies where China excels, China's autonomous driving start-up landscape is also highly competitive, driven by rapid technological advancements and significant investment. Meanwhile, international adaptation of autonomous driving technology is hindered by security and safety concerns as well as geopolitical tensions. China's advancing lead in EVs and autonomous driving should be a concern to policymakers and automakers globally.

Also writing about China and its technological lead in EVs, *Anders Hove* examines how ongoing innovation is enabling Chinese players to introduce groundbreaking new battery chemistries that have the potential to further expand the country's lead. Chinese firms are advancing in emerging battery technologies due to a combination of a thriving and ultra-competitive domestic market plus years of accumulated battery manufacturing know-how. Fierce domestic competition, not only within China but even within the same province or region of China, means companies have to move fast and not rest on a few early successes like Tesla with the Model 3 and Y. Further, China's vertically integrated or clustered supply chains enable rapid interactions with key suppliers,

driving innovation and helping develop the local knowledge and skills base that can help scale up production. These twin advantages make it difficult for other world regions to fully catch up technologically to China. Still, there are potential strategies to consider, and Europe is taking the lead in adapting successful industrial policies from the China EV playbook, such as requiring technology transfer and licensing as a condition for receiving subsidies. Domestic content requirements for critical minerals and batteries are on the books or under discussion. Often, such policies do not need to explicitly exclude Chinese players. To the contrary, Europe and other regions are now in the position of needing to work together with Chinese suppliers to ensure access to the latest technology and processes.

NORTH AMERICAN BATTERY METALS ENTER A NEW ERA OF UNCERTAINTY

Milo McBride

Across the US and Canada, efforts to onshore battery metal production face risks in a new policy, trade, and market environment. Despite significant efforts to re-shore parts of this supply chain during the Biden administration, and the stated interest in mineral security from the second Trump administration, recent political signals suggest the outlook for re-shoring is precarious. Billions of dollars in grants and low-interest-rate loans pledged for electric vehicles (EV)- or battery-supply-chain-related projects have been impounded by the new administration. Calls to repeal subsidies for EV battery metals have become more pronounced. Despite the Trump Administration's pledges for minerals financing and expedited permitting, Washington's increasingly aggressive trade policy could further hamper US efforts to get back into the battery and EV game, and pose new risks for regional mineral production.

Currently, various battery metals are at different development stages in North America. Lithium extraction from hard rock, geothermal brines, and oilfield brines is a notable area where US companies have made progress, with some examples of chemical conversion also coming online. Graphite and anode-related products have seen fewer developments, but there are important hard rock plays in Canada and material production across the United States. Other notable battery products including nickel and cobalt remain almost entirely developed in Canada. Phosphate and manganese chemicals are in short supply, which could hinder efforts to localize cathode material production. Developers of new mines and processing facilities will now need to navigate higher levels of policy uncertainty and political risk surrounding the business environment in the US and its trade connections with Canada.

Background

Over the past couple of years, North America's EV supply chain has emerged on top of an already intertwined trade architecture, especially for the automobile manufacturing sector. Under the original North American Free Trade Agreement and subsequently under the United States-Mexico-Canada Agreement (USMCA), the Inflation Reduction Act's (IRA's) EV subsidy, legislatively known as '30D', readied this existing framework for automobile production in the 21st century.[2] The IRA's subsidy requires that EVs and batteries be produced in North America, and that minerals be produced in free-trade-aligned countries—which include both Mexico and Canada.[3]

Alongside the IRA's demand-pull subsidy for North American batteries and metals, both the US and Canada initiated a wave of additional policies that could benefit regional mineral production. Alongside a modest production tax credit for minerals, the IRA issued a range of investment tax credits for supply chain development, and expanded the Energy Department's low-interest-rate loan authorizations, which benefited multiple mineral projects.[4] Additional grants were allocated through the Infrastructure Investment and Jobs Acts (IIJA) as well as through the Defense Production Act, including to mines in Canada.[5] North of the border, Ottawa responded with its own green industrial policy in the 2023 budget that includes more lucrative tax credits for mining, as well as government funding for projects and infrastructure adjacent to mines.[6]

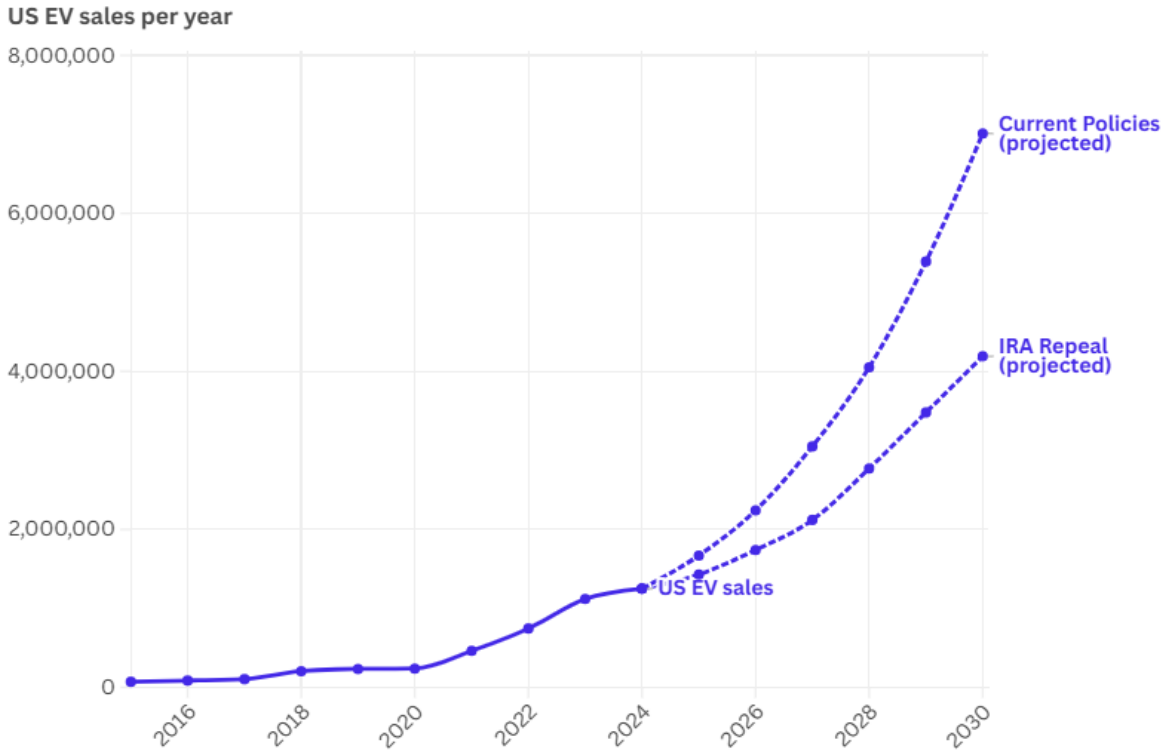
[2] *2024 USMCA Autos Report to Congress*, United States Trade Representative, 1 July 2024, https://ustr.gov/sites/default/files/2024%20USMCA%20Autos%20Report%20to%20Congress_0.pdf.
[3] 'Section 30D new clean vehicle credit', *Federal Register*, 17 April 2023, at https://www.federalregister.gov/documents/2023/04/17/2023-06822/section-30d-new-clean-vehicle-credit.
[4] Meaghan Connors, 'The Inflation Reduction Act as a mining finance alternative', *Mayer Brown*, 5 March 2024, https://www.mayerbrown.com/en/insights/publications/2024/03/the-inflation-reduction-act-as-a-mining-finance-alternative.
[5] Milo McBride and Daniel Helmeci, *Minerals, Manufacturing, and Markets: Foreign Policy for US Energy Technology and Minerals*, Carnegie Endowment for International Peace, 26 February 2025, at https://carnegieendowment.org/research/2025/02/minerals-manufacturing-and-markets-foreign-policy-for-us-energy-technology-and-minerals/?lang=en.
[6] Government of Canada, 'A made-in-Canada plan: affordable energy, good jobs, and a growing clean economy', chapter 3 in *Budget 2023*, 28 March 2023, at https://www.budget.canada.ca/2023/report-rapport/chap3-en.html.

Despite these policy signals, a confluence of inconvenient market dynamics, looming political shifts, and cumbersome rulings on subsidies all impacted the scale of regional mineral project development. To start, mineral investments have been modest compared to those in battery manufacturing: in 2024, about $2.6 billion was invested in US mineral projects, while an astounding $43.6 billion went toward battery manufacturing.[7] Barring a few exceptions, North American mineral projects are the work of 'junior miners' that lack large internal cashflow and thus face elevated borrowing costs.[8]

These challenges were further complicated by the time it took to define relevant subsidies. It took several years after the legislation was passed to disburse mineral-related grants from the IIJA, choose recipients from 48C, which provides a 30 per cent investment tax credit, and clarify rulings to include mineral extraction in 45X, a comprehensive supply-side tax credit for domestic clean tech manufacturing.[9] Most notably, the US Treasury Department took nearly two years to rule on the intricacies of 30D, the previously mentioned subsidy for EV batteries and minerals.[10] This ruling signified some of the most difficult industrial policy decisions that the Biden administration faced, including the balancing of EV sales for the sake of climate ambition and industrial momentum, with the need to develop some semblance of domestic mineral, chemical, and battery production.

30D is a potent and innovative policy tool that carved out a market for North American minerals. Given that US battery energy stationary storage is a smaller market than EVs and that its 30D equivalent does not extend to minerals, US EV sales are thus the main driver of the region's battery metals demand. But American EVs face an uncertain future, with a delta of about 3 million annual car sales in 2030 between the scenario where the IRA is left intact and the scenario where it is repealed (Figure 1).[11]

Figure 1: US EV outlook: IRA vs. IRA repeal



Source: Princeton REPEAT Project, Argonne National Laboratory. https://zenodo.org/records/15001499; https://www.anl.gov/esia/reference/light-duty-electric-drive-vehicles-monthly-sales-updates-historical-data

[7] *Clean Investment Monitor*, accessed 8 March 2025, at https://www.cleaninvestmentmonitor.org/.
[8] *The Big Green Machine*, accessed 9 February 2025, at https://www.the-big-green-machine.com/.
[9] Daniel T. Kiely et al., 'Treasury and IRS issue final regulations on Section 45X', *Mayer Brown*, November 2024, at https://www.mayerbrown.com/en/insights/publications/2024/11/treasury-and-irs-issue-final-regulations-on-section-45x; Paul Ciampoli, 'DOE announces over $3 billion to support US battery manufacturing sector', *Public Power*, 20 September 2024, at https://www.publicpower.org/periodical/article/doe-announces-over-3-billion-support-us-battery-manufacturing-sector.
[10] 'Clean vehicle credits under sections 25e and 30d; transfer of credits; critical minerals and battery components', *Federal Register*, 6 May 2024, at https://www.federalregister.gov/documents/2024/05/06/2024-09094/clean-vehicle-credits-under-sections-25e-and-30d-transfer-of-credits-critical-minerals-and-battery.
[11] Jesse Jenkins, 'Potential impacts of electric vehicle tax credit repeal on US vehicle market and manufacturing', *Princeton University ZERO Lab*, 10 March 2025, at https://zenodo.org/records/15001499.

The Trump administration's U-turn on nearly all Biden-era energy and industrial policies will further challenge investor and industry confidence. As of writing, grants and loans for clean tech projects have been impounded, including those related to bolstering mineral supply security.[12] Despite the majority of EV and battery developments going to Republican districts and states, conservatives in Congress and the Trump administration are threatening to repeal Biden's various green industrial policies.[13] Two battery gigafactory projects have already been cancelled—Kore Power's $1.2 billion facility in Arizona and Freyr's $2.6 billion plant in Georgia.[14]

In addition to pulling back on Biden-era EV policies, President Trump has imposed an unprecedented and severe tariff regime which could dampen automobile demand at a critical moment when it is needed stimulate the production of regional battery metals. Alongside implementing minimum 10 per cent tariffs across the board and between 54 to 63 per cent tariffs on China (exact estimates vary), 25 per cent tariffs have been imposed on all steel, aluminium, and auto part imports (excluding USMCA compliant auto parts from Canada and Mexico).[15] This will likely increase sticker prices for passenger vehicles at a moment of already precarious consumer confidence.[16] Uncertainty adds to the challenges: US tariff policy is increasingly unpredictable and subject to arbitrary changes: the Trump administration's previously-announced tariffs have been walked back or altered and others proved more drastic than markets had priced in.[17]

Although modest tariffs have already been applied to US battery metal imports, it is unclear if or when more severe levies will be issued. The 2 April US tariff announcement exempted key battery metals including lithium cobalt, manganese, and natural graphite, but this could be subject to change.[18] In particular, there is currently a Section 232 trade investigation underway regarding copper, and media reports indicate that a similar investigation could be instigated to cover minerals more broadly.[19] Although not as severe as the 2 April tariffs, previous measures have impacted regional battery material costs. They include raising Chinese import fees 20 per cent (totalling to 45 per cent), issuing a 10 and 25 per cent levy on Canadian and Mexican minerals respectively, and the aforementioned unilateral 25 per cent tariff on aluminium, which is used as a current collector in battery cells.[20] It remains unclear whether tariffs of this size are significant enough to stimulate domestic production, but it is possible that they will solely

[12] Emily Pontecorvo, 'Flabbergasted by the funding freeze? here's what could happen next', *Heatmap*, 24 February 2025, at https://heatmap.news/politics/funding-freeze-what-happens-next; Jake Pearson and Anjeanette Damon, 'The courts blocked Trump's federal funding freeze. Agencies are withholding money anyway', *Pro Publica*, 10 February 2025, at https://www.propublica.org/article/trump-administration-funding-freeze-workarounds; Lindsay McKenzie, 'Trump spending freezes sow confusion among researchers', *American Institute of Physics*, 30 January 2025, at https://ww2.aip.org/fyi/trump-spending-freezes-sow-confusion-among-researchers.

[13] Jeff Young, 'Trump's removal of EV support could risk thousands of red state jobs', *Newsweek*, 23 January 2025, at https://www.newsweek.com/trumps-removal-ev-support-could-risk-thousands-red-state-jobs-2019326 ; Jarrett Renshaw et al., 'Trump's transition team aims to kill Biden EV tax credit', *Reuters*, 14 November 2025, at https://www.reuters.com/business/autos-transportation/trumps-transition-team-aims-kill-biden-ev-tax-credit-2024-11-14/ ; 'Production underway at dozens of US electric vehicle manufacturing sites after historic levels of investment', *Environmental Defense Fund*, 8 January 2025, at https://www.edf.org/media/production-underway-dozens-us-electric-vehicle-manufacturing-sites-after-historic-levels.

[14] Sara Samora, 'Freyr Battery cancels $2.6B Georgia battery factory plans', *Manufacturing Dive*, 19 February 2025, at https://www.manufacturingdive.com/news/freyr-battery-cancels-georgia-battery-plant-pivot-solar-module-cell/740250/#:~:text=Freyr%20Battery%20called%20off%20plans,month%20to%20Coweta%20County%20officials.; Julian Spector, 'Kore Power has a new plan after canceling $1.2B battery plant in Arizona', *Canary Media*, 7 February 2025, at https://www.canarymedia.com/articles/energy-storage/kore-power-has-a-new-plan-after-canceling-1-2b-battery-plant-in-arizona.

[15] *Trump Tariff Tracker,* Atlantic Council*,* accessed 4 April 2025, https://www.atlanticcouncil.org/programs/geoeconomics-center/trump-tariff-tracker/; Nathan Janzen and Claire Fan, 'U.S. reciprocal tariffs spare Canada, Mexico for now, but trade risks remain', *RBC Thought Leadership*, 6 April 2025, at https://thoughtleadership.rbc.com/u-s-reciprocal-tariffs-spare-canada-mexico-for-now-but-trade-risks-remain/; Joshua Kagan, 'Tariff Update: List of Products Covered by the Autos Tariffs', *Kelley Drye & Warren LLP*, 3 April 2025, at https://www.kelleydrye.com/viewpoints/blogs/trade-and-manufacturing-monitor/tariff-update-list-of-products-covered-by-the-autos-tariffs; David E. Bond et al**.**, 'President Trump expands steel and aluminum tariffs to all countries; effective March 12, 2025', *White & Case LLP*, 16 February 2025, at https://www.whitecase.com/insight-alert/president-trump-expands-steel-and-aluminum-tariffs-all-countries-effective-march-12.

[16] 'US Consumer Confidence tumbled again in March', *The Conference Board*, 25 March 2025, at https://www.conference-board.org/topics/consumer-confidence/press/CCI-Mar-2025.

[17] Stephan E. Becker et al**.**, 'U.S. Tariffs on Non-USMCA-Compliant Products Take Effect; Increased Tariff Rate on China Imposed', *Pillsbury Winthrop Shaw Pittman LLP*, 10 March 2025, at https://www.pillsburylaw.com/en/news-and-insights/tariffs-usmca-compliant.html; Howard Schneider, 'Fed's Powell says larger-than-expected tariffs likely to boost inflation, slow growth', *Reuters*, 4 April 2025, at https://www.reuters.com/markets/us/feds-powell-weigh-amid-tariff-fray-market-drop-2025-04-04/.

[18] 'Executive Order 14656', *The White House*, 2 April 2025, at https://www.whitehouse.gov/wp-content/uploads/2025/04/Annex-II.pdf.

[19] Ewa Manthey, 'Industrial Metals Monthly: US copper tariff risks pull more metal to the US', *ING Think*, 19 March 2025, at https://think.ing.com/articles/industrial-metals-monthly-us-copper-tariff-risks-pull-more-metal-to-the-us/; 'US floats idea of launching critical minerals tariff probe', *Mining.com*, 3 April 2025.

[20] Ian Saccomanno and Samuel Scoles, 'US Tariffs on Canada and Mexico Enter into Effect; Tariff on China Rises from 10% to 20%', *White & Case LLP*, 4 March 2025, at https://www.whitecase.com/insight-alert/us-tariffs-canada-and-mexico-enter-effect-tariff-china-rises-10-20; David E. Bond et al**.**, 'President Trump expands steel and aluminum tariffs to all countries; effective March 12, 2025', *White & Case LLP*, 16 February 2025, at https://www.whitecase.com/insight-alert/president-trump-expands-steel-and-aluminum-tariffs-all-countries-effective-march-12.

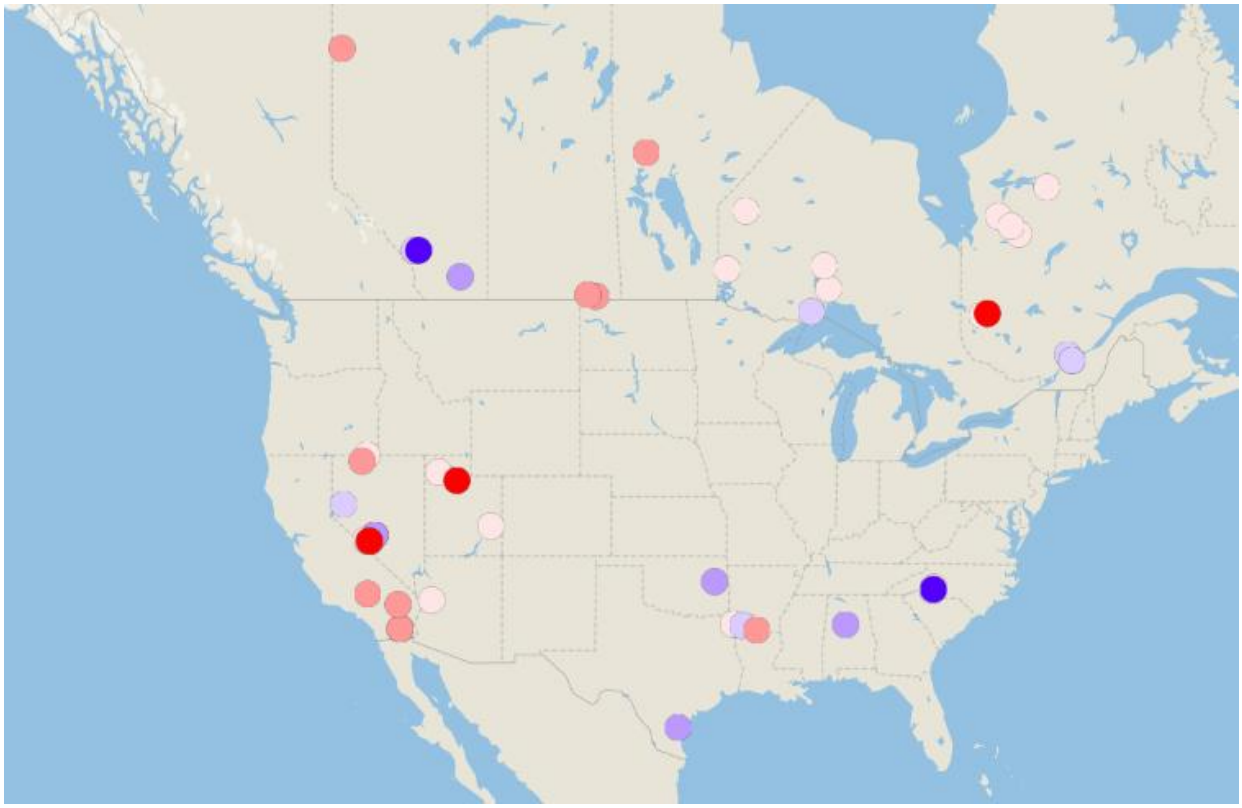
impose economic pain on domestic producers without providing a boost for new projects at home—especially if domestic demand for EVs also weakens. Support measures for domestic minerals will take time, and clarity is still lacking on exactly how they will be implemented. The White House's Executive Order to 'Increase American Mineral Production' could help certain mining projects through expedited permitting on federal lands, the use of foreign financing agencies for domestic projects, and federal government as a mineral off-taker.[21] But investment and finance depends on a thriving domestic EV market, which is now under threat. Escalating tariffs, inflation concerns, elevated interest rates, and general market uncertainty will stall investment in both the upstream and downstream. Progress on local mineral supply may not completely grind to a halt, but momentum will slow.

The following section illustrates where North American mineral production and refining are emerging and takes stock of which facilities—and which metals—are closest to domestic maturity.

Regional lithium production has moved forward

Lithium (Figure 2) is likely the greatest area of success in North American battery metal development. The US and Canada are well positioned as future lithium players due to diverse and rich reserves, presence of legacy lithium and chemical producers, and potential synergies in the oil and gas and geothermal sectors. As a result of these dynamics and supportive policy developments, an array of projects is relatively close to reaching commercial start. Several lithium hubs have started to emerge across the US and elsewhere in North America: direct lithium extraction (DLE) in southern Arkansas and California as well as hard rock mines in Nevada, Quebec, and Ontario. Fewer lithium processing facilities are in development, but there are projects in southern Quebec, as well as opportunities across Nevada, Oklahoma, and North Carolina.

Figure 2: North American lithium production



Note: red = extraction; purple = processing; dark colours = operational; medium = pilot/under construction; light = planned
Source: Author's visualization based on data from The Big Green Machine, Jay Turner, Wellesley College. https://www.the-big-green-machine.com/

There are three active lithium production sites in North America, a dozen at pilot stage or under construction, and about 20 additional projects at some more preliminary planning stage. The three projects currently operational are American Battery Technology's DLE site in Utah, Albemarle's Silver Peak mine in Nevada, and Soyana's mine in Quebec. Nevada's hard rock mines, Thacker Pass and Rhyolite Ridge, both received significant funds from the Biden-era Energy Department, but have faced

[21] 'Executive Order 14241, Immediate measures to increase American mineral production', *The White House*, 21 March 2025, at https://www.whitehouse.gov/presidential-actions/2025/03/immediate-measures-to-increase-american-mineral-production/.

challenges in the path to commercial development. Thacker Pass has faced ongoing pushback from local communities, and a prominent investor in Rhyolite Ridge has recently pulled out, citing bearish sentiments in the lithium market.[22] There are also new mine projects in Quebec and Ontario that are at early planning stages.

Critical to the long-term potential of North American lithium will be whether various DLE projects using geothermal, continental, and oilfield brines can scale up amid the chaotic policy backdrop and a weak minerals price environment. California's Salton Sea has three geothermal DLE projects, one of which is moving to demonstration stage.[23] The Smackover formation in Arkansas has four projects that include US oil majors and chemical giants. E3's DLE project in Alberta has received ample state backing and is progressing towards commercial demonstration with an unprecedented potential at about 200,000 tons lithium carbonate equivalent (about 100 or 200 times the average lithium mine).[24]

Critical questions for DLE will be where these projects land on the cost curve and whether US projects will struggle without access to Chinese sorbents, a filtration technology that the Chinese government recently restricted for export.[25] Most DLE technologies require sorbents to pull lithium molecules from brine, but these systems are mostly produced in China and have become the latest target of Beijing's export control regime, potentially thwarting some American DLE pilot projects seeking to scale.[26]

Despite some co-location with DLE sites, there are fewer lithium refineries than upstream projects. At present, Livent's facility in North Carolina is the only operational, stand-alone lithium refinery in North America, but another four or five facilities are in either pilot or construction phase across Oklahoma, Texas, Nevada, and Alabama, with nine additional refineries in the planning stage including many across Canadian provinces. Overall, there is a surprising balance of lithium carbonate and hydroxide production (especially when including DLE chemical outputs), implying that North American chemical producers are expecting downstream development of both lithium-nickel-cobalt-manganese/aluminium and lithium-iron-phosphate (LFP), because these two chemicals are used as different feedstocks to different batteries.[27] But the lithium chemical market also faced headwinds in 2024: Piedmont Lithium's Tennessee-based hydroxide facility was cancelled, and Albemarle announced an indefinite pause on building a massive new facility in South Carolina.[28] In sum, there are many North American lithium facilities in development, the majority in planning stages, but economic and policy uncertainty about market demand put these projects at risk.

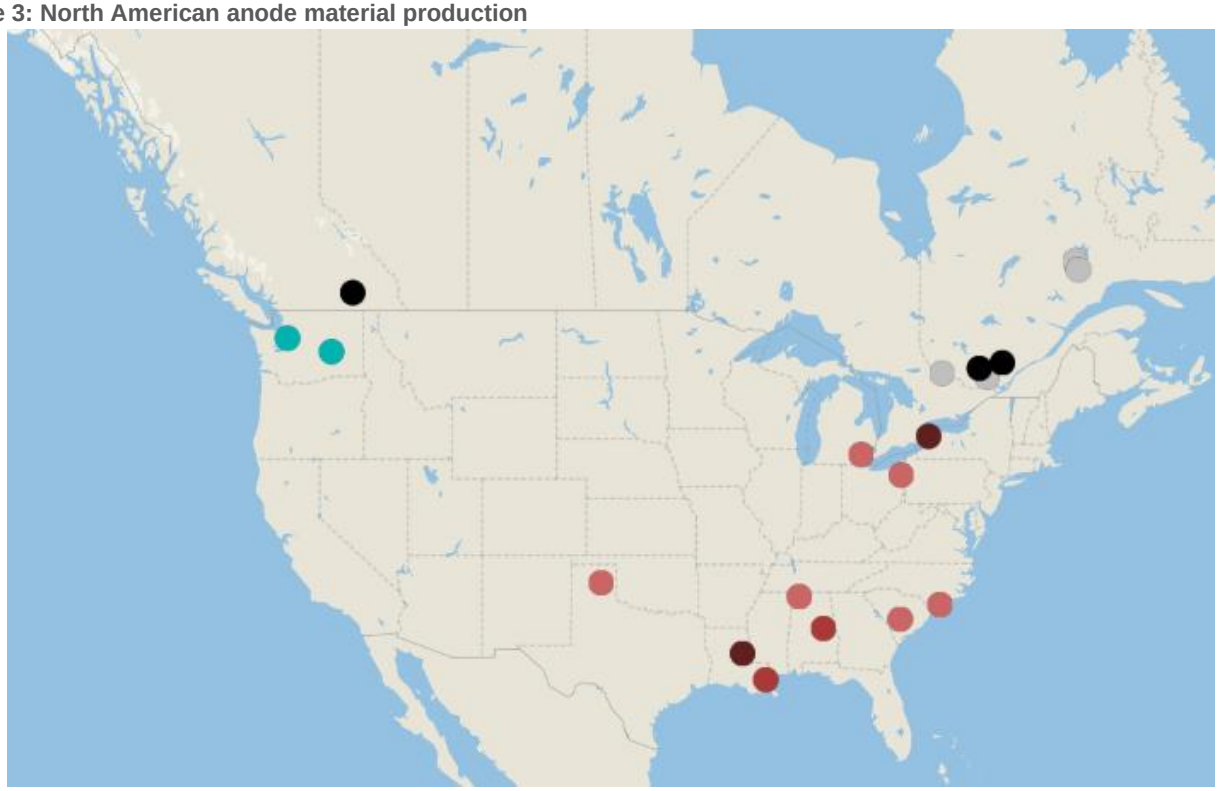
Graphite development is split between US and Canada

Compared to lithium, there are fewer anode material projects across North America (Figure 3), but some key developments are emerging for natural graphite flake, synthetic graphite, and anode material production, as well as next-generation silicon materials. The most notable trend is the stark split between US and Canadian developments: the entirety of North American natural graphite mines are in Canada, with a notable concentration in Quebec. Three mines are at early operational stages, while another five remain in planning. Two graphite processing facilities are online in the US: Syrah Resource's Louisiana factory, which uses Mozambique flake as input for active anode material, and Anovion's synthetic graphite factory in New York state

Across key battery hubs like the American Southeast and the Rust Belt are an additional eight facilities in the planning stage aiming to produce synthetic graphite and related anode materials. In Washington State, several firms are pioneering silicon-based anodes that come with their own technology risks and rewards.[29]

[22] Cecilia Jamasmie, 'Sibanye walks away from Rhyolite Ridge lithium project on weak prices', *Mining.com*, 26 February 2025, at https://www.mining.com/sibanye-walks-away-from-rhyolite-ridge-lithium-project-on-weak-prices/.
[23] Kori Suzuki, 'Imperial Valley's first lithium project moving forward again after court denies legal challenge', *KPBS*, 6 February 2025, at https://www.kpbs.org/news/environment/2025/02/06/imperial-valleys-first-lithium-project-moving-forward-again-after-court-denies-legal-challenge.
[24] 'E3 Lithium provides update on the Clearwater project and 2025 demonstration plant development plans', *Business Wire*, 25 February 2025, at https://www.theglobeandmail.com/investing/markets/markets-news/Business%20Wire/31092296/e3-lithium-provides-update-on-the-clearwater-project-and-2025-demonstration-plant-development-plans/.
[25] Ernest Scheyder and Lewis Jackson, 'Exclusive: Chinese lithium company halts tech exports as trade tensions build', *Reuters*, 18 February 2025, at https://www.reuters.com/technology/chinese-lithium-company-halts-tech-exports-trade-tensions-build-2025-02-18/.
[26] Jiayi Cen, 'Navigating the complex landscape of lithium extraction technologies', *Idtech*, 19 September 2024, at https://www.idtechex.com/en/research-article/navigating-the-complex-landscape-of-lithium-extraction-technologies/31726#:~:text=It%20also%20requires%20minimal%2Fno,the%20next%205%2D10%20years.&text=However%2C%20companies%20are%20gravitating%20towards%20adsorption%2Dbased%20DLE%20methods.
[27] 'Low-cost lithium carbonate incentivizes LFP battery switch', *Bloomberg New Energy Finance*, 16 June 2020, at https://about.bnef.com/blog/low-cost-lithium-carbonate-incentivizes-lfp-battery-switch/.
[28] 'Piedmont Lithium scraps Tennessee project, shifts approach on Gaston County mine', *Charlotte Business Journal*, 16 August 2024, at https://www.bizjournals.com/charlotte/news/2024/08/15/piedmont-lithium-carolina-gaston-tennessee-mine.html; 'Albemarle to cut staff, pause expansions amid falling lithium prices', *Reuters*, 17 January 2024, at https://www.reuters.com/markets/commodities/lithium-producer-albemarle-cut-workforce-lower-spending-2024-2024-01-17/.
[29] Matt Blois, 'Silicon could make car batteries better—for a price', *Chemical Engineering News*, 4 February 2024, at https://cen.acs.org/energy/energy-storage-/Silicon-make-car-batteries-better/102/i4.

Figure 3: North American anode material production



Note: black = natural graphite extraction (dark black = operational, grey = planned); red = graphite material processing (dark = operational, medium = pilot/under construction, light = planned); green = silicon anode
Source: Author's visualization based on data from The Big Green Machine, Jay Turner, Wellesley College. https://www.the-big-green-machine.com/

Despite these positive developments, the graphite supply chain faces several challenges. First, the economics of competing with China's world-renowned graphite industry are daunting. Despite the Trump administration's 20 per cent increased fee on Chinese products and the Biden administration's looming a 25 per cent tariff on graphite (to take effect on January 1, 2026), such a levies alone are unlikely to suffice, given present US costs.[30] The North American Active Anode Material Producers, an industry group, recently lobbied the US Congress for a 920 per cent tariff—a comically large number that, even when pricing in generous profit margins, implies a gargantuan cost differential with Chinese producers.[31] The 10 per cent tariff on Canadian products, including natural graphite flake, would add costs to American end-stage producers, since most anodes use a blend of synthetic and natural graphite inputs.[32] Finally, questions remain amid a looming trade war with China. China's Ministry of Commerce has now twice tightened its export flows on graphite, implying that this metal would be a logical next step in an outright ban in the escalating tit-for-tat high-tech trade war.[33] But even so, Chinese graphite could be circumvented at a premium, simply incurring costs to North American battery producers.

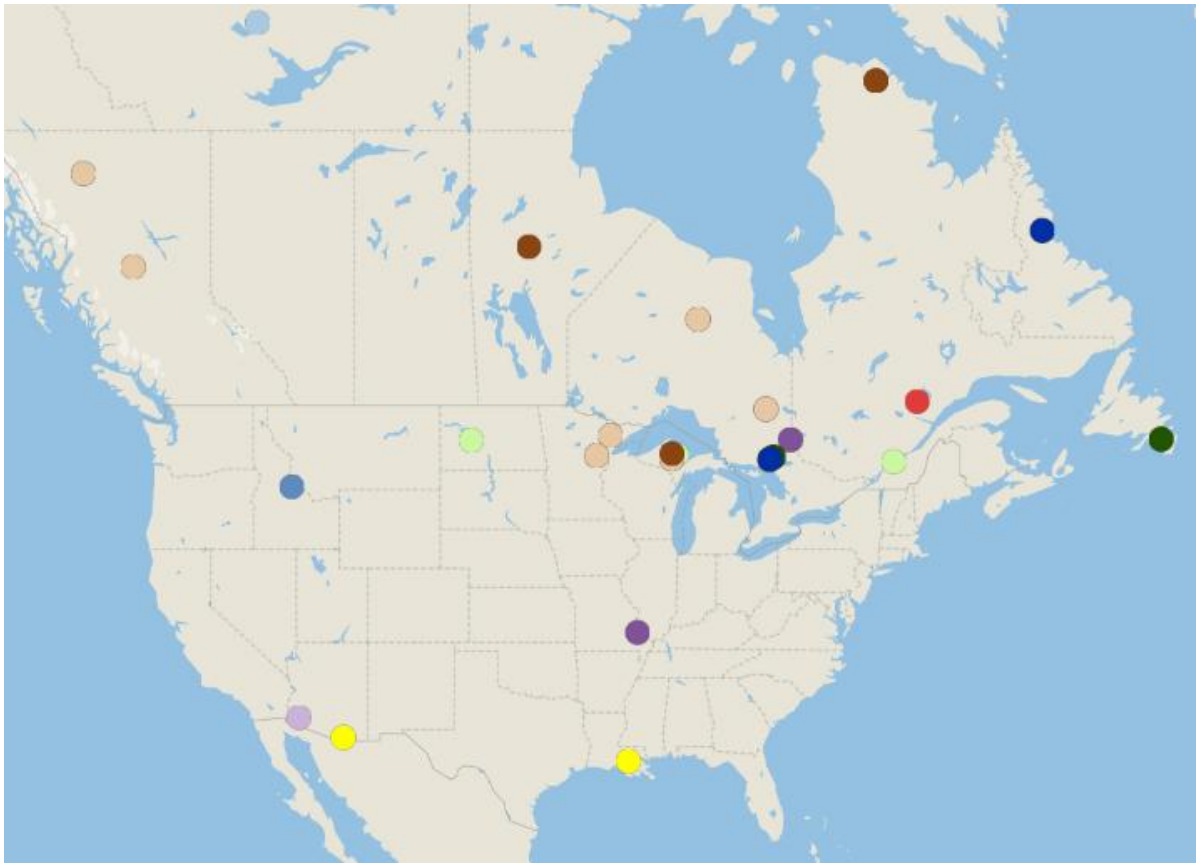
Other battery metals are (currently) a Canadian story

Other important battery metals (Figure 4) remain far less developed than lithium and graphite. Metals like nickel and cobalt remain almost entirely centred in Canada for extraction and processing, with some modest early projects in the United States. This is likely due to Canada's legacy nickel and cobalt industry with metal majors like Vale and Glencore, which have numerous operational projects across Ontario and Quebec, alongside several new developments in the pipeline. The only operational nickel mine in the US is the Lundin project in Michigan, which produces comparatively small quantities of the metal and is slated to close

[30] David E. Bond et al., 'United States finalizes section 301 tariff increases on imports from China', *White & Case*, 17 September 2024, at https://www.whitecase.com/insight-alert/united-states-finalizes-section-301-tariff-increases-imports-china.
[31] 'US graphite miners ask Washington to impose 920% tariff on Chinese rivals', *Reuters*, 18 December 2024, at https://www.reuters.com/markets/commodities/us-graphite-miners-ask-washington-impose-920-tariff-chinese-rivals-2024-12-18/.
[32] 'Rising synthetic graphite costs may push battery makers to rely on natural material', *FastMarkets*, 12 January 2022, at https://www.fastmarkets.com/insights/rising-synthetic-graphite-costs-may-push-battery-makers-to-rely-on-natural-material/#:~:text=Most%20anode%20technologies%20used%20by,graphite%2C%20in%20varying%20percentages%20depending.
[33] Isaac B. Kardon and Milo McBride, *Rocks vs. Chips: How the U.S.-China Competition is Playing Out for High-Tech Advantage*, Carnegie Endowment for International Peace, 14 February 2025, at https://carnegieendowment.org/research/2025/02/rocks-vs-chips.

later this decade. Two additional nickel mines are in early planning stages in northern Minnesota, while a cobalt mine in Idaho has struggled to overcome financial challenges, despite funding from the Defense Production Act.[34] Although the US and Canada are both home to cobalt chemical refineries, only Canada can produce the Class 1 nickel sulphate needed for battery cathode materials.

Figure 4: North American nickel, cobalt, phosphate, and manganese



Note: brown = nickel extraction; green = nickel processing; blue = cobalt extraction; purple = cobalt processing; yellow = manganese processing; red = phosphate extraction; dark colours = operational; light colours = planned
Source: Author's visualization based on data from The Big Green Machine, Jay Turner, Wellesley College. https://www.the-big-green-machine.com/

Often-neglected battery materials like battery-grade phosphoric acid and manganese sulphate are in short supply, with only one mine in Canada being planned to produce high-grade phosphate and two refineries in the US to supply manganese sulphate. Manganese is a globally traded metal with applications in the steel sector; the critical chokepoint is in refining manganese for battery-grade use. While these projects are a niche, they are nonetheless important considerations if North America is to indigenize LFP battery production, especially given the rising importance of manganese sulphate to extend the energy density of LFP technology in so-called lithium-manganese-iron-phosphate (LMFP) chemistries, which offer good energy density without the need for cobalt.[35]

The outlook for North American LFP and LMFP manufacturing and regional demand for its minerals and chemicals remains unclear. With low cost and increasingly strong performance, LFP now accounts for over 40 per cent of the global EV battery

[34] 'DOD enters agreement to expand domestic manufacturing and strengthen US cobalt supply chains', Department of Defense, 15 June 2023, at https://www.defense.gov/News/Releases/Release/Article/3429442/dod-enters-agreement-to-expand-domestic-manufacturing-and-strengthen-us-cobalt/#:~:text=The%20Office%20of%20the%20Assistant,14017; Ernest Sheyder and Eric Onstad, 'Cobalt miner Jervois in rescue deal to better compete with China', *Reuters*, 2 January 2025, at https://www.reuters.com/markets/commodities/cobalt-miner-jervois-rescue-deal-after-struggle-compete-with-china-2025-01-02/.
[35] Jian Zhao, 'Lithium manganese iron phosphate (LMFP) batteries receiving renewed attention in China, expected to be installed mainly in middle-class EVs', *Mitsui*, 19 September 2023, at https://www.mitsui.com/mgssi/en/report/detail/__icsFiles/afieldfile/2023/09/19/2308t_zhao_e.pdf.

market and is well positioned as an optimal chemistry for mass EV and stationary adoption over the coming years.[36] But nearly the entirety of LFP production comes from Chinese companies, leaving North America dependent on start-ups like Mitra Chem or Nano One and joint ventures with Chinese LFP incumbents like CATL and Gotion High Tech. In January, the Chinese government threatened to ban the export of LFP cathode equipment to the US, potentially thwarting the adoption of the technology there.[37]

Conclusion

North America's battery metals stand at a critical juncture. Massive policy support and amicable bilateral trade relations between the US and Canada have made it possible for a transborder supply chain to emerge. But fractious politics, antagonistic diplomacy, and protectionist trade regimes will challenge the scale of these developments. What's at risk is not merely a regional mineral ecosystem with national security applications, but a myriad of new extraction and processing technologies that could help North America develop sustainable and competitive supply chains related to a critical industry of the future.

THE US BATTERY VALUE CHAIN: A FORK IN THE ROAD

Ray Cai

The US battery industry stands at a critical inflection point: as electrification accelerates, the underdeveloped domestic and allied battery value chain must expand to address substantial supply–demand gaps and reduce dependence on foreign producers, which currently enjoy scale and cost advantages. Despite the momentum generated by recent US legislation, rising bipartisan interest, and private sector commitments, the current policy landscape and competing political priorities present major uncertainties that complicate progress across the value chain. Stakeholders must balance capacity expansion goals with challenges ranging from US federal funding freezes to new trade restrictions and domestic content requirements. As a result, a period of recalibration might commence as the industry works to reconcile evolving market and political dynamics.

Momentum in a shifting landscape

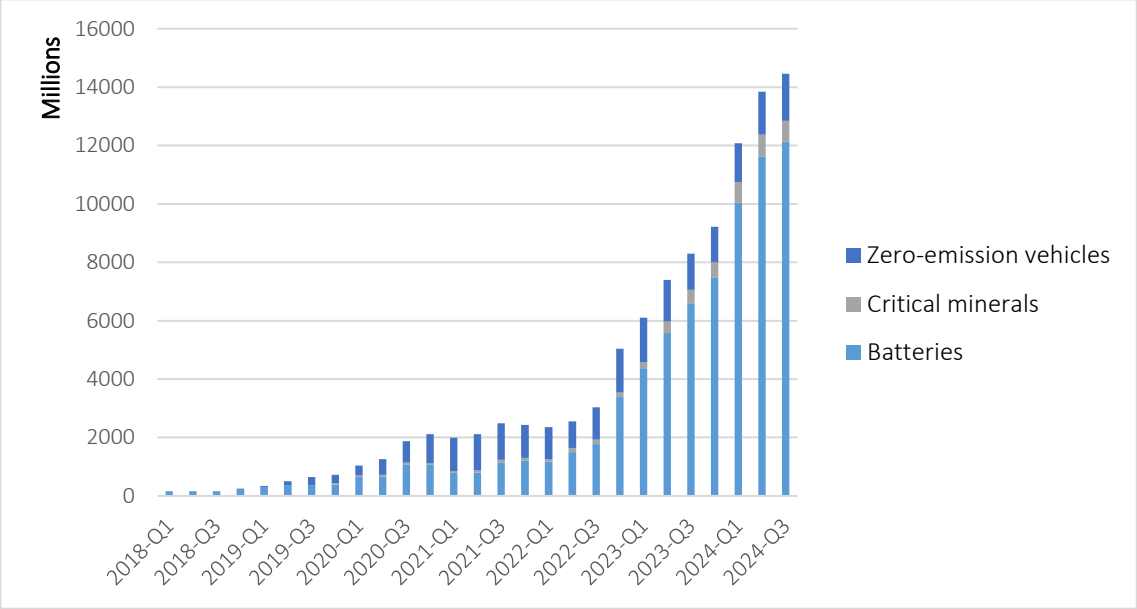
Batteries will likely play an increasingly vital role in the US energy system, as the broader trend of electricity demand growth is generally expected to persist.[38] By 2030, US demand for EV batteries could increase by over 500 per cent from 2022 levels to 911 GWh, while demand for grid-scale battery energy storage systems could grow six-fold to 119 GWh.[39] Key segments of the US value chain remain underdeveloped, however, and the country remains a net importer of batteries and battery inputs. For instance, by 2030 the US is projected to produce only 0.8 million metric tons of cathodes and 500,000 metric tons of anodes annually, compared to projected demand of 1.3 million metric tons and 700,000 metric tons, respectively.[40] This supply–demand mismatch has led to increasing reliance on imports, especially from China. US lithium-ion battery imports grew 486 per cent from 2018 to 2023, with China supplying over 70 per cent of batteries and 56 per cent of battery inputs in 2023.[41] By absolute value, Chinese imports have increased more than tenfold from pre-pandemic levels.[42]

Rising investments—buttressed by the landmark Infrastructure Investment and Jobs Act (IIJA) and the Inflation Reduction Act (IRA)—have started reshaping the US battery value chain. With a total of 116 projects and $87 billion in investments related to

[36] Mike Takahashi, 'Iron and phosphate to unlock the mass-market EV at last', *IDTechEx*, 16 January 2025, at https://www.idtechex.com/en/research-article/iron-and-phosphate-to-unlock-the-mass-market-ev-at-last/32411; 'LFP now commands 40% of the global EV battery market', *Adamas Intelligence*, at https://www.adamasintel.com/lfp-now-commands-40-of-the-global-ev-battery-market/.
[37] Matt Blois, 'China threatens to stop export of iron-based cathode technology', *Chemical Engineering News*, 10 January 2025, at https://cen.acs.org/energy/energy-storage-/China-threatens-stop-export-iron/103/web/2025/01#:~:text=China's%20Ministry%20of%20Commerce%20has,could%20struggle%20to%20catch%20up.
[38] Cy McGeady, *Powering the Commanding Heights: The Strategic Context of Emergent US Electricity Demand Growth*, Center for Strategic and International Studies, 28 October 2024, at https://www.csis.org/analysis/powering-commanding-heights-strategic-context-emergent-us-electricity-demand-growth.
[39] Ashna Aggarwal et al., *Measuring Gaps in Supply and Demand for EV Battery Materials in the United States*, RMI, 2023, at https://rmi.org/insight/measuring-gaps-in-supply-and-demand-for-ev-battery-materials-in-the-united-states/; *Energizing American Battery Storage Manufacturing*, Solar Energy Industries Association, November 2023, at https://seia.org/wp-content/uploads/2023/11/FINAL20SEIA20Energizing20Battery20Storage20Manufacturing20Whitepaper-Nov202023_0.pdf.
[40] William Alan Reinsch et al., *Friendshoring the Lithium-Ion Battery Supply Chain: Battery Cell Manufacturing*, Center for Strategic and International Studies, 6 June 2024, at https://www.csis.org/analysis/friendshoring-lithium-ion-battery-supply-chain-battery-cell-manufacturing.
[41] David Coffin and Jeff Walling, 'Electrifying the global BEV landscape: top suppliers and consumers of BEVs and BEV batteries', *Journal of International Commerce and Economics*, 1 , June 2024, at https://www.usitc.gov/publications/332/journals/jice_electrifying_the_global_bev_landscape.pdf.
[42] Joseph Webster, 'What US tariffs on Chinese batteries mean for decarbonization—and Taiwan', Atlantic Council, 13 May 2024, at https://www.atlanticcouncil.org/blogs/energysource/what-us-tariffs-on-chinese-batteries-mean-for-decarbonization-and-taiwan/.

battery manufacturing announced, the battery sector has been the top-funded clean tech sector in the US since the passage of the IRA in 2022.[43] Combined, batteries and critical minerals received 67 per cent of all clean tech investments from 2018 to 2024 (Figure 1).[44] A gigafactory building boom has increased forecast battery manufacturing capacity by two-thirds, with planned cell manufacturing capacity expected to surpass 1,250 GWh by 2030.[45] Some countries in the broader battery supply chain, including South Korea, Japan, and Australia, have also increased investments either directly in US projects or in domestic resource extraction, processing, and component manufacturing to service the growing US market.

Figure 1: Total US battery-related investments since 2018, by quarter



Source: Rhodium Group and MIT CEEPR, Clean Investment Monitor, accessed 8 March 2025, at https://www.cleaninvestmentmonitor.org/database

Closing the gap

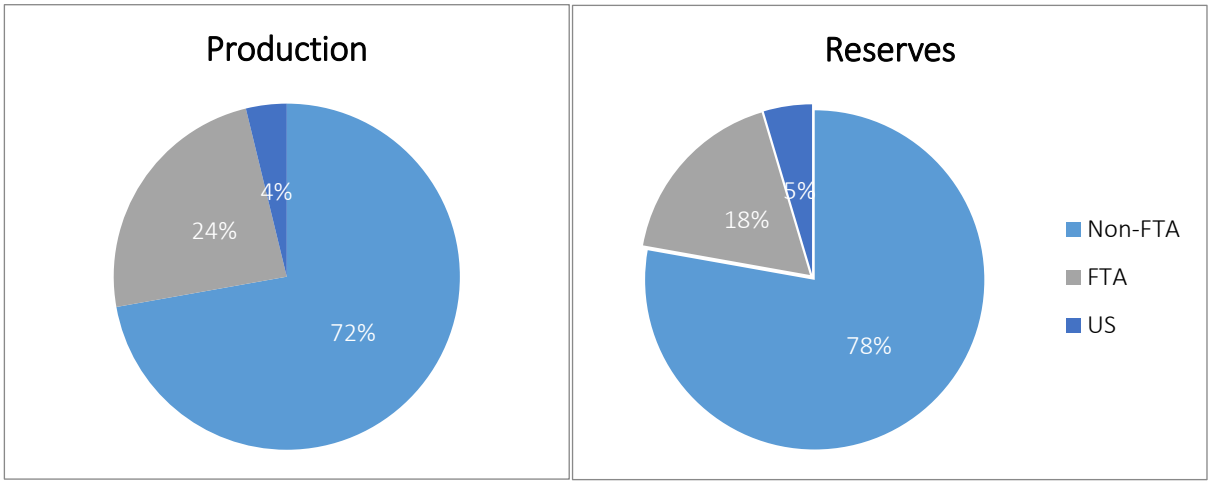
Despite recent growth, the US and its free trade agreement (FTA) partners are still at a significant competitive disadvantage to Chinese firms, whose decades of experience have enabled significant technological and cost advantages through economies of scale and supply chain integration. Lagging know-how, uncertain demand, high energy costs, capital risk aversion, equipment shortages, policy instability, and other structural disadvantages stemming from the lack of a robust domestic clean tech manufacturing ecosystem continue to hinder the learning rate of US firms. The scale of this challenge is reflected in workforce projections, with the US battery manufacturing value chain needing to more than quadruple its workforce from 2022 levels by 2030 to reach adequate scale.[46] Production projections further highlight the issue; based on 2023 trajectories, the US is expected to produce only 10 per cent of global battery capacity for battery-powered EVs in 2027, up modestly from 6.6 per cent in 2021.[47]

US firms now need to navigate increased tension between global market realities and ongoing policy efforts to reduce dependence on China. This challenge is particularly acute in the upstream and midstream segments of the value chain, with the US currently possessing negligible domestic mineral extraction and processing capacities (Figure 2).[48] According to 2024 estimates, US firms

[43] 'Tracking the North American clean energy supply chain', *The Big Green Machine*, accessed 8 March 2025, at https://www.the-big-green-machine.com/.
[44] *Clean Investment Monitor*, accessed 8 March 2025, at https://www.cleaninvestmentmonitor.org/database.
[45] 'One year on, Biden's IRA has changed the battery landscape', *Benchmark Minerals*, 15 August 2023, at https://source.benchmarkminerals.com/article/one-year-on-the-ira-has-changed-the-battery-landscape-in-the-us; *Energy Technology Perspectives 2024*, International Energy Agency, February 2025, at https://iea.blob.core.windows.net/assets/34511d5d-8dc8-42a2-8faf-b83606cffced/EnergyTechnologyPerspectives2024.pdf.
[46] *Building a Robust and Resilient U.S. Lithium Battery Supply Chain*, National Energy Technology Laboratory, US Department of Energy, March 2023, at https://netl.doe.gov/sites/default/files/2023-03/Li-Bridge%20-%20Building%20a%20Robust%20and%20Resilient%20U.S.%20Lithium%20Battery%20Supply%20Chain.pdf.
[47] 'Electrifying the global BEV landscape: top suppliers and consumers of BEVs and BEV batteries', *Journal of International Commerce and Economics,* US International Trade Commission, June 2024, at https://www.usitc.gov/publications/332/journals/jice_electrifying_the_global_bev_landscape.pdf.
[48] Melissa Barbanell, *Overcoming Critical Minerals Shortages Is Key to Achieving US Climate Goals*, World Resources Institute, 3 May 2023, at https://www.wri.org/insights/critical-minerals-us-climate-goals.

produce only about 1 per cent of global cathodes today and are projected to account for just 2 per cent of global anode supply by 2030. US cell production is set to reach 800 GWh by 2030, but domestic cathode supply is only expected to meet a third of this demand, with FTA partners also likely unable to bridge the gap on current trajectories.[49] Potential key mineral producers like Indonesia and Argentina lack FTA status, while friend-shoring initiatives like the Mineral Security Partnership still need to be refined.[50]

Figure 2: Lithium-ion battery minerals production and reserves as percentage of total volume



Source: US Geological Survey
Note: includes lithium, nickel, manganese, cobalt, and graphite at 2021 levels[51]

Trade restrictions and compliance standards are further complicating this imbalance. The Biden administration's increase of Section 301 tariffs on Chinese battery imports from 7.5 per cent to 25 per cent in 2024—combined with potential trade actions under consideration by the Trump White House and Congress that could add up to 150 per cent in tariffs—will likely reduce Chinese imports.[52] Meanwhile, the IRA and IIJA domestic content requirements, along with the 'foreign entity of concern' (FEOC) designation—which restricts companies linked to adversarial governments from participating in certain domestic industries—could create near-term supply bottlenecks, stranded asset risks, or other market distortions.[53] These evolving standards may force US manufacturers—currently sourcing less than 30 per cent of cell value domestically—to choose between accessing government incentives or pursuing cost-effective procurement contracts and joint ventures with foreign partners. In addition, China's expanding export restrictions on critical minerals, along with Chinese firms' intensifying efforts to relocate production to US trade-compliant third countries, will further complicate an already challenging strategic landscape for US firms.[54]

Progress under policy uncertainty

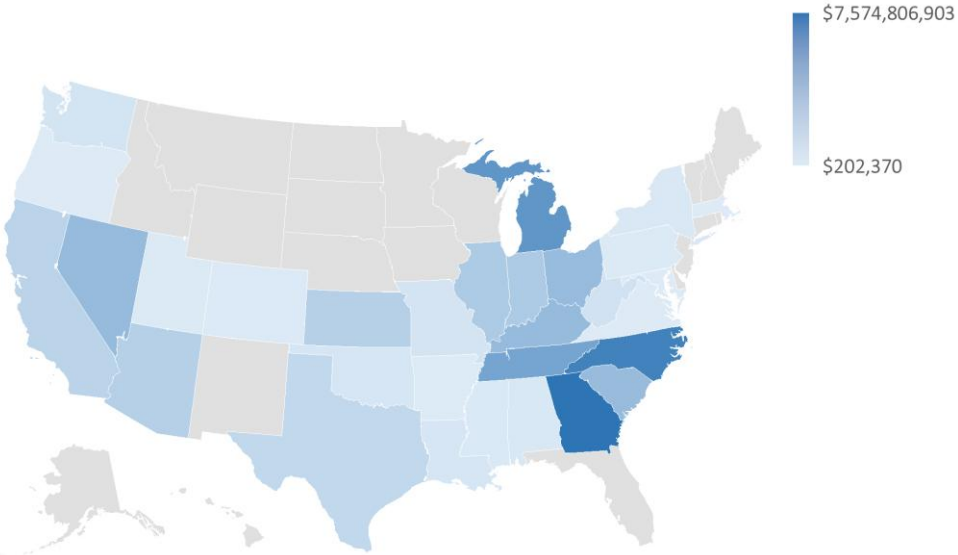
Policy remains pivotal for the US battery value chain. Combined with strong private sector momentum, a growing bipartisan consensus recognizing battery supply chains as a national security priority and economic growth driver provides some reassurance for the long-term expansion of US battery production. In the near term, however, policy shifts could significantly impact market dynamics and investment decisions.

[49] Agnes Chang and Keith Bradsher, 'Can the world make an electric car battery without China?', *New York Times*, 16 May 2023, at https://www.nytimes.com/interactive/2023/05/16/business/china-ev-battery.html; 'One year on, Biden's IRA has changed the battery landscape', *Benchmark Minerals*, 15 August 2023, at https://source.benchmarkminerals.com/article/one-year-on-the-ira-has-changed-the-battery-landscape-in-the-us; 'How is the post-IRA cathode landscape evolving?', *Benchmark Minerals*, 6 May 2024, at https://source.benchmarkminerals.com/article/how-is-the-post-ira-cathode-landscape-evolving.
[50] Hyun Soo Cho and Jane Nakano, *The Minerals Security Partnership under the South Korean Leadership*, Center for Strategic and International Studies, 28 January 2025, at https://www.csis.org/analysis/minerals-security-partnership-under-south-korean-leadership.
[51] Brad Simmons, *US Efforts to Decarbonize and De-risk the Battery Supply Chains: Are They Fundamentally Incompatible?*, Oxford Institute for Energy Studies, May 2024, at https://www.oxfordenergy.org/wpcms/wp-content/uploads/2024/05/US-efforts-to-decarbonize-and-de-risk-the-battery-supply-chains.pdf.
[52] Brain Martucci, 'US battery market faces possible "significant tariff impacts": Clean Energy Associates', *Utility Dive*, 14 January 2025, at https://www.utilitydive.com/news/us-battery-storage-lfp-nmc-market-possible-significant-tariff-impacts-clean-energy/737317/.
[53] Brad Simmons, *US Efforts to Decarbonize and De-risk the Battery Supply Chains: Are They Fundamentally Incompatible?*, Oxford Institute for Energy Studies, May 2024, at https://www.oxfordenergy.org/wpcms/wp-content/uploads/2024/05/US-efforts-to-decarbonize-and-de-risk-the-battery-supply-chains.pdf.
[54] 'US facing challenges in its attempts to diversify its graphite supply chain', S&P Global, 20 March 2024, at https://www.spglobal.com/mobility/en/research-analysis/us-graphite-supply-chain-expansion-dimming-import-needs.html.

Along with calling for an end to policies supporting EVs, such as tax credits for new vehicle purchases, the new administration's recent flurry of executive orders called for an immediate pause on disbursement of funds under the IRA and IIJA.[55] Although federal courts have ordered funding to resume, tens of billions in funds effectively remain frozen as the overhaul of federal agencies in charge of disbursement, including the Department of Energy, continues.[56] Nearly 44 per cent of the roughly $107.57 billion awarded by the Department of Energy's Loan Program Office, for instance, had not been finalized and committed at the time of the freeze.[57] Some estimates find that if clean vehicle tax credits (IRA Section 30D and 45W) and the Environmental Protection Agency's tailpipe emissions standards are repealed, from 29 per cent to 72 per cent of current battery manufacturing capacity and 100 per cent of planned facilities could be at risk of closure.[58]

The legal basis of the Office of Management and Budget's attempt to withhold funds already appropriated by Congress through 'impoundment' is disputed; as such, these efforts will continue to be subject to court challenges.[59] At the same time, competing priorities within the Republican party will likely complicate Congressional attempts to modify or repeal the IRA. With only a two-seat Republican majority in the House, legislative compromise will likely be needed. Efforts to roll back government spending will continue, yet support for certain incentives—such as the 45X tax credit, which incentivizes US manufacturing or production of solar, wind, inverters, battery components, and certain critical minerals—could persist among Republican lawmakers representing swing districts or constituents benefiting from IRA-driven investments.[60] Although no Republicans in Congress voted for the IRA, the majority of investments have flowed to Republican-held districts, contributing to emerging battery manufacturing clusters in the South and Midwest (Figure 3).[61] As recently as March 2025, 21 Republican representatives signed a letter in support of preserving the IRA, citing the law's economic and energy security benefits.[62] Precedents also suggest that projects that have commenced significant work should be grandfathered in and protected from any rollback of incentives.[63]

Figure 3: Battery-related investments since 2018 by state



Sources: Rhodium Group, MIT CEEPR; *Clean Investment Monitor*, accessed 8 March 2025, at https://www.cleaninvestmentmonitor.org/database

[55] 'Unleashing American energy', The White House, 20 January 2025, at https://www.whitehouse.gov/presidential-actions/2025/01/unleashing-american-energy/.
[56] Steve Holland et al., 'Trump aid freeze stirs chaos before it is blocked in court', *Reuters*, 29 January 2025, at https://www.reuters.com/world/trump-orders-pause-all-federal-grants-loans-2025-01-28/.
[57] Jeff St. John, 'Trump freezes funding under Inflation Reduction Act and infrastructure law', *Canary Media*, 21 January 2025, at https://www.canarymedia.com/articles/policy-regulation/trump-orders-freeze-on-inflation-reduction-act-infrastructure-law-funding.
[58] Jesse D. Jenkins, 'Potential impacts of electric vehicle tax credit repeal on US vehicle market and manufacturing', *REPEAT Project*, Princeton, NJ, March 2025, at https://zenodo.org/records/15001499.
[59] Josh Chafetz, 'Congress's Constitution', *University of Pennsylvania Law Review* 160, 3 (2012), 715–78, http://www.jstor.org/stable/41511291.
[60] Emma Dumain, 'Some House Republicans warn against climate law repeal', *E&E News*, 7 August 2024, at https://www.eenews.net/articles/some-house-republicans-warn-against-climate-law-repeal/.
[61] Kelsey Tamborrino and Josh Siegel, 'Big winners from Biden's climate law: Republicans who voted against it', *Politico*, 23 January 2023, at https://www.politico.com/news/2023/01/23/red-states-are-winning-big-from-dems-climate-law-00078420.
[62] Josh Siegel and James Bikales, 'House Republican support grows for keeping clean energy tax breaks', *Politico*, 10 March 2025, at https://www.politico.com/news/2025/03/10/house-republican-clean-energy-tax-breaks-00218126.
[63] Keith Martin, 'Cost of capital: 2025 outlook', *Norton Rose Fulbright*, 24 January 2025, at https://www.projectfinance.law/publications/2025/january/cost-of-capital-2025-outlook/.

At the same time, public and private efforts at the state and local levels to cultivate battery manufacturing as an employment and economic growth engine could face further friction from efforts to decouple from Chinese counterparts. Ford's repeatedly stalled efforts to launch battery plant in partnership with China's CATL—amid political pushback centered around national security concerns— illustrate the broader challenge.[64] The evolving enforcement of standards such as 'foreign entity of concern' qualifications, along with the creation of new restriction criteria under the America First Investment Policy, will be key to determining whether these partnerships can move forward.[65]

In this context, each segment of the battery value chain is likely to face its own set of policy uncertainties, while domestic demand drivers for battery manufacturing may split along sectoral lines. Even as EV uptake remains uncertain under the Trump administration, grid-scale battery storage is poised for continued expansion. With large variable solar and wind power fleets coming online, California and Texas are driving a nationwide battery storage boom. Total storage deployments have already doubled since 2022 and are expected to surpass 251.3 GWh between 2024 and 2028 in the US.[66] As artificial-intelligence-driven load growth accelerates, demand for grid-scale energy storage will create further tailwinds for battery energy storage systems. Similarly, the outlook for each battery technology will also continue to diverge; nascent solutions such as sodium-ion batteries and thermal long-duration storage, for instance, will need to contend with obstacles and growth drivers distinct from those for lithium iron phosphate batteries.

Looking ahead: the balancing act

Competing forces are shaping the future of the US battery industry; it could be too soon to determine which will prevail. While demand growth and recent investment surges have generated robust momentum, US firms remain at a disadvantage against entrenched Chinese competitors. Trade policies targeting imports could strengthen domestic capacity but also risk creating bottlenecks in critical components where capacity remains insufficient. At the same time, growing bipartisan support for battery manufacturing as a national security priority and economic driver could bolster long-term resilience, yet recent policy shifts and uncertainties may slow development in the short term as investors adopt a more cautious approach. The next few years will be pivotal in determining whether US stakeholders can reconcile these tensions and forge a viable path toward creating a competitive and resilient battery industry.

SIX WAYS TO REDUCE THE NEED FOR BATTERY MINERAL MINING

Will Atkinson

The impacts of battery minerals mining are real and far-reaching, and the global adoption of electric vehicles (EVs) and battery storage should go along with efforts to limit these impacts. However, there is reason for optimism: this analysis argues that even as battery demand surges, continuing past trends of efficiency, innovation, and circularity will cause the demand for mined minerals to peak within a decade. They could even allow the world to avoid mineral extraction altogether by 2050. These advances enable a transition from a linear extraction model to a circular loop, with compounding benefits for climate, security, health, and wealth.

There are six solutions to mitigate the need for mineral mining:[67]

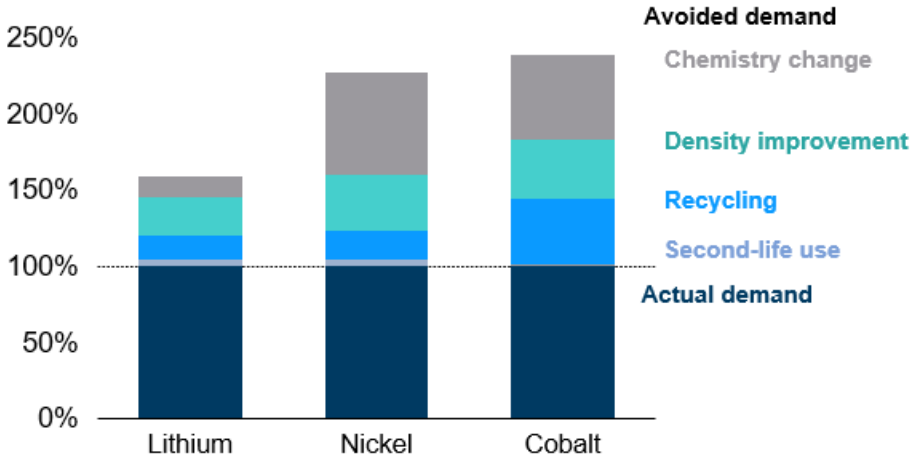
- **Deploying new battery chemistries**: switching to battery chemistries that use fewer or no critical battery minerals
- **Making batteries more energy-dense:** packing more energy into a kilogram of battery, requiring fewer minerals to do the same work

[64] Sarah Vogelsong, 'Youngkin halted Ford battery plant efforts in Virginia over concerns about China', *Virginia Mercury*, 12 January 2023, at https://virginiamercury.com/2023/01/12/youngkin-halted-ford-battery-plant-plans-in-virginia-over-concerns-about-china/; Jon King, 'Chinese company involved in Ford's Marshall EV battery plant added to Pentagon blacklist', *Michigan Advance*, 11 January 2025, at https://michiganadvance.com/2025/01/11/chinese-company-involved-in-fords-marshall-ev-battery-plant-added-to-pentagon-blacklist/.
[65] 'America First investment strategy', The White House, 21 February 2025, at https://www.whitehouse.gov/presidential-actions/2025/02/america-first-investment-policy/.
[66] 'US battery storage capacity expected to nearly double in 2024', US Department of Energy, Energy Information Administration, 9 January 2024, at https://www.eia.gov/todayinenergy/detail.php?id=61202; Brian Martucci, 'US energy storage deployments jumped 86% year over year to 10.5 GWh in Q2: ACP/WoodMac', *Utility Dive*, 8 October 2024, at https://www.utilitydive.com/news/us-battery-energy-storage-deployments-jump/729248/.
[67] Daan Walter et al., *The Battery Mineral Loop*, RMI, 2024, at https://rmi.org/insight/the-battery-mineral-loop; Amory Lovins, *Six Solutions to Battery Mineral Challenges*, RMI, 2022, at https://rmi.org/insight/six-solutions-to-battery-mineral-challenges/.

- **Recycling battery mineral content:** recovering battery minerals at the battery's end of life to avoid the need for new minerals
- **Extending battery lifetime**: using and reusing batteries for longer, to avoid having to make new batteries with new minerals
- **Improving vehicle efficiency**: improving vehicle efficiency and right-sizing cars for purpose to reduce the battery size per vehicle, and therefore mineral demand
- **Improving mobility efficiency**: improving urban planning, logistics efficiency, active modes, public transit, and electric micromobility to encourage alternatives to EV mineral demand

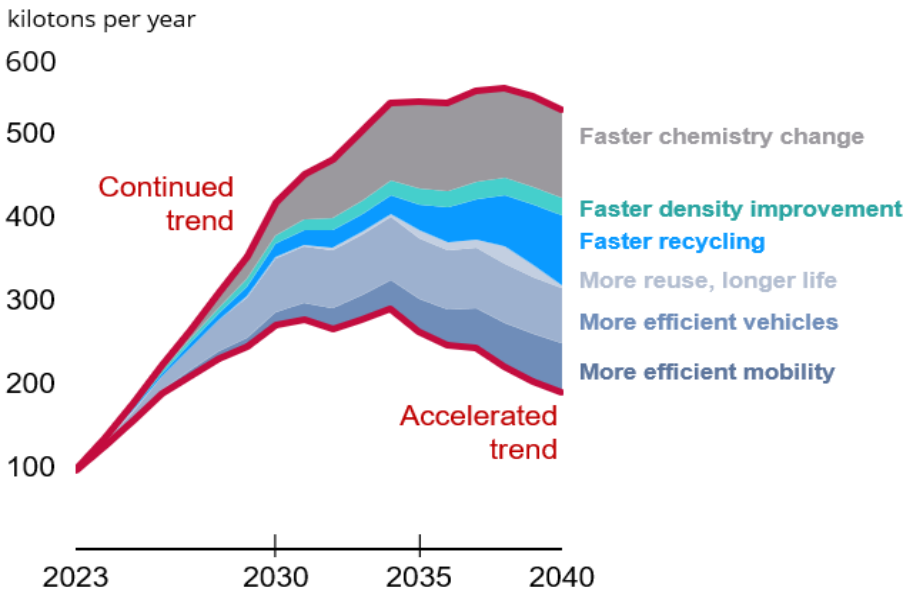
Change is already underway. Without the past decade of improvements—to chemistry mix, energy density, and recycling—lithium, nickel, and cobalt demand would be 60 to 140 per cent higher than they are today (see Figure 1). The majority of global lithium-ion batteries are already recycled today.

Figure 1: Virgin battery mineral demand avoided in 2023 due to improvements since 2015



According to RMI's analysis, peak demand for battery minerals (lithium, nickel, and cobalt) is only a decade away even under current trends (see Figure 2 for lithium). If these trends accelerate, using all six solutions listed above, net demand (total demand minus recycled supply) could reach near zero before 2050. At that point, it will be possible to meet virtually all battery mineral demand through recycling.

Figure 2: Projected net lithium demand under continued trends and accelerated solutions



Source: RMI

With accelerated trends, mineral mining will be a one-off effort. End-of-life batteries will become the new mineral ore, limiting the need for battery mineral mining in the long term. After a battery is used for 10–15 years, its mineral content can be collected and recycled at 90–94 per cent efficiency. So improving overall battery and transport efficiency by 6–10 per cent per decade is enough to offset recycling losses. Such a closed-loop supply chain means the world can continue to derive value from battery minerals for centuries. Over the next 20 years, global mineral extraction will suffice not just to power the energy system of 2050, but also through to 2100 and beyond.

Accelerated progress means the world needs to mine a cumulative 125 million tons of battery minerals. This quantity alone can enable circular battery self-sufficiency, implying minerals demand is 17 times smaller than the amount of oil currently extracted and processed for road transport every year—and, at today's commodity prices, about 20 times cheaper as well.

The world also has enough minerals. Known reserves of lithium, cobalt, and nickel are twice the level of total global demand for newly mined minerals. Announced mining projects are already sufficient to extract almost all the minerals the world needs.

Countries can move from oil dependence to circular independence. Most economies would grind to a halt if oil imports were to stop. EVs powered by renewables face no such short-term risk, especially when paired with battery recycling and (re)manufacturing. However, China currently leads the battery circularity race to the top. China's largest battery manufacturer, CATL, expects battery recycling to lead to mineral independence in China by 2042. The West is trying to catch up, while the Global South can benefit from the batteries in used vehicle imports.

Benefits are maximized when all six solutions succeed. The more holistically policymakers and other stakeholders approach demand through efficient batteries, vehicles, and mobility, the broader the benefits for all. To accelerate action, participation of all stakeholders is essential. Policymakers, the private sector, and research labs all have a role to play in capturing the opportunity for self-sufficiency.

Novel chemistries: sodium could change the game for lithium

In less than a decade, the world has halved the nickel and cobalt demand per battery—and the primary driver has been the growth of lithium iron phosphate (LFP) batteries that need no nickel and cobalt (see Figure 1). This progress has caused dramatic shifts in future outlooks, such as those from Bloomberg New Energy Finance. In just one year, LFP technology and cost improvements doubled the projected LFP share of commercial vehicle batteries (from around 40 per cent to around 80 per cent), with significant increases for passenger vehicles as well. That change led to a drop in projected nickel demand by 25 per cent to 35 per cent.[68]

Other promising chemistries, such as sodium-ion, can reduce reliance on lithium. Today, sodium-ion batteries are nascent but have high potential for applications such as grid storage and micromobility, which do not require very high energy density or charging performance compared to cars and trucks. These sectors are just a fraction of the battery market, and sodium will have to compete there with second-life batteries—but sodium is taking off in cars as well. In just the past year, several battery and car companies made significant product announcements that included sodium-ion batteries.

Energy density is steadily improving

As battery sales increased in past decades, the energy density of batteries rose, driven by growing R&D budgets and economies of scale. On average, for every doubling of cumulative battery demand, the average energy density of lithium-ion battery cells improved by about 6 per cent.

After adjusting for chemistry change, RMI's analysis finds that energy density has improved by about 4 per cent per cumulative doubling of battery demand. This has led to improvements of around 25 per cent in just the past decade. As cumulative battery deployment doubles at least another 4 to 5 times before 2050, density can be expected to rise by another 25 per cent.

Our modelling outlooks are rooted in that history of improvement. It is easy to underestimate the pace of progress by assuming static policy or technology. As Oxford researchers have shown, continued trends are often more accurate when it comes to these technologies.[69]

[68] Colin McKerracher et al., 'Electric vehicle outlook 2024', *Bloomberg New Energy Finance*, 2024, at https://about.bnef.com/electric-vehicle-outlook/.

[69] Rupert Way et al., 'Empirically grounded technology forecasts and the energy transition', *Joule* 6, 1–26, 2022, at https://doi.org/10.1016/j.joule.2022.08.009.

Battery recycling: 10x more common than sometimes assumed

Some sources claim that only 5 per cent of lithium-ion batteries are recycled.[70] However, this figure is outdated and misconstrued; end-of-life recycling rates are actually more than 10 times higher. A 2023 study led by Argonne National Laboratory found that even in 2019—more than five years ago, when there was only a fraction of today's recycling facilities and innovation—59 per cent of end-of-life lithium-ion batteries were recycled globally.[71] China is leading, but progress is global; this number was 54 per cent in the United States.

There is already enough global recycling capacity to recycle all projected end-of-life batteries through 2030, as well as their production scrap.[72] Not all countries have enough capacity, but more and more are seeking to improve this for their security. Thankfully for newer entrants, there is still room for long-term growth: to recycle all batteries in the long term even as demand surges, roughly 80 per cent of the required recycling capacity has yet to be built.

Recycling innovations are progressing fast. Recycled minerals meet and can even exceed the purity and performance of mined minerals, and emerging technologies like direct recycling are cutting costs, economically and environmentally. Numerous start-ups are improving mineral recovery rates, charting a path for even the harder-to-recycle materials such as graphite. For lithium, nickel, and cobalt, Bloomberg New Energy Finance estimates that 80 per cent to 95 per cent of minerals can be recovered with current recycling processes—or as much as 99.6 per cent for nickel and cobalt, as claimed by CATL in China.[73]

Recycling economics are also improving, but just as in the early days of solar energy or battery manufacturing, support is needed to help scale and drive down cost. Triple-bottom-line accounting shows that there is already a strong argument for recycling investments if companies systemically incorporate recycling's full social and environmental benefits.[74] Policy support can help, such as through increased targets and certification for recycled content (such as in China and the EU), as well as greater incentives for companies to incorporate recycling into battery design and manufacturing.

Longer lifetimes: reuse and recycling go hand in hand

Battery lifetimes are getting longer. They vary widely by sector (from high-use trucks to longer-lived cars) but will continue to increase as innovation booms. There are even some reported cases of 'million-mile batteries' coming to market, which may well last for multiple decades. Second-life use is also increasing, particularly for grid storage. Europe has strong warranty and repairability laws, and other regions could follow.

Reuse is complementary to recycling. Extended lifetimes can help to right-size battery manufacturing investments and reduce a battery's environmental footprint. This can potentially reduce consumer cost per kWh of energy delivered, particularly in regions that may not have sufficient manufacturing today.

Extended lifetimes can save a million tons of mined lithium through 2050.[75] Reuse can delay recycling, but it delays new demand even more, thus reducing the pressure on manufacturing build-out and its environmental impact.

Efficient vehicles: trends toward inefficiency can be reversed

Inefficiency is a real risk. According to the International Energy Association, the share of EV sales from larger vehicles (such as SUVs and pickup trucks) has been increasing from 2018 to 2023 across regions.[76] This share has risen seven-fold in the United States and five-fold in Europe, with increases in China and elsewhere as well.

This could delay peak mineral demand and circularity, but there is ample room to reverse these trends. Across RMI's six solutions, vehicle efficiency has some of the most immediate benefits—even if only a fraction of the full potential is realized by simply returning to 2018 car size shares. Reducing battery needs (for a given range) can also help to overcome EV profitability concerns.

[70] Hannah Richie, 'What share of lithium-ion batteries are recycled?', *Sustainability by Numbers*, 26 June 2024, at https://www.sustainabilitybynumbers.com/p/battery-recycling-myth.
[71] Linda Gaines et al., 'Tracking flows of end-of-life battery materials and manufacturing scrap', *Batteries*, 9(7), 2023, at https://www.mdpi.com/2313-0105/9/7/360.
[72] *Global EV Outlook 2024*, International Energy Agency, April 2024, at https://www.iea.org/reports/global-ev-outlook-2024.
[73] 'Avoiding a crunch in critical minerals through technology, recycling and global collaboration: Robin Zeng', CATL, 18 January 2024, at https://www.catl.com/en/news/6201.html.
[74] Monkgogi Buzwani et al., *Battery Recycling: How Accounting for Social and Environmental Benefits Boosts Returns*, RMI, 5 June 2024, at https://rmi.org/battery-recycling-how-accounting-for-social-and-environmental-benefits-boosts-returns/.
[75] Will Atkinson et al., *Q&A: Six Charts for Six Solutions to the Battery Mineral Challenge*, RMI, 6 September 2024, at https://rmi.org/qa-six-charts-for-six-solutions-to-the-battery-mineral-challenge/.
[76] *Global EV Outlook 2024*, International Energy Agency, April 2024, at https://www.iea.org/reports/global-ev-outlook-2024.

In our accelerated case on feasible improvements, we follow the International Energy Association case where per-vehicle battery demand drops nearly 30 per cent by 2030. Companies and governments can realize this potential by prioritizing sleek, right-sized vehicles that are more affordable, more efficient, and more enjoyable.

Efficient mobility: systemic solutions have the broadest benefits

Further upstream, more efficient mobility can significantly curb mineral demand for trucks and cars. A 2023 report describes how efficient mobility could reduce demand for lithium in the United States by 18 per cent to as much as 66 per cent.[77] RMI research has found that such improvements are both possible and necessary to reach climate goals.[78]

Much more is possible in developing countries, where smart design of new infrastructure can provide major benefits.[79] And as the city of Paris has proven, major progress can come from just a few years of smart policy. Our report uses global analysis by the International Council on Clean Transportation, which implies potential reductions of about 15 per cent to 20 per cent by 2050.[80]

Reducing mineral demand is just the beginning. Deeper efficiency measures come with greater benefits across emissions, health, security, and environmental justice. Switching to EVs avoids the harms of fossil fuels, while more efficient vehicles with smaller batteries can reduce the impacts of the battery production process. More efficient mobility can unlock further improvements, from access to active health benefits.

Looking ahead

Research on this topic can become quickly out of date. In 2024, battery demand growth surged even more than expected, with costs dropping 20 per cent and estimated energy storage additions nearly doubling.[81] Battery innovation continues across chemistries, from manganese to solid state, while density and durability improve as well. It is crucial to stay current in these times of change.

Based on these trends, several other analyses—such as recent reports published by the International Council on Clean Transportation, the World Economic Forum, and the Union of Concerned Scientists—have supported RMI's battery minerals demand findings while strengthening the literature on policy, case studies, and remaining needs.[82]

There is much uncertainty ahead, including geopolitics as well as the threat of low mineral prices on recycling economics. But as we have found, there is also uncertainty on the upside—with the strong track record and even stronger potential of all six solutions. It is time to take advantage by doubling down across efficiency, innovation, and circularity.

[77] Thea Riofrancos et al., *Achieving Zero Emissions with More Mobility and Less Mining*, Climate and Community Institute, January 2023, at https://climateandcommunity.org/research/more-mobility-less-mining/.
[78] David Roberts, 'How much can urban land use policy do for the climate?', *Volts*, 10 April 2023, at https://www.volts.wtf/p/how-much-can-urban-land-use-policy.
[79] Felix Creutzig et al., *Demand, Services and Social Aspects of Mitigation: Supplementary Material*, Intergovernmental Panel on Climate Change, 2022, at https://www.ipcc.ch/report/ar6/wg3/downloads/report/IPCC_AR6_WGIII_Chapter05_SM.pdf.
[80] Arijit Sen et al., *Vision 2050*, International Council on Clean Transportation, 27 November 2023, at https://theicct.org/publication/vision-2050-strategies-to-reduce-gap-for-global-road-transport-nov23/.
[81] Will Atkinson and Chiara Gulli, *The Energy Transition in 2025*, RMI, 8 January 2025, at https://rmi.org/the-energy-transition-in-2025-what-to-watch-for/.
[82] Eyal Li et al., *Electrifying Road Transport with Less Mining: A Global and Regional Battery Material Outlook*, International Council on Clean Transportation, December 2024, at https://theicct.org/publication/ev-battery-materials-demand-supply-dec24/; Jessica Dunn et al., *Making the Most of Electric Vehicle Batteries*, Union of Concerned Scientists, 15 December 2024, at https://www.ucsusa.org/resources/making-most-electric-vehicle-batteries; Pramoda Gode et al., *Powering the Future: Overcoming Battery Supply-Chain Challenges with Circularity*, World Economic Forum, 16 January 2025, at https://www.weforum.org/publications/powering-the-future-overcoming-battery-supply-chain-challenges-with-circularity/.

NAVIGATING US TARIFF RISKS: MEXICO'S INDUSTRIAL POLICY FOR ELECTRIC VEHICLES

Francisco Parés Olguín, Aditya Ramji, Roland Hwang, and Juan C. Garcia Sanchez

North America's automotive industry is at a turning point, reshaped by the electric vehicle (EV) transition, post-pandemic supply chain shifts, and rising US protectionism. Mexico, as the largest vehicle supplier to the US, with a 15 per cent market share, is deeply embedded in these changes.[83] However, this success comes with a major vulnerability—Mexico's overwhelming dependence on US markets and automakers.

In 2024, US automakers accounted for 42 per cent of Mexico's light-duty vehicle (LDV) production, and nearly 70 per cent of total output was exported to the US, while only 12 per cent remained for domestic consumption (see Table 1).[84] At the same time, nearly 65 per cent of LDVs sold within Mexico were imported—mainly from China, the US, Brazil, and India. Although the auto sector contributes 5 per cent to Mexico's GDP and attracts 20 per cent of foreign direct investment, its high exposure to external markets leaves it vulnerable to global policy shifts. A proposed 25 per cent US tariff on Mexican auto imports now threatens to destabilize this export-driven model, putting market share and future investment at risk.

Table 1: Light-duty vehicle production in Mexico by automaker and market destination, 2024

		Destination (%)						
Automaker	Production (units)	Mexico	USA	Canada	Germany	Rest of Europe	Latin America & Caribbean	Rest of world
General Motors (USA)	889,072	3%	83%	11%	0%	0%	1%	1%
Nissan (JPN)	669,941	34%	47%	7%	0%	0%	10%	2%
Ford Motor (USA)	386,776	1%	92%	0%	1%	1%	4%	0%
Chrysler (USA)	419,426	4%	85%	9%	0%	0%	1%	0%
Volkswagen (DEU)	382,312	14%	55%	11%	8%	8%	3%	0%
Toyota (JPN)	245,007	3%	90%	5%	0%	0%	2%	0%
Honda (JPN)	194,612	5%	89%	7%	0%	0%	0%	0%
KIA (KOR)	270,700	24%	62%	5%	0%	0%	7%	3%
Audi (DEU)	144,223	1%	27%	7%	29%	30%	2%	4%
Mazda (JPN)	209,303	34%	53%	7%	0%	0%	7%	0%
BMW (DEU)	95,151	0%	34%	2%	12%	35%	1%	15%
Mercedes Benz (DEU)	57,487	1%	21%	2%	15%	37%	2%	22%
JAC (CHN)	25,391	100%	0%	0%	0%	0%	0%	0%
Total	3,989,401	13%	69%	7%	2%	3%	4%	2%

Note: Darker shading indicates a higher percentage of each automaker's production allocated to a specific destination market
Source: National Institute for Statistics and Geography (Instituto Nacional de Estadística y Geografía, INEGI), Administrative Registry of the Light Vehicle Automotive Industry, 2025, https://www.inegi.org.mx/datosprimarios/iavl/ - tabulados)

This article explores Mexico's pathways for safeguarding its EV industry, highlighting key strategies to mitigate risks, leverage investment opportunities, and secure Mexico's role in the evolving global EV supply chain.

Background

North America's trade and industrial integration was formalized by the 1994 ratification of the North American Free Trade Agreement (NAFTA) between Canada, Mexico, and the US. Over time, Mexico grew into a major LDV supplier, accounting for 24 per cent of regional production and 57 per cent of regional exports.[85]

[83] *Annual Energy Outlook 2023*, US Energy Information Administration, 2023 at: https://www.eia.gov/outlooks/aeo/tables_ref.php; *UN Comtrade Database*, United Nations, accessed 8 March 2025, at https://comtrade.un.org/.
[84] *Administrative Registry of the Light Vehicle Automotive Industry*, Instituto Nacional de Estadística y Geografía (Mexican National Institute for Statistics and Geography), accessed 8 March 2025, at https://www.inegi.org.mx/datosprimarios/iavl/#tabulados.
[85] *Vehicle Production Database*, International Organization of Motor Vehicle Manufacturers, accessed 8 March 2025, at https://www.oica.net/category/production-statistics/; *UN Comtrade Database*, United Nations, accessed 8 March 2025, at https://comtrade.un.org/.

However, efficiency-driven free trade policies reshaped supply chains, transferring US manufacturing capacity abroad and ushering in deindustrialization and economic discontent in US communities once sustained by manufacturing. The US now relies on imports for 47 per cent of its vehicle market, leaving it uniquely vulnerable to supply chain disruptions—a stark contrast to China, the EU, and Japan, where imports account for just 5 per cent, 22 per cent, and 9 per cent, respectively.[86]

Capitalizing on these economic shifts, the first Trump administration made restoring domestic manufacturing a priority—framing it as critical to economic security and national competitiveness. The administration imposed aggressive trade measures, including Section 232 tariffs on steel and aluminium and Section 301 tariffs on Chinese imports. (Section 232 refers to tariffs related to national security, whereas Section 301 refers to violations of trade agreements or other discriminatory trade measures.) Also during the first Trump administration, NAFTA was renegotiated into the US-Mexico-Canada Agreement (USMCA), introducing stricter labour regulations and raising North American content requirements from 62.5 per cent to 75 per cent.[87]

There is an overall shift toward industrial policy and away from free trade that has remained across US administrations since 2017, with varying degrees of protectionism and trade-offs between near-shoring and on-shoring. This materialized in the Inflation Reduction Act of 2022, which incentivized regional EV supply chains and promoted 'shared prosperity' with allied nations,[88] aiming to implement these policies while reinforcing the US's position both economically and geopolitically.

Navigating the uncertainties of a second Trump administration requires Mexico to strike a delicate balance—reducing dependence on the US market by strengthening its domestic industry and expanding trade partnerships in Latin America and other emerging economies while enhancing its productive capacity to remain globally competitive.

Industry response to tariff threats

The automotive industry has responded with growing concern to the escalating trade tensions. Before the Trump administration, major auto manufacturers such as General Motors and Ford were already shifting supply chains closer to the US to qualify for Inflation Reduction Act tax credits, while Mexican suppliers faced increasing pressure to localize EV components to comply with USMCA regulations.

Now, with the threat of 25 per cent tariffs, major automakers, including Ford, General Motors, Stellantis, and Nissan North America, are actively reassessing their production and investment strategies. The US auto industry has also highlighted that tariffs could cause significant disruptions to the auto industry, going beyond relocation of production and impacting jobs and vehicle cost.[89] Meanwhile, Mexico's auto industry associations, in collaboration with President Sheinbaum's administration, are working toward easing trade restrictions and securing exemptions that can minimize the impact.

Mexico's policy response

In response to mounting trade pressures, Mexico is pursuing both defensive and proactive strategies to safeguard its automotive sector. The Sheinbaum administration has moved swiftly to negotiate tariff exemptions for essential EV components while simultaneously launching a new domestic industrial policy drive. This includes the Plan Mexico initiative, a comprehensive national industrial strategy to reduce import dependence, and the Taruk and Olinia projects, Mexican-designed EVs tailored to domestic needs, signalling a shift toward self-sufficiency in vehicle production.[90] In addition, the Made in Mexico campaign aims to boost domestic consumption and reduce reliance on exports, particularly as tensions with the US continue to escalate.[91]

These policies not only address Mexico's vulnerability to protectionist measures but also anticipate growing global scrutiny over overcapacity—an issue particularly directed at China, where calls for increased domestic consumption have emerged as a solution

[86] *Vehicle Production Database*, International Organization of Motor Vehicle Manufacturers, accessed 8 March 2025 at https://www.oica.net/category/production-statistics/; *UN Comtrade Database*, United Nations, accessed 8 March 2025, at https://comtrade.un.org/.
[87] *USMCA Automotive Sector Report*, US Department of Commerce, International Trade Administration, accessed 8 March 2025, at https://www.trade.gov/usmca-auto-report.
[88] 'Building resilient supply chains, revitalizing American manufacturing, and fostering broad-based growth', The White House, June 2021.
[89] Peter Johnson, 'Ford CEO warns Trump tariffs will "blow a hole" in US auto industry', *Electrek*, 11 February 2025, at https://electrek.co/2025/02/11/ford-ceo-warns-trump-tariffs-will-blow-a-hole-in-us-auto-industry/.
[90] Emilia Quintana, *Plan México: Claudia Sheinbaum's Vision for Sustainable Growth and National Prosperity*, Wilson Center, 22 January 2025, at https://www.wilsoncenter.org/article/plan-mexico-claudia-sheinbaums-vision-sustainable-growth-and-national-prosperity; 'Olinia project is presented, and the state of Sonora is a strong candidate to host the facility', Coordinación de Estrategia Digital Sonora, 6 February 2025, at https://codeso.mx/en/olinia-project-is-presented-and-the-state-of-sonora-is-a-strong-candidate-to-host-the-facility/#:~:text=The%20initiative%20includes%20the%20manufacture,safety%20and%20lower%20operating%20costs.
[91] Eliza Galeana 'Walmart joins "Hecho en México" campaign', *Mexico Business News*, 27 February 2025, https://mexicobusiness.news/ecommerce/news/walmart-joins-hecho-en-mexico-campaign.

to surplus production. By proactively stimulating local demand, Mexico is positioning itself to mitigate future trade friction while fostering a more resilient EV market. Beyond trade policy, recent bilateral agreements with the US—including measures to curb migration, drug trafficking, and arms smuggling—highlight the broader geopolitical stakes at play in Mexico's efforts to maintain economic stability.[92]

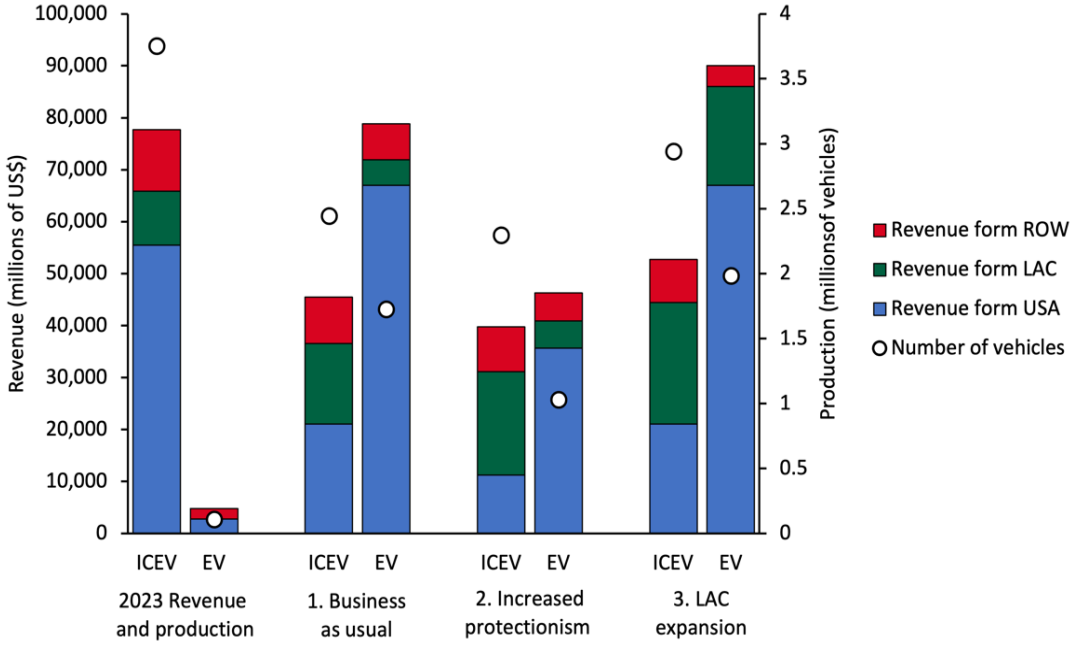
Effects of US tariffs on Mexico's light-duty vehicle sector

Mexico's domestic EV market remains limited due to limited charging infrastructure, the relatively small domestic market for new vehicles, and the high cost of EVs, making its automotive industry heavily reliant on US demand. While Mexico supplies 15 per cent of the US LDV market, its EV production lags, holding only a 10 per cent share of US battery-powered EV and plug-in hybrid EV sales in 2023.[93]

To assess how trade policy shifts could reshape Mexico's automotive sector, we simulated potential outcomes using the Model for International EV Trade.[94] These simulations integrate vehicle demand forecasts and country-specific LDV price data.[95]

Figure 1 presents Mexico's projected automotive revenue and LDV production for 2035, disaggregated by major markets and trade scenarios. For comparison, it also includes 2023 production volumes and revenue. This analysis highlights how different policy environments could impact Mexico's role in global supply chains, influencing both production volumes and sales revenues.

Figure 1: Mexico's projected LDV production and revenue under three 2035 scenarios



Notes: ICEV = internal combustion engine vehicle; LAC = Latin America and the Caribbean; ROW = rest of world

Scenario 1, business as usual, assumes pre-Covid trade flows (2017–2019) remain stable, there is no rollback of current US emission standards, and supply chains remain the same based on past trade patterns. To maintain its current global market share under this scenario, Mexico must shift 46 per cent of its 2023 internal combustion engine vehicle (ICEV) production to EVs while expanding overall production by 8 per cent.

[92] José de Córdoba, 'Trump agrees to help stop flow of weapons into Mexico, Sheinbaum says', *Wall Street Journal*, 3 February 2025, at https://www.wsj.com/livecoverage/stock-market-today-dow-sp500-nasdaq-live-020-03-2025/card/trump-agrees-to-help-stop-flow-of-weapons-into-mexico-sheinbaum-says-P4fs5qLlBXGjcmOYzOiM.

[93] Francisco Parés Olguín et al., 'Shifting manufacturing: electric vehicle supply strategy using the Model for International EV Trade', *Journal of Cleaner Production*, 484, December 2024, at https://www.sciencedirect.com/science/article/pii/S095965262403806X.

[94] Ibid.; Pablo Busch et al., 'Future of global electric vehicle supply chain: exploring the impact of global trade on electric vehicle production and battery requirements', *Transportation Research Record*, 2678(11), 29 May 2024, at https://doi.org/10.1177/03611981241244797.

[95] *Global EV Outlook 2024: Trends in Electric Mobility and Infrastructure*, International Energy Agency, 2024, at https://www.iea.org/reports/global-ev-outlook-2024; *Roadmap v1.9 Documentation*, International Council on Clean Transportation, 2023, at https://theicct.github.io/roadmap-doc/versions/v1.9/; *UN Comtrade Database*, United Nations, accessed 8 March 2025, at https://comtrade.un.org/.

Scenario 2, increased protectionism, models a scenario where countries prioritize domestic production and consumption, reducing global trade flows. While Mexico's domestic market grows relative to Scenario 1, its total production contracts by 20 per cent and revenues decline by 30 per cent due to a 46 per cent loss in US market share—a figure aligned with recent estimates of the impact of a 25 per cent tariff.[96]

Scenario 3, expansion into Latin America and the Caribbean (LAC), redirects production from North America to Latin America, strengthening Mexico's domestic market and regional supply role. This scenario increases Mexico's total LDV production by 22 per cent compared to Scenario 1, leveraging rising demand in LAC. While US revenues remain dominant, they now account for 61 per cent of total earnings—a 10 per cent decrease from Scenario 1. However, most of the growth in Mexico and LAC remains in ICEVs, as EV adoption continues to lag the rest of the world.

What becomes evident from all scenarios is that despite ICEVs continuing to constitute the lion's share of Mexico's vehicle production, most of its industry's profit growth depends on its ability to supply projected EV demand—under the assumption that EVs continue to reach higher price points than ICEVs.

Beyond diversification, LAC's preference for smaller vehicles and slower EV adoption would reduce lithium-ion battery intensity, easing supply chain pressures as technologies advance and costs decline. This extended transition timeline allows Mexico to initially focus on high-income USMCA markets, producing large and luxury EVs for faster and more resilient returns. Over time, Mexican industry could gradually shift to lower-cost EVs for LAC, positioning itself as a supplier across the full EV spectrum—from premium to affordable models with smaller batteries.

This analysis highlights the stark contrast between two possible futures—one where tariffs are implemented without a strategic response (Scenario 2) and another where Mexico proactively diversifies its market and strengthens its supply chain (Scenario 3).

The following section explores key pathways to reduce the risks from US trade policy shocks, enhance Mexico's EV supply chain resilience, and support the region's transition to clean transportation.

Pathways for Mexico's EV supply chain resilience

As US protectionist policies threaten Mexico's export-driven automotive industry, it will be essential for Mexico to strengthen its domestic EV sector, diversify trade partnerships, and secure investment in its supply chain. A balanced strategy can maintain competitiveness, reduce dependency on any single market, and position Mexico as a leading EV manufacturing hub in the Americas.

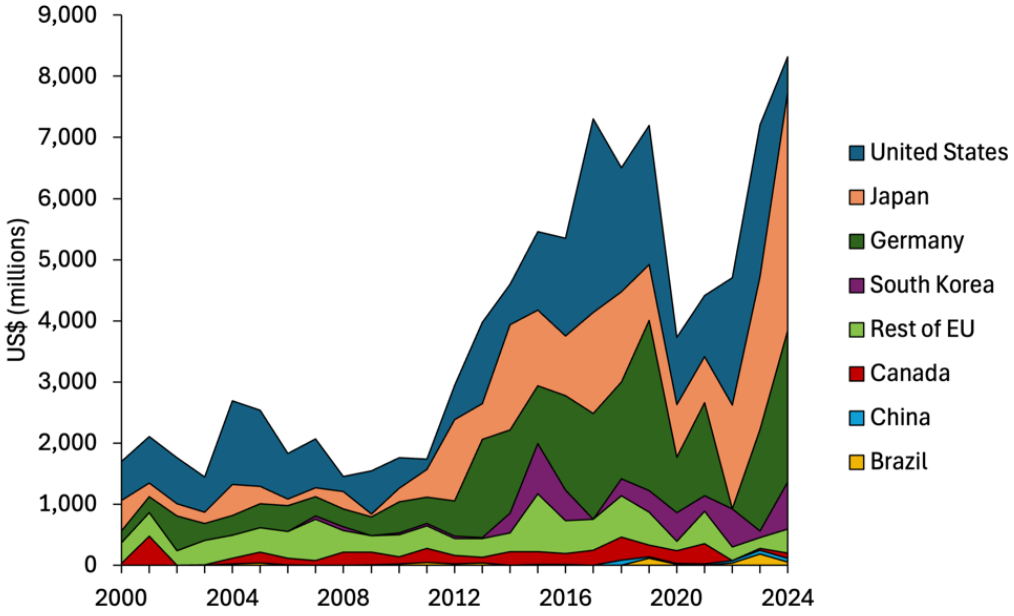
A much-speculated possibility is that Mexico could seek greater Chinese investment, deepening its ties with Chinese EV and battery manufacturers. On paper, this partnership appears strategically sound—it could accelerate industrialization, diversify manufacturing, and provide access to advanced technology and cost-efficient supply chains. For China, Mexico offers proximity to the US and LAC, a well-established automotive sector, and a skilled workforce, all of which could help expand its global footprint. This alignment has driven over $10 billion in announced investments from BYD, SAIC, JAC, and CATL.[97] However, foreign direct investment in Mexico's automotive sector remains dominated by incumbent producers—as shown in Figure 2—as these projects remain stalled due to geopolitical pressure and the threat of US trade retaliation, underscoring the risks of closer ties with China amid rising global economic tensions.

US policymakers increasingly view China's growing presence in the region as an economic and strategic challenge. The fear is that China's influence on its largest trading partner and closest neighbour could undermine US industrial aspirations. In response, the US has escalated economic countermeasures on its neighbours, aiming to limit China's reach in North American supply chains. Navigating this geopolitical tension carefully will be critical to ensure that Mexico does not become collateral damage in the rivalry between the world's two largest economies—echoing many other great-power conflicts where the brunt of geopolitical rivalries is borne by proxy locations.

[96] Joshua P. Meltzer, 'Trump's 25% tariffs on Canada and Mexico will be a blow to all 3 economies', Brookings Institution, 3 February 2025, at https://www.brookings.edu/articles/trumps-25-tariffs-on-canada-and-mexico-will-be-a-blow-to-all-3-economies/.

[97] Adriana Alarcón, 'JAC expands production, expresses interest in Olinia project', *Mexico Business News*, 24 February 2025, at https://mexicobusiness.news/automotive/news/jac-expands-production-expresses-interest-olinia-project; Francisco Parés Olguín et al., 'Shifting manufacturing: electric vehicle supply strategy using the Model for International EV Trade', *Journal of Cleaner Production*, 484, December 2024, at https://www.sciencedirect.com/science/article/pii/S095965262403806X.

Figure 2: Foreign direct investment in Mexico's automotive sector by major country or region



Source: Secretaría de Economía, DataMexico, http://datamexico.org/
Note: Includes manufacture of automobiles and trucks, motor bodies and trailers, and motor vehicle parts, as well as repair and maintenance of cars and trucks

Strengthening Mexico's domestic market and supply chains

As trade dynamics shift and geopolitical pressures mount, expanding Mexico's domestic market and productive capacity has emerged as a crucial strategy for reducing dependency on both the US market and Chinese capital. Recent policy announcements already signal a move toward greater self-sufficiency and long-term industrial resilience.

Key pillars of this strategy are the Olinia initiative and the Made in Mexico programme, which emphasize localized EV production to decrease reliance on exports and foster a sustainable domestic market. These initiatives prioritize the development of smaller, affordable EVs designed to meet the specific needs of Mexico and the broader LAC region, where urbanization trends and income levels favour compact, cost-effective transportation solutions.

Beyond vehicle production, positioning Mexico as a full-spectrum EV supply chain hub would further enhance its competitiveness. Sonora's lithium reserves present an opportunity to develop an integrated ecosystem encompassing mineral extraction, battery manufacturing, and recycling—securing a domestic supply of critical materials while strengthening Mexico's role in the broader EV supply network.

The country's evolving industrial policy has also placed greater emphasis on localizing key EV components, such as batteries, motors, and inverters, aligning with USMCA requirements and deepening integration into North America's EV supply chains. Investments in domestic lithium-ion battery gigafactories and bilateral battery recycling initiatives—leveraging the US–Mexico second-hand vehicle exports flow, the largest in the world[98]—could reduce reliance on Chinese imports while ensuring a steady supply of materials for long-term EV production.

At the same time, maintaining fuel-efficient ICEV production for emerging markets—particularly in LAC—offers a strategic hedge during the uncertain EV transition. With affordability and infrastructure barriers slowing EV adoption in many developing economies, demand for ICEVs is expected to persist in the near to medium term. Mexico's well-established automotive sector and skilled workforce—built on decades of NAFTA and USMCA investment—position it to supply these markets while expanding its EV industry. A dual-track approach supports EV growth while ensuring continued vehicle exports to regions that may take longer to electrify.

[98] Francisco Parés Olguín et al., 'US-Mexico second-hand electric vehicle trade: battery circularity and end-of-life policy implications', *Transportation Research Part D: Transport and Environment*, 125, December 2023, at https://doi.org/10.1016/j.trd.2023.103934; Alissa Kendall et al., *Electric Vehicle Lithium-ion Batteries in Lower- and Middle-income Countries: Life Cycle Impacts and Issues*, UC Davis: Institute of Transportation Studies, 2023, at https://escholarship.org/uc/item/7m2536mp.

However, a robust domestic EV market cannot develop without the adoption of strong vehicle standards that ensure a supply of EVs. Expanding public charging networks and introducing consumer incentives can stimulate local demand, making EVs more accessible to Mexican consumers and accelerating the country's transition to electrified transportation.

Taken together, these strategies lay the groundwork for Mexico to build a resilient EV industry—one capable of withstanding external shocks while adapting to the unique needs of emerging markets, including its own.

Strategic trade diversification

To ensure long-term competitiveness, reducing extraordinary reliance on the US market will be key. Strengthening domestic demand and expanding into LAC present natural opportunities, given geographic proximity, cultural ties, and existing trade agreements. These efforts can enhance Mexico's economic stability while mitigating risks from shifting US trade policies.

That said, managing relations with the US will remain essential. Reinforcing Mexico's role in the North American EV supply chain by demonstrating its indispensability to US manufacturers can help navigate tariff risks. Adjusting trade balances—such as reducing reliance on Chinese imports—may also help prevent further economic countermeasures.

Broadening foreign investment beyond traditional North American partners can also enhance Mexico's resilience. Offering incentives to automakers outside the Detroit Three could attract a more diverse set of manufacturers and supply chain investments, boosting technological capabilities and market stability.

A phased expansion into LAC markets offers a strategic hedge during the uncertain EV transition. In the short term, Mexico can leverage its automotive sector to supply fuel-efficient ICEVs to LAC, where affordability and infrastructure barriers slow EV adoption. At the same time, this approach lays the groundwork for future EV market development. Over time, producing smaller, cost-effective EVs tailored to the region's needs will create a viable alternative to US exports and position Mexico as a regional leader in EV manufacturing.

Strengthening regional trade agreements—such as deepening ties with Mercosur—would further expand export markets and reduce exposure to US trade fluctuations. A broader trade network would also enhance Mexico's leverage in global automotive negotiations, reinforcing its position as a key player in EV supply chain resilience.

By strategically diversifying trade relationships and investing in EV production, Mexico can mitigate economic risks, unlock new growth opportunities, and support the transition to clean transportation—not just in North America, but across the broader Americas.

Charting a path for others

Mexico stands at a crossroads, balancing short-term adaptation to US protectionist policies with long-term strategies to diversify markets, investment sources, and supply chains. A robust industrial policy—one that attracts a broader range of investors, expands domestic production capacity, grows the domestic market, and deepens trade ties with LAC—would bolster resilience and lessen reliance on any single market. The choices Mexico makes in navigating its EV transition will not only shape its automotive future but also serve as a case study for other global manufacturing hubs facing similar geopolitical and economic pressures. As global supply chains realign in response to evolving trade policies, Mexico's approach could offer a blueprint for nations striving to maintain competitiveness while mitigating risks in an increasingly fragmented global economy.

LOW ADDED VALUE IN THE REAR-VIEW MIRROR: ALIGNING NATIONAL AND SUBNATIONAL EV STRATEGIES IN MEXICO

Carlos Guadarrama

Plan Mexico, the country's new industrialization strategy, aims to position Mexico as an electric vehicle (EV) leader. Two key projects are outlined in the plan: a new Stellantis EV plant and the manufacturing of low-cost Mexican EVs, named Olinia, by 2026.

A striking feature of Plan Mexico is the selection by federal policymakers of regions to specialize in different industries. Stellantis plans to set up shop in the State of Mexico. The three Olinia models—targeting personal mobility, neighbourhood mobility, and last-mile delivery—will be built in Puebla, also home to Volkswagen's largest automotive plant.[99] But the Mexican president is also considering the state of Sonora to host some of the manufacturing plants for Olinia.[100] This would align with the northern state's own Plan Sonora, which incorporates the intention to become an EV manufacturing hub. Sonora's efforts raise the importance of examining the impact of EV markets and policies at the subnational level, to identify challenges and opportunities for development that the rising EV industry could provide across different regions of the country.

For both Mexico and Sonora, the creation of a new EV manufacturing industry will require coordinated industrial policy efforts at both national and local levels. As a first step to develop the industry, Mexico will need to bring in battery manufacturing technologies, in which China currently leads, while navigating trade and geopolitical tensions with the US. This article explores the historical context, current strategies, and prospects of Mexico and Sonora in the automotive and EV sectors, focusing on the potential to drive local economic growth through technology and coordinated industrial policy.

The automotive sector in Mexico: from growth engine to low-added-value trap

The North American Free Trade Agreement catalysed the development of an automotive value chain in Mexico, geared towards integration with the US. However, despite the growth in vehicle exports, Mexico's role in the automotive value chain has been limited to low-complexity and labour-intensive activities. High-value-added activities such as research and development, design, and advanced manufacturing have remained concentrated in the US. This has constrained Mexico's ability to move up the value chain and achieve higher levels of technological advancement and economic growth.[101]

In Sonora, the arrival of Ford's assembly plant in 1986 supported the shift from an agriculture-based economy to an industrial one. This transformation led to increased productivity, higher wages, and economic growth. However, decades later, Sonora has been stuck in traditional manufacturing sectors, making it vulnerable to competition from other low-cost hubs. To revitalize its economy and to live up to its nearshoring potential, Sonora needs to diversify its industrial base and invest in high-value-added sectors such as EV manufacturing.[102] For that, sound and coordinated industrial policies will be needed to jumpstart growth in a decarbonizing world.

Federal and state industrial policy: Plan Mexico and Plan Sonora

Plan Mexico aims to modernize the country's industrial base through innovation, sustainability, and the development of high-tech industries. In the plan, the Mexican government has identified the EV sector as a key strategic industry, given the country's existing industrial base, infrastructure, and skilled workforce. Plan Mexico, however, has been a top-down national strategy, in which, through the identification of *polos de bienestar* (poles of well-being)—namely, the promotion of regional growth and social well-being through targeted investments and job creation—the plan selects regions within Mexico that have specific comparative advantages and encourages them to expand in those areas. Sonora's key comparative advantage, according to the national government's outline, is in electricity generation, not in the automotive industry. Subsequently, however, the federal government signalled that Sonora could indeed be an EV manufacturing hub.

[99] *Plan México*, Government of Mexico, 2025, at https://www.planmexico.gob.mx

[100] 'Olinia Project is presented and the state of Sonora is a strong candidate to host the facility', Consejo para el Desarrollo Sostenible de Sonora (Council for the Sustainable Development of Sonora), 6 February 2025, at https://codeso.mx/en/olinia-project-is-presented-and-the-state-of-sonora-is-a-strong-candidate-to-host-the-facility/.

[101] Mateo Crossa, 'Contorting transformations: uneven impacts of the U.S.–Mexico automotive industrial complex', *Competition & Change*, 26(5), 2022, at https://doi.org/10.1177/10245294211045453.

[102] Douglas Barrios et al., *There is a Future after Cars: Economic Growth Analysis for Hermosillo*, Harvard University Center of International Development, 2018, at http://www.tinyurl.com/2boa29wz; Andres Fortunato, *Nearshoring in Hermosillo: Analysis of Economic Growth Opportunities*, Harvard University Growth Lab, 2025, at https://growthlab.hks.harvard.edu/sites/projects.iq.harvard.edu/files/2025-01-glwp-240-nearshoring-in-hermosillo.pdf.

This would align with Sonora's own plans to leverage the state's existing strengths in the automotive sector and its vast supplies of natural resources to develop a competitive EV manufacturing hub. Lithium is at the heart of Plan Sonora. The state has Mexico's largest lithium reserves, which had attracted the Sonora Lithium Project, operated by Bacanora Lithium and partially owned by China's Ganfeng Lithium. The project aimed to produce 35,000 tonnes of lithium per year, having positioned Sonora as a crucial player in the global lithium market.[103]

However, the nationalization of lithium in 2022 added layers of complexity to the development of the EV industry in Sonora.[104] The Mexican government declared lithium a strategic mineral, giving the state control over all activities along the value chain. In 2023, Ganfeng's concession was cancelled, driving the lithium company to file arbitration against Mexico a year later.[105]

Beyond the dispute, the Sonora Lithium Project focused on lithium carbonate production and not manufacturing of cathodes, anodes, electrolytes, or separators. Merely extracting lithium to export it to China, the leading and dominant refiner, would have contributed little to national development. For Sonora, relying on mineral exports could have been detrimental by prompting de-industrialization. To maximize lithium's potential, strategic investments in downstream industries are needed. Accordingly, Plan Sonora includes a strategic focus on the development of EV manufacturing, adding value across mining and battery manufacturing, as part of a broader effort to create a sustainable industrial ecosystem in the state. The plan aims to leverage not only its lithium reserves but also its untapped solar photovoltaic potential. In addition, the plan includes modernizing its ports and establishing scientific parks to drive technological innovation and research.[106]

Sonora cannot do it all alone, however. The success of its EV industry will rely on coordinated efforts between national and local governments, as well as strategic partnerships with foreign investors and research institutions to facilitate technology transfer and industrial upgrading. Lessons can be learned from Indonesia, which, having the biggest nickel reserves in the world, has banned its raw exports to encourage refining and manufacturing industries to establish in the country—though that approach has its critics.[107] For clean energy value chains, the US and others have started to adopt similar industrial policy instruments as China, albeit with less success.[108] Mexico can also take a page from its Asian peers' playbook.

Quid pro quo: forcing technology transfer for market access

While both Plan Mexico and Plan Sonora are ambitious, Mexico currently lacks the battery manufacturing technology to kick-start an EV value chain.

Tesla had considered building a gigafactory in the state of Nuevo León to produce batteries and EVs for the North American market. However, concerns about water consumption in the region and ongoing trade tensions with the US have delayed the decision. Other potential investors include global automotive companies such as General Motors, Ford, and Volkswagen, all of which have manufacturing facilities in Mexico. But relying solely on the US or legacy companies for technology transfer may keep Mexico in the low-value-added segment of the automotive industry. Cooperation with Chinese companies offers an alternative for Mexico to escape these traps.

BYD, for instance, has been establishing a presence in Mexico to leverage the country's proximity to the US market and its free trade agreement. The company has expressed interest in building a factory in Mexico to produce 150,000 cars a year, but the plans have not materialized.[109] In that context, Mexico is in a position to replicate the industrial policy instruments that China itself once used to force technology transfer and climb up the automotive value chain.[110] By offering access to its market and free trade

[103] *Technical Report on the Prefeasibility Study for the Sonora Lithium Project, Mexico*, Ausenco, April 2016, at https://bacanoralithium.com/_userfiles/pages/files/documents/technicalreportontheprefeasibilitystudyforthesonoralithiumprojectmexico_compressed.pdf.

[104] Vlado Vivoda et al., 'Lithium nexus: energy, geopolitics, and socio-environmental impacts in Mexico's Sonora project', *Energy Research & Social Science*, 108, February 2024, at https://doi.org/10.1016/j.erss.2023.103393.

[105] 'China's Ganfeng files arbitration against Mexico over lithium deposit', *Reuters*, 24 June 2024, at https://www.reuters.com/markets/commodities/miners-including-chinas-ganfeng-launch-arbitration-case-against-mexico-over-2024-06-24/.

[106] *Plan Sonora de Energías Sostenibles*, Government of the State of Sonora, 2024, at https://codeso.mx/wp-content/uploads/2024/07/ESP_PLAN-SONORA-FINAL-ABR1.pdf

[107] 'Just because Indonesia has nickel doesn't mean it should make EVs', *The Economist*, 9 January 2025, at https://www.economist.com/leaders/2025/01/09/just-because-indonesia-has-nickel-doesnt-mean-it-should-make-evs.

[108] Linxiao Zhu, *Reconfiguring Globalisation: A Review of Tariffs, Industrial Policies, and the Global Solar PV Supply Chain*, Oxford Institute for Energy Studies, December 2024, at https://www.oxfordenergy.org/publications/reconfiguring-globalisation-a-review-of-tariffs-industrial-policies-and-the-global-solar-pv-supply-chain/.

[109] 'Does made in Mexico mean made by China?', *The Economist*, 5 January 2025, at https://www.economist.com/the-americas/2025/01/05/does-made-in-mexico-mean-made-by-china.

[110] Jie Bai et al., *Quid Pro Quo, Knowledge Spillover, and Industrial Quality Upgrading: Evidence from the Chinese Auto Industry*, National Bureau of Economic Research Working Paper No. 27644, November 2022, at https://www.nber.org/papers/w27644.

agreement, Mexico could require Chinese EV manufacturers to establish joint ventures with local companies. In the upstream industry, the nationalized lithium company LitioMX can enter partnerships with foreign companies as long as the Mexican government retains 51 per cent of the ownership. Together with other industrial policy instruments, this would promote Mexican companies acquiring the necessary skills and knowledge to produce advanced EV components such as batteries and electric drivetrains.

Additionally, Mexico could implement local content requirements, mandating that a certain percentage of EV components be sourced from domestic suppliers. This would stimulate the growth of a local supply chain and create opportunities for Mexican companies to participate in the EV value chain.

All that, however, would have to be done carefully and strategically amid growing tensions with its neighbour to the north. The US has expressed concerns over China's use of Mexico as a back door to access the American market, which could potentially disrupt the United States-Mexico-Canada Free Trade Agreement. The influx of Chinese EVs, in fact, has been a bipartisan concern dating back to President Biden, who raised tariffs on Chinese EVs to 100 per cent. Importantly, those tariffs did not apply to vehicles manufactured in Mexico. In fact, to address American concerns, Mexico has previously imposed tariffs on other Chinese imports, such as steel and aluminium, and implemented rules requiring substantial transformation of these materials within Mexico before export to the US. That said, the rules of the game are changing with the new US administration. President Trump has threatened to impose a 25 per cent tariff on all Mexican imports, including all cars that are not manufactured in the US, unless Mexico curbs illegal migration and drug trafficking.[111]

If these tariffs are implemented—they were initially set to take effect on February 4, then first delayed until March 4 and subsequently until April 2—the auto manufacturing industry will be significantly affected. About 80 per cent of Mexican exports go to the US, of which 40 per cent come from the automotive industry. Because of its strong ties with the US economy, Sonora is among the 10 states in Mexico that will be most affected by the implementation of tariffs.[112]

As geopolitical tensions with the US continue to rise, the window of opportunity shrinks for Mexico to leverage EV value chains to accelerate and upgrade industrialization. But the opportunity still exists. Increasing its success rates will require better industrial policy coordination between national and local governments, with a focus on forging foreign partnerships while developing local capabilities that will contribute to national and local development in the long run.

Conclusions

EVs present an opportunity for Mexico and Sonora to escape low-added-value traps by moving up the automotive value chain. The production of high-value components such as batteries can enhance technological capabilities and economic growth.

Both Mexico and Sonora are in a privileged position to capitalize on the growing EV market, even with all the challenges mentioned above. Mexico and Sonora have an established automotive industry—albeit one focused on low-added-value activities—and Sonora has Mexico's largest lithium reserves. However, lithium has been nationalized, which demands coordination between federal and state governments for an EV industry to develop, especially if Sonora plans to become an EV manufacturing hub. Lithium mining is complex and, as most commodities, vulnerable to market fluctuations. Economies of scale can be exploited by developing vertical value chains. But that takes time, and the federal government is prioritizing Puebla over Sonora. In addition, China dominates the refining segment.

Thus, the development of an EV value chain must start with foreign partnerships and the import of advanced battery manufacturing technologies, and related technical knowhow, that are absent in the country.

For that, Mexico is also in a privileged position, but seizing this opportunity amid current geopolitical tension will be no easy task. As the US's biggest trading partner with a free trade agreement, Mexico is an attractive destination for EV value chains. However, the constant threat of 25 per cent tariffs on Mexican exports to the US diminishes the attractiveness. Tesla, for example, is backing out from its announced investments. Moreover, there have been bipartisan concerns in the US regarding the influx of Chinese EVs in the American market, and companies potentially using Mexico and the free trade agreement as a back door. In the past, Mexico has addressed these concerns by imposing tariffs on Chinese imports, and American public officials have recently claimed that Mexico has offered to repeat this action.[113] But tariffs alone, beyond increasing consumer prices, will do little to develop value

[111] 'Does made in Mexico mean made by China?', *The Economist*, 5 January 2025, at https://www.economist.com/the-americas/2025/01/05/does-made-in-mexico-mean-made-by-china.
[112] 'Some Mexican states could be vulnerable to a potential tariff on U.S. exports', *S&P Global*, 27 February 2025.
[113] 'EU dice que México propuso igualar los aranceles estadounidenses a China', *Animal Politico*, 28 February 2025, at https://animalpolitico.com/internacional/mexico-igualar-aranceles-eu-china.

chains. Instead, licensing, joint ventures, and explicit technology transfer requirements can help Mexico develop an advanced manufacturing base over time, while allowing Chinese companies to invest in Mexico while minimizing geopolitical tensions with the US.

That said, both Mexico and Sonora must do more than forcing technology transfer through joint ventures and local content requirements, like Chinese industries have done in their own automotive industry. Industrial policy will have to involve research and development, building advanced manufacturing facilities, and developing a skilled workforce. Indeed, developing local capabilities to adopt, adapt, learn, and innovate in battery manufacturing will be crucial for Mexico's ability not only to manufacture low-cost EVs by 2026 but also to become a significant player in the global market. All of this requires coordinated efforts at the national and local levels, to promote technology transfer and innovation rather than simple assembly. Plan Mexico and Plan Sonora are good first steps, but improved coordination and better long-term incentives will be necessary—particularly to face together the geopolitical and trade uncertainty from the US.

CHINA'S ROLE IN THE ELECTRIC VEHICLE INDUSTRY IN BRAZIL

José Carlos Díaz Silva

Electromobility is one of the most important industrial developments brought about by the energy transition. Competition in EV manufacturing is now strategic, both technologically and geopolitically, and that competition has profound impacts on regional trade and political relationships. In electromobility, China has taken a clear position of leadership over the countries of the West. Since 2023, BYD, now the world's largest electric vehicle (EV) manufacturer, has been selling more cars globally than Tesla, the leading non-Chinese EV carmaker and exemplar of US EV technology. The reason that China is leading the global trade in EVs is that China controls the entire EV supply chain: extraction and refining of minerals such as lithium, battery production, car assembly, and distribution.

Meanwhile, Western companies depend on China to manufacture their cars. Tesla does not produce most of the batteries for the cars it sells in many markets; instead, it buys low-cost lithium-iron-phosphate batteries from Chinese battery giant CATL. Volkswagen, the most important European carmaker, manufactures in China for export because it is much more efficient. However, this has not prevented the German carmaker from finding itself in crisis, as it has been unable to compete at the same level as its Asian peers.

Given its widening lead in EV technology, China has begun not only to export EVs, but also to invest in local EV production in countries such as Brazil. Brazil is one of its strategic partners, not only in the context of the informal group formed by Brazil, Russia, India, China and South Africa (BRICS), but also commercially and politically since the return of Lula da Silva to the government. Internal and external factors have combined to encourage the construction of the electromobility industry in Brazil. On the one hand, government policies favour public investment and offer incentives for private initiatives in the EV space. On the other hand, the arrival of Chinese companies, especially BYD, is resulting in not only assembly but also establishment of a complete production chain to supply the South American markets.

Government policies

Brazil's EV industry traces its origins to 2006, with the creation of the Electric Vehicles programme, through the association of the Brazilian and Paraguayan co-owned hydroelectric company Itaipu Binacional with the Swiss energy company Kraftwerke Oberhasli AG and the Italian automaker Fiat. This partnership resulted in the research and development of EV models, such as the Palio Weekend Electric.[114] Under the same programme, the country's first 100 per cent electric bus was introduced in 2010.[115] Although not yet commercialized, the first piloted electric aircraft in Latin America was also developed in 2015, as well as the production of batteries and a battery recycling programme.

The Brazilian government also supported the nascent EV industry at an early stage. In 2010, the Brazilian Ministry of Science invested US$5 million in the creation of a network of institutes for the development of EV-related technologies.[116] In addition, in partnership with Brazilian companies such as Petrobras, the Innovation Projects Fund of the federal government was financed, which meant an outlay of US$55 million between 2010 and 2014.

[114] Gilmar Masiero et al., 'The global value chain of electric vehicles: a review of the Japanese, South Korean and Brazilian cases, *Renewable and Sustainable Energy Reviews*, 80, December 2017, at https://doi.org/10.1016/j.rser.2017.05.148.
[115] 'Veículos elétricos', *Itaipu*, accessed 9 March 2025, at https://www.itaipu.gov.br/tecnologia/veiculos-eletricos.
[116] Roberto Marx and Adriana Marotti de Mello, 'New initiatives, trends and dilemmas for the Brazilian automotive industry: the case of Inovar Auto and its impacts on electromobility in Brazil', *InderScience Online*, 14 April 2014, at https://doi.org/10.1504/IJATM.2014.060751.

Later, in 2015, Brazil exempted electric and hybrid vehicles from import tariffs, which previously were set at 35 per cent on battery-powered EVs and 7 per cent on hybrids.[117] However, in 2024, the Lula da Silva government gradually reinstated tariffs to protect domestic production.[118] In the case of battery EVs, in January 2024 the tariffs were raised to 10 per cent, then in July to 18 per cent, and they will be reinstated at the original 35 per cent by 2026. On the same dates, hybrids will be taxed at 15 per cent and 25 per cent and will reach the same value as electrics in 2026.

Brazil, the US, and Europe all have large domestic markets. The US and Europe have initiated protective tariffs and other measures targeting China; but given their high cost relative to Chinese players, and China's centrality to the EV supply chain globally, these measures do little to bolster their position in global EV markets. In contrast, in pursuing low-cost production through partnership with Chinese automakers, Brazil expands EV adoption domestically and through exports to Latin America. Moreover, Brazil's EV trade links are with Asia and not with the West, and Brazil is partnering with the technological leader in the EV industry, which will seek to invest more in the nation to get around protectionist measures. In contrast to US President Trump's trade war—which seeks to protect established companies like Tesla, which have lost the competitive battle with Chinese players—the tariffs in the South American country protect a nascent industry, to take advantage of its economies of scale and the region's markets.

In 2024 the government launched the Green Mobility and Innovation Program (Mover), which includes extensive tax incentives over four years.[119] Fiscal support was set at US$681 million in 2024, and US$759 million in 2025, with a target of US$2 billion by 2028. In addition to national and governmental projects, foreign companies have invested in the country, especially Chinese companies such as BYD and Great Wall, and to a lesser extent Stellantis, Volkswagen, BMW, Toyota, and General Motors. Foreign investment has allowed Brazil to move away from the US sphere of influence, and move closer to China, which has the technology to consolidate the state's policy of industrialization in electromobility.

BYD, now the world's largest EV company, has been present in Brazil since 2015, when it signed an agreement with Marco Polo, a domestic company, to build an electric bus plant. That same year, the Chinese company put into operation a factory dedicated to the production of 100 per cent electric buses in Manaus, with an assembly capacity of 1,000 units per year. The facility also produces the batteries needed for the buses. In 2017, a factory to produce solar panels was opened in the city of Campinas. In 2020, a lithium ferrophosphate refining plant was inaugurated in Manaus. Subsequently, in 2024, more than US$600 million was invested in a plant with a capacity of 150,000 EVs per year.[120] Investments made by the Asian company will exceed US$1.1 billion by the end of 2024, an amount close to the investment projects expected to be fulfilled by other companies by 2028. This manufacturer has consolidated a complete supply chain in Brazil, which will allow it to export its vehicles throughout Latin America.

Other Chinese players have also been actively growing their Brazilian presence. In 2021, Chinese carmaker Great Wall bought Mercedez Benz's Iracemápolis factory for conventional cars, which had been closed in 2020 due to the Covid pandemic, and adapted it for the production of EVs. In 2022 Great Wall announced it would invest US$1.64 billion in the plant through 2032. In 2022, Geely-owned Volvo announced a US$284 million EV investment in Brazil, to last through 2025.[121]

The development of the Brazilian supply chain places the country as a strategic area for securing entry into Latin American markets. Companies from all over the world are interested in expanding their presence in Brazil. In 2024 Stellantis announced a US$6 billion investment in Brazil and the rest of South America. Investment plans from other companies are also in the pipeline. Toyota will invest US$2 billion until 2030.[122] Volkswagen is to invest US$1.8 billion between 2024 and 2028.[123] However, the company's current crisis creates uncertainty, raising doubts about the viability of the capital expenditure; besides, it has

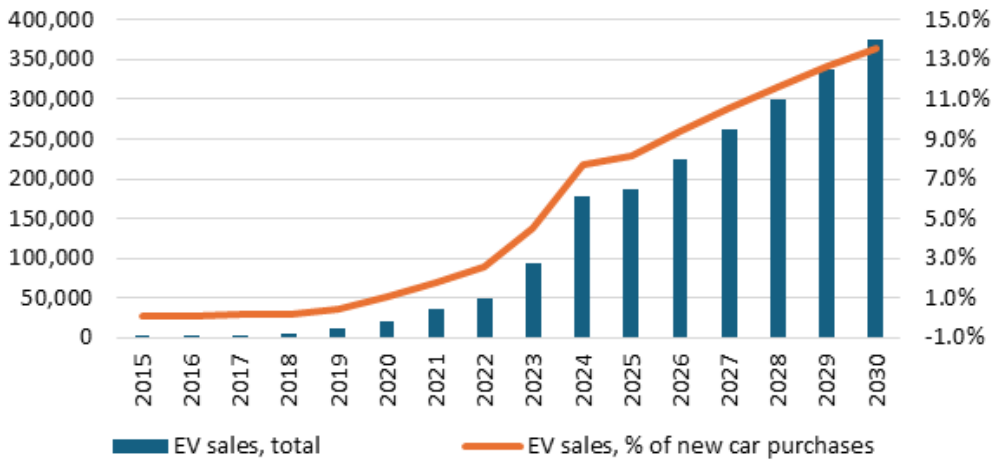
[117] 'Governo zera imposto de importação para carro elétrico e a hidrogênio', *Auto Esporte*, 27 October 2015, at https://g1.globo.com/carros/noticia/2015/10/governo-zera-imposto-de-importacao-para-carro-eletrico-e-hidrogenio.html.
[118] 'Brazil Electric Vehicle Market Databook 2024: global firms are investing in the development of manufacturing facilities to by-pass tax tariffs implemented on the import of electric vehicles in Brazil', *Business Wire*, 16 August 2024, at https://www.businesswire.com/news/home/20240816677947/en/Brazil-Electric-Vehicle-Market-Databook-2024-Global-Firms-are-Investing-in-the-DVEelopment-of-Manufacturing-Facilities-to-By-pass-Tax-Tariffs-Implemented-on-the-Import-of-Electric-Vehicles-in-Brazil---ResearchAndMarkets.com.
[119] 'Brazil automotive green mobility investment', US Department of Commerce, International Trade Administration, 18 April 2024, at https://www.trade.gov/market-intelligence/brazil-automotive-green-mobility-investment.
[120] *Global EV Outlook 2024*, International Energy Agency, 2024, at https://www.iea.org/reports/global-ev-outlook-2024.
[121] 'Volvo to invest $283 mln in Brazil by 2025', *Reuters*, 2 February 2022, at https://www.reuters.com/business/autos-transportation/volvo-invest-283-mln-brazil-by-2025-2022-02-02/.
[122] 'Toyota confirms $2 billion investment in Brazil by 2030', *Reuters*, 5 March 2024, at https://www.reuters.com/business/autos-transportation/toyota-confirms-2-billion-investment-brazil-by-2030-2024-03-05/.
[123] 'Volkswagen to invest a further $1.8 billion in Brazil in next five years', *Reuters*, 2 February 2024, at https://www.reuters.com/business/autos-transportation/volkswagen-add-18-bln-brazil-investment-plan-local-media-2024-02-01/.

announced that its electric production will remain in Europe, leaving a gap in the market. The other German manufacturer, BMW, announced in 2022 its investment plan until 2030, amounting to US$1.7 billion.[124] Finally, General Motors will invest US$1.4 billion until 2028.[125] While Western investment projects are looking to strengthen, BYD has already consolidated its position.

The results

Due to all of these factors, EV sales in Brazil have grown significantly, and the country is exporting more and more of its cars to the Southern Cone and Central America. Between 2023 and 2024, EV sales increased from 93,000 to 177,000, a growth of 88.8 per cent (see Figure 1). This has been accompanied by a higher share of sales, with 7.7 per cent of the total in 2024, compared to 4.5 per cent in 2023, so the displacement of conventional cars increased by 7 per cent. Furthermore, the data seem to indicate a turning point, whereby EVs, especially hybrids, can be expected to start gaining ground on conventional cars.

Figure 1: EV sales in Brazil and their share of total new car purchases, historical and projected (2015–2030)



Source: Elaborated with ANFAVEA data (https://anfavea.com.br/site/)

As shown in Figure 1, EV sales are expected to double in the next five years, accounting for 14 per cent of total purchases. This means that adoption would be steady, but not total. The most likely scenario is the coexistence of EV vehicles with diesel, ethanol, and flex fuel vehicles. Chinese carmakers also manufacture these types of cars, which means that Asian brands will be able to absorb this market.

BYD's EV market share, as a consequence of its investments, was 70 per cent in 2024,[126] putting it in absolute dominance in Brazil, which has a potential vehicle demand of 2.3 million units per year. The second largest brands are Great Wall Motors and Geely.[127] The best-selling models in the same year were the BYD Dolphin (24 per cent of the total) and the BYD Seagull (17 per cent of the total). Part of the rise of Chinese brands is that they offer much cheaper EVs compared to Western ones. For example, while the Tesla Model 3 costs around US$43,000, the BYD Dolphin costs around US$18,000, lower than a typical internal combustion engine (ICE) model of around US$20,000. The lower production costs and economies of scale of Asian companies have won the competition. This is a result of the industrial policy and large investments made in Brazil, which already has an entire production chain for EVs and electric buses, with BYD extracting lithium, refining it, producing batteries, and assembling vehicles. Because of this integration, and their overall leading position in EV technology, Chinese companies have been able to seize the opportunities offered by policymakers and the growing market.

[124] 'BMW Group announces $1.7 billion (USD) investment to build electric vehicles in the U.S. and signs agreement with Envision AESC for the supply of battery cells to plant Spartanburg', BMW Group, 19 October 2022, at https://www.press.bmwgroup.com/global/article/detail/T0404837EN/bmw-group-announces-1-7-billion-usd-investment-to-build-electric-vehicles-in-the-u-s-and-signs-agreement-with-envision-aesc-for-the-supply-of-battery-cells-to-plant-spartanburg?language=en.
[125] 'GM invierte miles de millones en la electrificación de su negocio brasileño', *Electrive*, 25 January 2024, https://www.electrive.com/es/2024/01/25/gm-invierte-miles-de-millones-en-la-electrificacion-de-su-negocio-en-brasil/.
[126] Brazil's EV sales hit record high in 2024', *Argus Media*, 6 January 2025, at https://www.argusmedia.com/en/news-and-insights/latest-market-news/2644370-brazil-s-ev-sales-hit-record-high-in-2024.
[127] Frik Els, 'Brazil's 2024 EV market is a tariff test case', *Adamas Inside*, 3 September 2024, at https://www.adamasintel.com/brazils-2024-ev-market-is-a-tariff-test-case/.

The growing presence of Asian brands in Brazil also brings difficulties—for example, in regulating the exploitation of workers. It was recently reported that a BYD contractor used slave labour in the construction of one of its factories in Camaçari.[128] Ironically, the plant at issue will be one of the most modern in the region, producing more than 150,000 EVs per year.[129] Despite the above, the company will continue its operations by severing its ties with the construction company. We must be aware that cost-cutting or unethical business practices are common among large companies, whether in the West or the East, and the challenge for governments will be to ensure labour regulations that prevent the rampant exploitation of the workforce.

A significant share of Brazil's imported EVs come from China. As shown in Table 1, more than 80 per cent of the EV imported volume came from Asia in 2024, while Germany and the United States together account for only 6.8 per cent. China is Brazil's main supplier of imported cars, regardless of the origin of the brands, given that Western companies such as Tesla, General Motors, and Volkswagen manufacture vehicles in China for export markets. In 2024, 149,000 EVs were imported from China; the same figure in 2023 was 46,000—EV shipments more than tripled in one year. This shows the appetite for EVs; therefore, we can expect that when Brazilian EV factories begin to produce at full capacity, EV imports will fall and be replaced by domestic production.

Table 1: Brazil's EV imports (batteries and hybrids) from top 10 importers, 2024

	Volume		Value	
Country	Units	% of total imports	Million US$	% of total imports
China	149,519	79.40%	3,039	67.90%
China, Hong Kong	6,881	3.70%	136	3.00%
Germany	5,868	3.10%	233	5.20%
USA	4,560	2.40%	242	5.40%
Thailand	4,269	2.30%	126	2.80%
Slovakia	4,232	2.20%	211	4.70%
Japan	3,900	2.10%	98	2.20%
Mexico	2,906	1.50%	120	2.70%
South Africa	1,322	0.70%	40	0.90%
Sweden	1146	0.60%	51	1.10%
Total top 10	184,603	98.00%	4,295	96.00%

Source: Elaborated with UNComtrade data (https://comtradeplus.un.org/)

While imports from China account for 80 per cent of the total in terms of volume, they account for less than 70 per cent of value. This difference is because Asian EVs are cheaper than their competitors, which mostly explains their success in penetrating the Brazilian and Latin American markets.

EV plants in Brazil produce not only for domestic consumption but also for export (see Table 2). In 2024, the country exported 27,000 EVs, 22 per cent more than in 2023, when the volume amounted 22,000 units. The main destinations were Argentina (39.2 per cent), Colombia (33.1 per cent), and Chile (9.6 per cent). Especially in the latter two countries, laws have been passed in favour of the transition to electromobility; therefore, Brazilian shipments are likely to grow steadily in the coming years. Moreover, their EVs have a presence throughout Latin America except for Mexico. For example, in 2024, 14,000 EVs were sold in Argentina, accounting for two-thirds of its exports; meanwhile, in Colombia, sales were 9,000, equivalent to the exported amount.

[128] 'BYD rescinde contrato con constructora acusada de trabajo esclavo en su fábrica en Brasil', *SWI*, 14 December 2024, at https://www.swissinfo.ch/spa/byd-rescinde-contrato-con-constructora-acusada-de-trabajo-esclavo-en-su-f%C3%A1brica-en-brasil/88638679.
[129] Manuel Pineda Curiel, 'BYD anuncia construcción de vehículos híbridos en Brasil', *Las Empresas Verdes*, 29 August 2024, at https://lasempresasverdes.com/byd-anuncia-construccion-de-vehiculos-hibridos-en-brasil/.

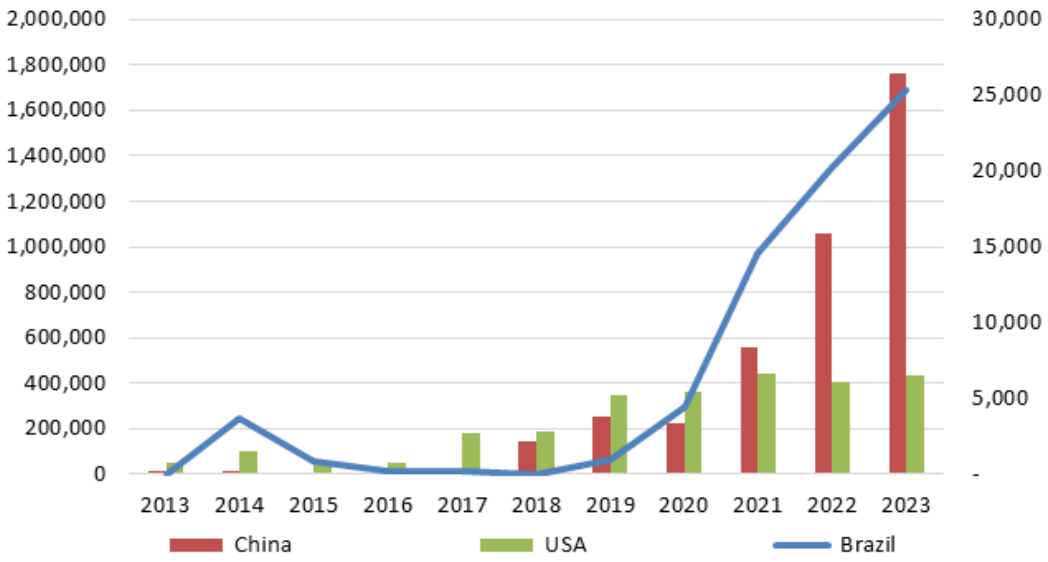
Table 2: Brazil's EV exports (batteries and hybrids) to top 10 export destinations, 2024

Country	Volume		Value	
	Units	% of total exports	Million US$	% of total exports
Argentina	10,888	39.2%	209	36.5%
Colombia	9,179	33.1%	202	35.2%
Chile	2,671	9.6%	48	8.3%
Ecuador	1,245	4.5%	28	5.0%
Uruguay	1,048	3.8%	24	4.2%
Peru	778	2.8%	18	3.1%
Paraguay	607	2.2%	14	2.4%
Guatemala	304	1.1%	7	1.3%
Costa Rica	213	0.8%	4	0.7%
Nicaragua	208	0.7%	5	0.9%
Total top 10	27,141	97.8%	559	97.6%

Source: Elaborated with UNComtrade data (https://comtradeplus.un.org/)

The link between Brazil and China will ensure that Asian brands will dominate EV markets in the Southern Cone and Central America, as a result of the policies implemented by Brazilian governments and the aggressive investment made by Chinese companies. This is compounded by the loss of competitiveness of US and European carmakers in this market. Figure 2 shows that China's and Brazil's exports have both grown since 2020, while those of the US began to decline in 2021. From the perspective of exports to Brazil or other countries, and overall economic influence, the US is going backwards, signalling its loss of regional hegemony vis-à-vis China.

Figure 2: EV exports China, USA and Brazil (right axis), 2013–2023



Source: Elaborated with UNComtrade data (https://comtradeplus.un.org/)

Conclusions

In conclusion, the EV industry in Brazil is in the process of consolidation, as a result of a combination of government industrial promotion policies and substantial investment by Chinese companies. The significant point is that not only are vehicles being assembled in Brazil, but the entire supply chain has been set up. Thus, the South American nation is moving away from the US sphere of influence and closer to China's, both economically and strategically. For Asian companies, Brazil will provide access to Latin American markets. This serves the government's policy goals to move away from US influence, as exemplified by recent moves to distance Brazil from the US positions in recent G20 meetings, and over the longer term through establishing the BRICS

as a counterweight to the US. The challenges for policymakers will be to put in place the necessary regulations to mitigate the effects associated with the use of resources and to avoid excessive exploitation of the workforce. In sum, the transition towards electromobility is advancing in Latin America, with Chinese companies gaining increasing influence and market share, in many respects displacing those of US and European players.

DOES BRAZIL HAVE A ROLE TO PLAY IN GLOBAL EV SUPPLY CHAINS?

Alexis Harmon

Brazil expects to bring Brazilian-made electric vehicles (EVs) to the mass market this year, but there's a catch—they're actually Chinese. Despite ranking eighth globally in automotive production, Brazil lacks a major domestic car brand. Traditionally dominated by American, European, and Japanese manufacturers, Brazil has been able to extract substantial value by incentivizing local production and serving as a regional export hub for Latin America.

Now, as the global EV industry expands, Brazil is seeking to define its role within the EV supply chain and production network, but new players are rapidly reshaping the market: Chinese EV manufacturers. Although EVs only represented about 6 per cent of new car sales in Brazil in 2024, the market is rapidly growing. EV sales doubled in 2023 and again in 2024—and 90 per cent of sales were Chinese brands.[130] Fortunately for Brazil's auto industry, the government's long-standing strategy of using tariffs to deter imports while offering subsidies to encourage domestic production is proving as effective as ever. Within just a couple of years of Chinese EV imports beginning to flood the market, all of China's largest EV manufacturers have begun opening factories in Brazil.

With Chinese EV and battery factories expected to come online in 2025, legacy Western automakers investing billions to stay competitive, and government policies supporting the industry, Brazil's auto sector appears ready to transition into the EV era. However, relatively high costs, geographic distance, and uncompetitive trade policies mean Brazil is likely to remain a regional leader rather than a global EV hub. Still, recent policies aimed at strengthening the battery supply chain could allow Brazil to move beyond its historic reliance on raw material exports and enter higher-value portions of the EV battery supply chain.

From Ford Model T assembly to BYD battery plants

Brazil has a remarkably long history of robust automotive manufacturing, but the industry is undergoing a transformation. A century ago, Henry Ford established Brazil's first car factory, an assembly plant putting together Model Ts. In 2021, after burning through almost $8 billion in a decade, Ford announced it would cease operations in Brazil altogether.[131] Just two years later, China's BYD—the world's largest EV manufacturer—took over one of Ford's former industrial parks to launch its first Brazilian plant. Not long before, Great Wall Motors took over a Mercedes-Benz plant to start producing its own EVs in Brazil.

These shifts reveal a broader trend: Western carmakers are struggling to stay competitive in Brazil, while Chinese manufacturers are stepping in aggressively. Despite being the largest auto market in Latin America, domestic demand has remained low and exports have been uncompetitive, leading to low factory utilization rates. While Brazilian auto factories can produce twice as many cars as the domestic market needs, exports rarely exceed 20 per cent of total production.[132] Compare this to Mexico, which is able to export over 80 per cent of the cars it makes.[133]

China's competitive EV pricing could change this dynamic. While Western automakers struggle to produce a single affordable EV, China has a variety of impressive options under $25,000. Legacy manufacturers like Stellantis and Volkswagen still dominate Brazilian sales today, but as EV adoption grows, their market position may be threatened if they cannot become more competitive.

What's clear is that Chinese automakers are here to stay. Although the initial flood of EV imports was a response to China's domestic overcapacity, they're now investing billions in local production and mines that can take a decade to come online, promising a long-term commitment.

[130] 'Latin America's EVs Get a Big Boost from Chinese Carmakers', *Bloomberg New Energy Finance*, February 2024, at https://about.bnef.com/blog/latin-americas-evs-get-a-big-boost-from-chinese-carmakers/.
[131] Anthony Boadle, 'How Ford burned $12 billion in Brazil', *Reuters*, 20 May 2021, at https://www.reuters.com/business/autos-transportation/how-ford-burned-12-billion-brazil-2021-05-20/.
[132] Anthony Boadle, 'How Ford Burned $12 Billion in Brazil', Reuters, 20 May 2021, at https://www.reuters.com/business/autos-transportation/how-ford-burned-12-billion-brazil-2021-05-20/.
[133] Anthony Boadle, 'How Ford Burned $12 Billion in Brazil', Reuters, 20 May 2021, at https://www.reuters.com/business/autos-transportation/how-ford-burned-12-billion-brazil-2021-05-20/.

For Brazilian President Luiz Inácio Lula da Silva, this is a welcome development. The reintroduction of tariffs on EVs should not be seen as Brazil pushing back against Chinese influence in the region; rather, it was a strategic move to encourage Chinese investment in local manufacturing. In fact, Lula has actively fostered ties with China, visiting the country soon after his inauguration and signing 15 trade agreements during his first year in office. Chinese companies are seizing the opportunity to expand their presence in Latin America, drawn by Brazil's welcoming investment environment and relative lack of geopolitical barriers. This stands in contrast to China's auto expansion in Mexico, where US pressure has stifled investment. Facing US President Trump's latest tariff threats, Mexico has restricted certain Chinese investments and raised tariffs on Chinese imports. Given the mutual benefits, Brazil appears likely to remain China's hub for automotive manufacturing in the region.

How Brazil is forging its own future

Government policies under President Lula are playing a crucial role in reindustrializing Brazil and boosting its EV supply chain. In 2023, Lula launched a program to support local auto production through credit lines, tax breaks, and domestic content incentives.[134] In 2024, he introduced tariffs to curb imported EVs and unveiled Mover, a green mobility initiative that rewards sustainable vehicle production. These policies seem effective so far, as automakers have already announced over $20 billion in new investments over the next decade.[135]

Lula's strategy is multifaceted. On one hand, he wants to leverage Brazil's massive critical mineral reserves to attract investment. Brazil has the second-largest nickel and graphite reserves, the third-largest manganese reserves, and substantial lithium and copper deposits. To take advantage of this wealth, Brazil is working to diversify not just its investors but its export markets. Lithium is a prime example. Although Brazil had 14 per cent of lithium reserves, it did not export any until a few years ago. Now, it has rapidly become a top global supplier of lithium, and the recent loosening of lithium export regulations is expected to attract $2.8 billion in investments through 2030.[136]

However, Lula is also wary of falling into the 'resource curse' trap, a particularly touchy issue in a country that some have called the 'worst example of premature industrialization in the world'.[137] With value-added manufacturing's share of GDP cut in half since 1990 and over half of Brazil's exports dominated by raw materials, Lula has vocally criticized policies that 'condemn our country to be an eternal exporter of raw materials'.[138] To avoid this, Lula has made his intention of upgrading into the midstream and increasing domestic processing of minerals clear. Alongside his efforts to boost raw exports in the short term, policies like Mover are ensuring processing capacity increases rapidly in the medium term and are gradually increasing the requirements for domestic content in batteries.

While this doesn't automatically lead to access to more technologically advanced and higher-value-added portions of battery supply chains, it does provide leverage to pressure manufacturers into establishing some level of battery production in Brazil. However, the BYD battery plant in the old Ford factory highlights the challenge of this; despite significant hype surrounding a major EV battery plant opening in Brazil, the facility appears to be primarily focused on battery pack assembly.[139] This process is less technologically advanced and has less value-add than other downstream parts of the battery supply chain.

Skipping from refined lithium straight to battery pack assembly is an achievement, but it leaves a crucial gap that Lula will want to address: the midstream. Brazil currently lacks midstream battery production assets, but its strengths in silicon and artificial graphite processing could provide a natural advantage in anode production.[140] Chile provides an instructive case. Despite not having its

[134] 'Lula anunciará incentivos para produção de carros com etanol', *Poder360*, February 2024, at https://www.poder360.com.br/governo/lula-anunciara-incentivos-para-producao-de-carros-com-etanol/.
[135] 'Investment in Brazil's auto sector to reach BRL 100 billion by 2029', *Agência Brasil*, February 2024, at https://agenciabrasil.ebc.com.br/en/economia/noticia/2024-02/investment-brazils-auto-sector-reach-brl-100-bi-2029.
[136] 'Brazil's critical minerals and the global clean energy revolution', Wilson Center, 2024, at https://www.wilsoncenter.org/article/brazils-critical-minerals-and-global-clean-energy-revolution.
[137] 'Why industrial decline has been so stark in Brazil', *Economist*, 5 March 2022, at https://www.economist.com/the-americas/2022/03/05/why-industrial-decline-has-been-so-stark-in-brazil.
[138] 'Manufacturing, value added (% of GDP)—Brazil', *World Bank Data*, accessed 19 March 2025 at https://data.worldbank.org/indicator/NV.IND.MANF.ZS?locations=BR; 'Brazil Exports and Imports of Product Groups 2022', *World Integrated Trade Solution*, World Bank, 2024, at https://wits.worldbank.org/CountrySnapshot/en/BRA/textview; Anthony Boadle and Lisandra Paraguassu, 'Lula promises to unite divided Brazil, seek fair global trade', *Reuters*, 31 October 2022. https://www.reuters.com/world/americas/lula-promises-unite-divided-brazil-seek-fair-global-trade-2022-10-31/.
[139] 'A glimpse of BYD battery factory in Manaus, Brazil', *Xinhua*, 14 March 2024, at https://english.news.cn/20240314/337eccb4cdbb4ca69a240110a5a61c5d/c.html.
[140] 'New industrial policy for a new world: Seizing Brazil's opportunities in the energy transition', Poder360, 28–31 January 2025, at https://static.poder360.com.br/2025/02/GP03-Brazil-geopolitical-brief-EMBARGOED-UNTIL-28th-31st-JANUARY-PLEASE-DONT-CIRCULATE.pdf.

own auto manufacturing industry, Chile leveraged its lithium resources to partner with BYD on lithium processing and crafted plans to produce lithium cathodes for EV batteries. Though plans for cathode production have since been indefinitely delayed, the original agreement shows that under the right conditions, BYD is willing to invest in more advanced parts of the battery value chain.[141]

Other than critical mineral exports, Lula's primary focus remains on the regional market, Brazil's traditional playing field. For example, any battery production that does occur in Brazil will likely supply domestically made cars rather than foreign manufacturers. However, in some cases—such as biofuels—Lula isn't even looking beyond the domestic context. Brazil's historical focus on ethanol and flex-fuel vehicles has delayed EV adoption and manufacturing. Since the 1970s, Brazil has promoted biofuels as a sustainable alternative to gasoline, making it a world leader in renewable fuels but slowing the transition to electrification. Even now, the powerful sugarcane lobby ensures that the majority of new vehicles remain flex-fuel rather than fully electric, and EV charging infrastructure lags behind other major markets.

How all these competing factors will play out is unclear. Sigma Lithium and CBL are strengthening Brazil's position in lithium processing, while Volkswagen and Stellantis are evaluating investments in local battery assembly. Despite delays due to human rights concerns, BYD's upcoming plant and lithium refining projects suggest that Brazil could expand its role in the battery supply chain within the next few years.[142] However, BYD has also made it clear that it sees plug-in flex-fuel hybrids as the future of Brazilian automaking. With hybrids dominating Brazilian EV purchases, this appears to be a good bet for now. Brazil's large domestic market allows it to diverge from global trends without immediate consequences, but delaying full EV adoption for too long could leave its manufacturers at a competitive disadvantage.

Is Brazil a natural home for US or EU friend-shoring?

Despite substantial government incentives, Brazil still struggles to compete on cost. Incredibly, it is estimated that despite tariffs a car imported from Mexico is 12 per cent cheaper than the same car produced locally in Brazil.[143] Relatively high labour costs and taxes, expensive energy, and logistical challenges mean that despite its robust auto industry, Brazil is an unlikely home for US or EU friend-shoring. They simply have better options, respectively Mexico and Central and Eastern Europe.

Labour costs are one of the most significant factors. Brazilian labour is only a dollar or two more expensive per hour than Mexico generally, but automotive manufacturing wages in Brazil are particularly strong. Although wages in Central and Eastern Europe are higher than in Latin America, they remain far lower than in Western Europe. Energy costs also favour Mexico, which benefits from cheap natural gas imports from the US, keeping industrial electricity prices significantly lower than in Brazil. In contrast, Brazil has some of the highest energy costs, largely due to its reliance on hydropower, which fluctuates based on drought conditions. However, despite high energy costs, Brazil's abundant clean energy has made it a more attractive destination for companies seeking to decarbonize their production processes.

Another key factor is supply chain integration and proximity to key markets. Mexico has a well-established automotive supply chain spanning across North America and benefits from the US-Mexico-Canada Agreement, which allows tariff-free exports to the US and Canada. Meanwhile, Central and Eastern European countries have deep integration with the EU's auto industry and enjoy free trade within the EU, making them the most logical choice for European exports. Brazil, however, is geographically distant from both the US and EU, leading to high logistics costs and long shipping times. Additionally, Mexico and Central and Eastern Europe are growing EV battery hubs, attracting investments from Tesla, Volkswagen, and CATL, whereas Brazil's battery production remains limited so far.

However, with trade agreements quickly evolving, nothing is certain. The US-Mexico-Canada Agreement is up for renegotiation in 2026, and President Trump's threatened 25 per cent tariffs on Mexico mean Mexico's advantages could be destroyed in favour of American-shoring. So far Trump's threats appear to be impermanent and focused on extracting political concessions, but his trade policy is anything but predictable. The newly signed EU–Mercosur trade deal could theoretically improve the competitiveness of Brazilian exports, the deal is more focused on enabling European auto imports into Brazil. With Brazilian tariffs removed

[141] 'China's BYD postpones plans for 2025 Chile lithium cathode plant', *Reuters*, 16 May 2024, at https://www.reuters.com/markets/commodities/chinas-byd-postpones-plans-2025-chile-lithium-cathode-plant-2024-05-16/.
[142] Annabelle Liang, 'Brazil shuts BYD factory site over "slavery" conditions', *BBC*, 24 December 2024, at https://www.bbc.com/news/articles/c8xj9jp57r2o
[143] Anthony Boadle, 'How Ford burned $12 billion in Brazil', *Reuters*, 20 May 2021, at https://www.reuters.com/business/autos-transportation/how-ford-burned-12-billion-brazil-2021-05-20/.

and European imports' competitiveness boosted, European automakers actually might be less likely to expand Brazilian investments, not more so. Instead, European partners seem more focused on enhancing trade and investment in critical raw materials.[144]

Overall, the best location for EV production depends on the target market. As a result, Mexico and Central and Eastern Europe are emerging as the dominant EV manufacturing hubs for North America and Europe, respectively. Ultimately, Brazil will struggle to attract US or EU EV investments on cost alone. Instead, its best path forward is solidifying its role as Latin America's EV leader and expanding critical mineral partnerships with Europe.

Conclusion

Brazil is not on track to become a global EV production hub, but it is slowly emerging as Latin America's dominant EV player. If it can successfully expand its mineral processing and midstream battery production, it may deepen its integration into global EV supply chains. However, without midstream investment, Brazil risks being locked into its traditional role as a raw material exporter. The next decade will be crucial in determining whether Brazil can turn its resource wealth into industrial strength.

THE ROLE OF AFRICA IN THE ELECTRIC VEHICLE VALUE CHAIN

Godwin Kafui Ayetor, Joseph Mashele, and Issaka Lawerh Tetteh

The quest to reduce emissions from transportation has led many countries to set targets and adopt policies that will ease the transition to electric vehicles (EVs). The goal to transition to EVs also intersects with several Sustainable Development Goals (SDGs). SDG 7, which aims for sustainable energy for all, could be achieved in the transport sector in part through EVs, which can significantly reduce reliance on fossil fuels. The electrification of the transport fleet also fulfils SDG 13, since it helps to combat climate change exacerbated by fossil-fuelled vehicles. SDG 11, which relates to air quality, also benefits from wider EV adoption.

Over 90 per cent of global car manufacturers have set electrification targets.[145] However, the different components of EVs compared to conventional ones indicate that the manufacturing dynamics will have to change, with a greater emphasis on batteries, electronics, and raw materials inputs.

This article explores Africa's role in the new-energy vehicle value chain and the existing manufacturing opportunities. It highlights the need for Africa to process its minerals before export, enhance labour skills, and establish intra-African component manufacturing and supply value chain linkages through infrastructural and industrial policies.

Raw materials

While some components of EVs are common to internal combustion engine (ICE) vehicles, the high demand for new raw materials and components for EVs has disrupted the vehicle production supply chain. Components such as high-voltage battery packs, power electronics, electric motor and e-axle, high-voltage electrical distribution, and thermal management systems are unique to new-energy vehicles, including battery EVs (BEVs) and fuel cell EVs. The high-voltage battery pack is the most significant contributor to the BEV value chain, requiring critical minerals such as lithium, cobalt, nickel, manganese, graphite, and tin.

Africa is endowed with most of these minerals, and it already plays a role in the battery manufacturing value chain, save for refining. The Democratic Republic of the Congo is the largest supplier of cobalt, accounting for 65 per cent of the market share in 2023.[146] South Africa is the world's largest manganese producer, accounting for 35 per cent of world production in 2023, followed by Gabon. There are currently lithium projects in Ethiopia, Mali, Namibia, Ghana, and the Democratic Republic of the Congo. In 2030, Africa's total lithium production in the base case will rise to 53 kilotons, according to the International Energy Agency.

[144] European Commission, 'EU–Mercosur partnership agreement: enhancing trade and investment in critical raw materials', fact sheet, 2024, at https://ec.europa.eu/commission/presscorner/api/files/attachment/880031/Factsheet%20EU-Mercosur%20Trade%20Agreement%20-%20Critical%20raw%20materials.pdf.

[145] *Global EV Outlook 2024*, International Energy Agency, 2024, at https://iea.blob.core.windows.net/assets/a9e3544b-0b12-4e15-b407-65f5c8ce1b5f/GlobalEVOutlook2024.pdf.

[146] *Global Critical Minerals Outlook 2024*, International Energy Agency, May 2024, at https://www.iea.org/reports/global-critical-minerals-outlook-2024.

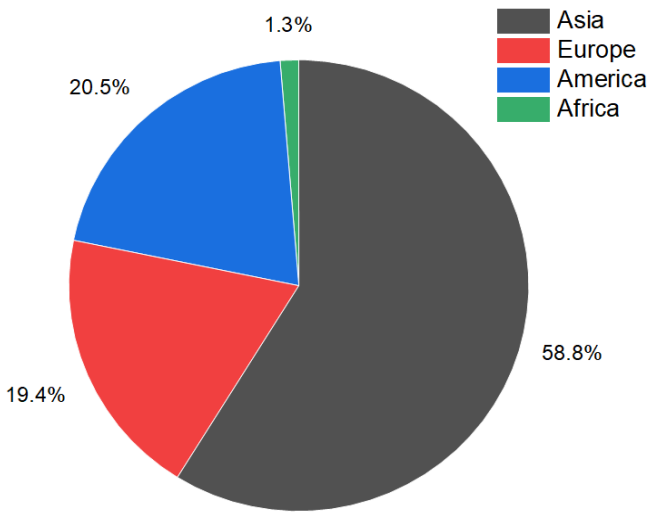
China has acquired stakes in most of the African mines and invested heavily in them. Chinese companies have at least 50 per cent equity ownership of seven lithium assets in Africa, which are expected to start production in 2027.[147] However, no value is added to the minerals through refining in Africa. Instead, most of the minerals are shipped to China for refining. China is responsible for half of the global processing of lithium and cobalt and hosts the entire graphite anode supply chain. China thus controls the downstream battery supply chain, including refining the battery minerals. China holds 85 per cent of battery cell production capacity, 90 per cent of cathode, and 98 per cent of anode material production.

Africa's role in world vehicle production

Chinese original equipment manufacturers have gained the edge in the new energy vehicle transition. In 2023, almost half of all BEVs worldwide were produced by Chinese automakers, despite China accounting for only 10 per cent of ICE vehicle sales.[148] According to the International Organization of Motor Vehicle Manufacturers, about 32.2 per cent of all vehicles produced in 2023 came from China.

Figure 1 shows total vehicle production in 2023 by region. Africa contributed to only 1.3 per cent of vehicle production in 2023. South Africa, Morocco, and Algeria are the only African countries with significant vehicle production capacities. South Africa produced 633,337 units, followed by Morocco (535,825 units) and Algeria (2,456 units).[149] The continent also has the lowest number of vehicles per thousand inhabitants, 49, compared to a world average of 209. However, it has one of the highest vehicle growth rates at 4 per cent, only second to that of Asia (8 per cent).[150]

Figure 1: World vehicle production



Source: International Organization of Motor Vehicle Manufacturers

Despite the enormous potential due to its mineral resources, raw materials, and youthful population, the contribution of Africa to vehicle manufacturing—conventional or EV—is insignificant, depriving the continent of job opportunities. For instance, the South African automotive manufacturing industry contributed about 2.9 per cent to the country's GDP in 2022. The industry employs 250,683 people directly or indirectly, including 214,305 jobs in the maintenance sector.[151] In comparison, Germany employs nearly 800,000 domestically in vehicle manufacturing. Secondly, Africa is yet to take advantage of the huge potential in its domestic market because most of the vehicles produced in Africa are exported to countries outside Africa. In 2022, South Africa exported 72.7 per cent of its vehicles to Europe, 10 per cent to Asia, 7.4 per cent to America, 3.5 per cent to Australasia, and only 6.4 per cent to other African countries. Similarly, nearly 80 per cent of vehicles produced in Morocco are exported to Europe.

[147] Magnus Ericsson et al., 'Chinese control over African and global mining—past, present and future', *Mineral Economics*, 33, 22 July 2020, at https://doi.org/10.1007/s13563-020-00233-4.
[148] Weidong Yan et al., 'Global mining outlook in 2024', *China Mining Magazine*, 33(1), 2024, at https://doi.org/10.12075/j.issn.1004-4051.20240052.
[149] *2023 Production Statistics*, International Organization of Motor Vehicle Manufacturers, 2024, https://www.oica.net/wp-content/uploads/By-country-region-2023.xlsx.
[150] *Motorization Rate 2020—Worldwide*, International Organization of Motor Vehicle Manufacturers, 2020, at https://www.oica.net/wp-content/uploads/Total-World-vehicles-in-use-2020.xlsx.
[151] Norman Lamprecht, *The Automotive Export Manual*, National Association of Automotive Component and Allied Manufacturers, 2023, at https://naacam.org.za/wp-content/uploads/2023/05/Automotive-Export-Manual-2023.pdf.

Africa's role in electric vehicle production

Europe, North America, and other regions that Africa exports its vehicles to have vehicle electrification targets. The Fit for 55 package seeks to end the sale of new ICE cars in Europe by 2035, similar to the vehicle electrification target for the United Kingdom.[152] African countries such as South Africa and Morocco (the only countries on the continent with significant vehicle production) rely predominantly on European countries to buy their cars; they will need to transition vehicle production to EVs by 2035 to meet the policy requirements of the international market.

Morocco has emerged as a prime example of Africa's EV production potential. Morocco is working to lead fully decarbonized automotive supply chains and exploit the growing global EV market by utilizing its strong manufacturing sector, automotive expertise, renewable energy potential, abundant raw material sources, strategic location, and free trade agreements. The country produces between 40,000 and 50,000 EVs annually,[153] with plans to reach 100,000 units by 2025, at which time EVs would contribute 22 per cent to Morocco's gross domestic product.[154] The models produced there include the Mini Evs, Fiat Topolino, Opel E-Rocks, and Citroën Ami.

Morocco has the potential to use EV supply chain policies to boost its EV production further. Already an industrial base, Morocco is home to 70 per cent of the world's phosphate reserves and is the tenth largest producer of cobalt globally—both are critically needed for global battery production.[155] Morocco is well positioned geographically due to its strategic location at the nexus of trade among Europe, Africa, and North America. Its free trade agreements with Europe and the United States have enabled it to enjoy subsidies on EVs sold in both regions. However, Morocco has yet to take advantage of the African Continental Free Trade Area to create an automotive supply value chain in Africa, which could further boost its exports.

Although the country has engaged in strategic partnerships with Chinese companies to establish battery gigafactories, the partnerships suggest a lack of infrastructure, a low technological base, lack of skilled workforce and labour market. Currently, Morocco produces batteries, seats, tires, engines, shock absorbers, and electrical wiring harnesses for conventional cars in collaboration with Chinese, Japanese, American, and Korean companies, leaving it vulnerable to changes in demand for these components as the transition to EVs intensifies. This transition along with increasing manufacturing automation could also decrease the importance of low-cost labour, a key advantage that has benefitted Morocco and Tunisia in expanding into automotive component manufacturing to date. For jobs in the industry today, training of only four weeks is sufficient for production line workers, many of whom lack a high school diploma.[156] Automation adoption can decrease that advantage due to its requirement for higher skills obtained from advanced education. Stronger technical educational training systems are necessary to produce the small number of highly trained technical staff needed.

Opportunities in the electric vehicle value chain

After Morocco, South Africa offers the next most significant opportunity to produce an EV supply value chain in Africa. Seven multinational original equipment manufacturers with completely-knocked-down automotive assembly plants already operate in South Africa. A wide range of components for ICE vehicles are manufactured locally, contributing to 40 per cent local value addition in vehicles produced in South Africa, with approximately 200 locally owned component manufacturing firms engaged in the automotive value chain. The biggest automotive component export is catalytic converters, which comprise almost 50 per cent of total component exports. Some components include engines, engine parts, tyres, radiators, transmission shafts, and cranks.[157] A transition to manufacturing EV components means only tyres could remain relevant for the South African automotive industry, which is why in 2024 Morocco surpassed South Africa as Africa's automotive production leader in value and quantity. Similarly, there is a decline in the purchase of catalytic converters and other conventional vehicle components due to the global shift to electromobility.

[152] *Electric Vehicles*, European Environmental Agency, 5 December 2024, at https://www.eea.europa.eu/en/topics/in-depth/electric-vehicles?activeTab=e3e6b879-fef4-4a88-9436-5f004698270.
[153] Sebastian Ibold, *Morocco's Role in the Global Electro-Mobility Revolution,* Deutsche Gesellschaft für Internationale Zusammenarbeit, June 2024, at https://changing-transport.org/wp-content/uploads/2024_Moroccos-Role-in-the-Global-Electro-Mobility-Revolution.pdf.
[154] Mohamed Sharki, 'Morocco's thriving auto industry braces for challenges ahead', *Al Majalla*, 3 June 2024, at https://en.majalla.com/node/318396/business-economy/morocco%E2%80%99s-thriving-auto-industry-braces-challenges-ahead.
[155] Chris Matthews et al., 'The case for Morocco to be a key player in regional Climate Compatible Growth', *Climate Compatible Growth*, 2021, at https://climatecompatiblegrowth.com/wp-content/uploads/4F-COP26-Policy-Brief.pdf.
[156] Shamel Azmeh et al., 'Automation and industrialisation through global value chains: North Africa in the German automotive wiring harness industry', *Structural Change and Economic Dynamics*, 63, 2022, at https://doi.org/10.1016/j.strueco.2022.09.006.
[157] *Electric Vehicles White Paper*, Republic of South Africa Department of Trade, Industry and Competition, 2023, at https://www.thedtic.gov.za/wp-content/uploads/EV-White-Paper.pdf.

Unlike Morocco, South Africa has been unable to capitalize on the transition to EVs. South Africa will not produce BEVs until 2026, when the announced incentive for production totalling US$26.4 million will come into force.[158] The incentive allows investors to claim an income tax allowance of 150 per cent of the cost of any plant machinery, buildings, and installations used chiefly for producing BEVs in South Africa.[159]

The local availability of raw materials such as manganese, a mature industrial base, and existing manufacturing technologies provide opportunities for South Africa to benefit from the EV transition. High localization potential exists for sub-components such as electric motors, high-voltage wiring harnesses, charge ports, thermal management systems, hoses, and pipes. To achieve this, South Africa needs to produce low-cost EVs, and develop an African manufacturing value chain to take advantage of products and minerals from other African countries. Low-cost EVs could open up another market on the African continent through the African Continental Free Trade Area. To accomplish this, there is a need for research, development, and establishing strategic global partnerships in the EV value chain cannot be overlooked.

Other African countries such as Egypt, Ghana, Kenya, Namibia, Nigeria, and Rwanda have significant potential for component manufacturing due to the potential of cheap labour and the availability of raw materials. However, the lack of refining capacity for raw materials, low uptake of EVs locally, and lack of adequate incentives, manufacturing capacity, and grid capacity pose significant challenges. For most of these countries, there are not enough incentives for local consumers to purchase EVs, let alone to drive investment in manufacturing.

Consequently, African countries that wish to manufacture components would likely need to emulate Morocco's automotive production strategy. In particular, Morocco's Industrial Acceleration Plan (2014–2020) stipulated measures for free trade and manufacturing incentives for small and medium enterprises (SMEs) to be involved in the automotive value chain. The Industrial Acceleration Plan created new dynamics by establishing relationships between SMEs and large industries. As a result, exported vehicles contain 63 per cent Moroccan-made components on average,[160] facilitated by the availability of a modified tax section, social security support, and funding to support small and medium enterprises linked to the value chain.[161] A state industrial fund was created supported by the banking sector to offer an integrated and competitive financing scheme. The Industrial Acceleration Plan created a culture of deal-making to attract foreign investment and expose Morocco to opportunities in the evolving global market, such as the anticipated offshoring of 85 million Chinese jobs.[162] The Industrial Acceleration Plan (2014-2020) has since been succeeded by the second generation of the Industrial Acceleration Plan (2021-2025). While guaranteeing the protection of the nation's natural resources, this plan will build on pertinent goals to support the industry in a number of ways, such as research and innovation and the development of new sectors [163].

In contrast, most automotive policies in Africa only focus on incentives for assemblers. A typical example is the Ghana Automotive Development Policy, which sought to make Ghana a fully integrated and competitive industrial hub for automotive assembling, to generate highly skilled jobs in automotive assembly and the manufacture of components and parts, with spill-over effects into other sectors of the economy.[164] The Ghana Automotive Development Policy failed to propel Ghana to become an assembly hub because it neglected to provide a market for selling the new vehicles assembled. The absence of asset-based vehicle financing schemes, lack of investment capital, and high interest rates meant that manufacturers had to expand their market outside Ghana. Like most African countries, Ghana created an industrial policy and left the industry to find its market without providing financial support or linkage to SMEs that could supply materials.

[158] Amanda Visser, 'SA rolls out tax incentives to boost electric vehicle manufacturing by 2026', *The Citizen*, 14 August 2024 at https://www.citizen.co.za/business/tax-incentives-electric-vehicles-manufacturing-by-2026/.
[159] Charles Wachira, 'How are South Africa's EV plans faring post election?', *Dialogue Earth, 1 August* 2024, at https://dialogue.earth/en/energy/how-are-south-africas-ev-plans-faring-post-election/.
[160] C. Jakkie, *Morocco*, ISS African Futures, 2025, at https://futures.issafrica.org/geographic/countries/morocco/.
[161] Karim El Aynaoui Pierre-Richard Agénor, *Morocco: Growth Strategy for 2025 in an Evolving International Environment*, Policy Center for the New South, 24 January 2015.
[162] Karim El Mokri, *Morocco's 2014–2020 Industrial Strategy and Its Potential Implications for the Structural Transformation Process*, Policy Brief 16/27, Policy Center for the New South, 2016, at https://www.policycenter.ma/publications/morocco%E2%80%99s-2014-2020-industrial-strategy-and-its-potential-implications-structural.
[163] Karroumi, Bilal and Sedqui, Abdelfettah (2024), 'Implementation of Frugal Innovation Approach in Moroccan Industry through the Development of a New Frugal System', *Journal of Engineering,* 2024 (1), 3971088.
[164] *The Ghana Automotive Development Policy*, Ghana Ministry of Trade and Industry, 2019, at https://rome.mfa.gov.gh/uploads/SectionImagWithLinks/3286_AUTO_Brochure_NEW.pdf.

Conclusions

The EV policies of most African countries suggest that they are setting themselves up as consumers and not producers. The few countries concentrating on assembly have ended up importing semi-knocked-down components only to assemble them locally, neglecting the significant opportunities for component manufacturing. For example, the emerging interest of African countries in manufacturing electric two- and three-wheelers is only to import already manufactured products from China in kits and assemble them locally.[165]

While significant manufacturing potential exists for EVs and components in Africa—because of the abundance of raw materials, cheap labour, low motorization, and in some cases proximity to Europe—limitations such as low refinery capacity, lack of policy and investment from governments, lack of incentives, high interest rates, low access to loans, a high skills gap, trade barriers, inadequate infrastructure and grid capacity are significant obstacles to overcome.[166] Infrastructural and manufacturing policies should be introduced to attract investment and regional partnerships to boost the capacity to refine raw materials and manufacture components. This will require establishing regional production and minerals supply value chains and removing trade barriers. The threat of automation can be counteracted through partnerships and improvement in the technical educational system to provide the needed skills locally. The African continent remains a vast unexplored market for the sale of vehicles, but for the initial cost. Localization of component manufacturing has the potential to reduce the cost by half and thereby increase the demand.

NEW PATHWAYS: SOUTHEAST ASIA'S STRATEGIC ROLE IN GLOBAL ELECTRIC VEHICLE SUPPLY CHAINS

Nancy Wei

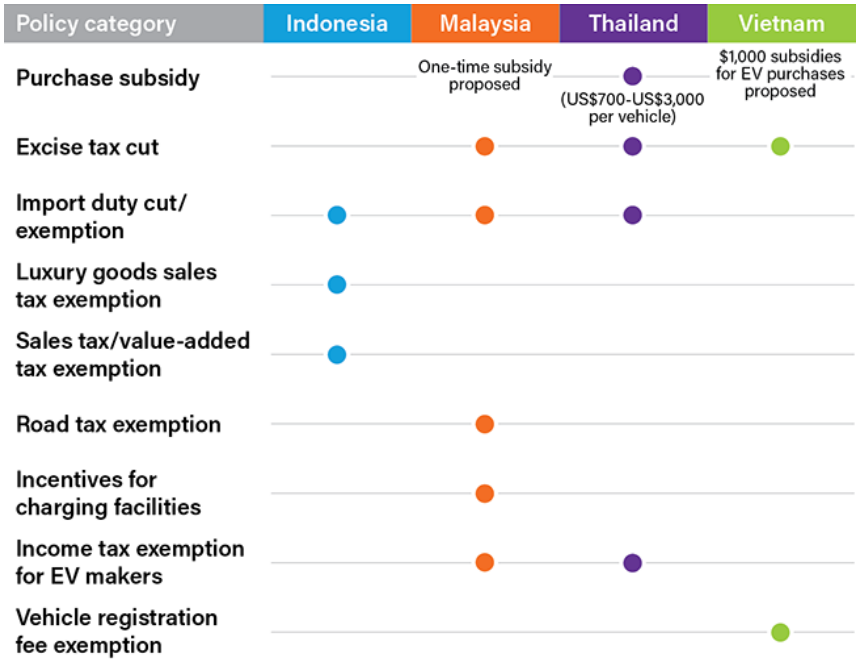
The Southeast Asian automobile market holds considerable growth potential for Chinese electric vehicle (EV) manufacturers due to its early stage of development and low vehicle ownership rates. With around 3 million vehicles produced annually and significant variations in vehicle ownership across the region, there is substantial untapped demand. The absence of dominant local brands allows Chinese companies to fill the demand for affordable vehicles, despite moderately high import tariffs. Forecasts suggest rapid growth in Southeast Asia, with the share of Chinese-made vehicles increasing from 12.7 per cent in 2023 to 20 per cent in 2025 and reaching 27 per cent by 2030.[167] During this time, EVs are expected to account for 60–70 per cent of Chinese automakers' sales in the region.

All Southeast Asian countries are engaging in the EV industry and partnering with China on EVs as part of their economic strategies. Given the dominance of Chinese EV manufacturers and the affordability of Chinese-produced EVs and batteries, China is becoming a crucial partner in this sector.

The strategic initiatives and policies of key Southeast Asian nations not only underline their commitment to sustainable transportation but also pave the way for enhanced collaboration and infrastructure development with China that aligns with the region's broader economic and environmental goals. Southeast Asia's relatively neutral geopolitical stance offers Chinese companies a variety of opportunities to expand, enabling China to enhance its EV presence from both a demand and supply perspective. Each nation is adopting a distinct strategy based on its specific circumstances (as shown in Figure 1), perceived strengths, and existing economic frameworks—some focusing on vehicle manufacturing, and others on upstream supply chains. It is too early to determine which countries' strategies will be more successful, and whether these strategies will lead to substantial technology transfer and the creation of high-value supply chains or will primarily focus on assembly. Regardless, Chinese EV companies are making investments that contribute to infrastructure development and stimulate economic growth.

[165] Godwin Kafui Ayetor et al., 'Feasibility of electric two and three-wheelers in Africa', *Green Energy and Intelligent Transportation*, 2(4), August 2023, at https://doi.org/10.1016/j.geits.2023.100106.
[166] Godwin Kafui Ayetor et al., 'The progress toward the transition to electromobility in Africa', *Renewable and Sustainable Energy Reviews*, 183, 2023, at https://doi.org/10.1016/j.rser.2023.113533.
[167] Betty Huang and Le Xia, 'EV sector: forging ahead amid intensifying headwinds', *BBVA Research*, 27 June 2024, at https://www.bbvaresearch.com/wp-content/uploads/2024/06/202406_Chinese-EV-sector_forging-ahead-amid-intensifying-headwinds.pdf.

Figure 1: EV support policies in Southeast Asia



Source: S&P Global Ratings, Eurasia Group

China's EV expansion into Southeast Asia: opportunities and symbioses

Chinese EV dominance in the global automotive sector reached a new milestone as BYD surpassed Tesla in quarterly deliveries at the end of 2024.[168] Because China's rapidly transforming domestic EV market faces intense price competition amid increasing overcapacity at home, Chinese brands are venturing into international arenas. With its proximity to China, a population exceeding 600 million, and a $3.4 trillion economy, Southeast Asia presents significant opportunities, especially given limited access to European and US markets.

The market potential is already large: Southeast Asia's market for cleaner cars is nearing a value of US$100 billion.[169] S&P Global estimated that policies and widening product offerings will likely result in more than 20 per cent compounded annual growth in EV sales and carmakers spending more than $20 billion in building EV production in key Southeast Asian markets over 2024–2026.[170] In 2023, Chinese EV exports to Southeast Asia surged by 120 per cent, but growth slowed to 3 per cent[171] in the first 11 months of 2024 due to a low market share in Vietnam and weak EV sales in Thailand and the Philippines. Despite this slowdown, Chinese EVs still made up over 70 per cent of all EVs sold across Southeast Asian countries. In Vietnam, where local manufacturer VinFast dominates with over 90 per cent market share, Chinese EVs have a limited presence. However, in the other countries, the share of Chinese EVs ranges from 80 per cent to 95 per cent, and their rising market share is largely displacing Japanese and South Korean automakers.

This expansion is part of a growing strategic and economic cooperation between the region's burgeoning markets and the technological capabilities of Chinese automotive companies. The region's dependence on imported crude oil highlights how electrification could address energy concerns and financial burdens in the long term. According to OPEC data in 2022, Thailand imports approximately 70 percent of its annual oil consumption. Although Indonesia is a net exporter of petroleum products, in 2022 Jakarta was compelled to raise fuel prices due to a fiscal deficit of 464.3 trillion rupiah ($29.77 billion), which triggered widespread protests.[172]

[168] Devjyot Ghoshal, 'Chinese automakers retain grip over Southeast Asia's booming electric car market', *Reuters*, 21 June 2024, at https://www.reuters.com/business/autos-transportation/chinese-automakers-retain-grip-over-southeast-asias-booming-electric-car-market-2024-06-21/.
[169] Hadi Azmi et al., 'Chinese EV makers race to Southeast Asia for growth amid US trade war', *South China Morning Post*, 30 June 2024, at https://www.scmp.com/week-asia/economics/article/3268491/chinese-ev-makers-race-southeast-asia-growth-amid-us-trade-war.
[170] 'EV makers to bet $20 billion on South and Southeast Asia', *S&P Global*, 29 October 2024, at https://www.spglobal.com/ratings/en/research/articles/241029-ev-makers-to-bet-20-billion-on-south-and-southeast-asia-13290200.
[171] Elif Binici, 'Chinese EV sales in Southeast Asia hit the brakes on gasoline demand growth, but not so fast', *Kpler*, 16 January 2025, at https://www.kpler.com/blog/chinese-ev-sales-in-southeast-asia-hit-the-brakes-on-gasoline-demand-growth-but-not-so-fast#:~:text=Despite%20a%20120%25%20rise%20in,the%20six%20Southeast%20Asian%20countries.
[172] Stanley Widianto and Gayatri Suroyo, 'Indonesia hikes fuel prices to rein in ballooning subsidies', *Reuters*, 3 September 2022, at https://www.reuters.com/world/asia-pacific/indonesia-hikes-fuel-prices-rein-ballooning-subsidies-2022-09-03/.

Amidst challenges posed by rising fuel costs, there was a notable surge in demand for EVs in the second quarter of 2023. Total EV sales in Southeast Asia experienced a year-on-year growth of 894 percent, the highest globally. This increase was propelled by consumers in Thailand, Vietnam, Indonesia, and Malaysia, as reported by Counterpoint Research. Much of this additional demand is being met by Chinese companies.[173] Data from Counterpoint indicated a significant rise in market share for Chinese automotive companies in Southeast Asia last year, increasing from 38 percent in 2022 to nearly 75 percent in 2023. For instance, China's share in Thailand's total new-auto market more than doubled to 11 percent in 2023, driven by EV leader BYD.[174] Thailand's EV imports tripled in the first half of 2023 to 33,000 units, with BYD accounting for approximately 30,000 units, surpassing competitors such as Nissan and Mazda.

From a larger geopolitical perspective, Southeast Asian countries do not necessarily perceive China as expansionist or a direct military threat.[175] Even nations with maritime disputes with China, such as the Philippines and Vietnam, have experienced fluctuating relations with China over time, influenced by changing leadership priorities. In the pursuit of economic security, these nations are increasingly expanding economic cooperation with China. In 2023, China emerged as the region's largest source of imports at 23.9 per cent, while the US accounted for just 7.4 per cent.[176]

Different country strategies in play

Each of the major countries of Southeast Asia has its own unique approach to EVs, and each has a different approach with respect to competition and cooperation with China. For example, Thailand is positioning itself as a future manufacturing hub for EVs, driven by policies such as the country's 30@30 initiative, which aims for zero-emission vehicles to account for at least 30 per cent of domestic production by 2030. This commitment to EVs is also economically strategic, aiming to lure investment from global automakers.

China's BYD, Great Wall Motors, and Hozon have leveraged Thailand's robust automotive sector.[177] In early 2024, Great Wall Motors' first international plant in Rayong, a key automotive centre, initiated its local EV production. Hozon started its in-country operations in late 2023, followed by BYD, which launched manufacturing at its Thai facility in July 2024. Together, these three companies are projected to achieve a combined annual production capacity of nearly 300,000 vehicles. Thailand's approach, therefore, emphasizes integration of its domestic supply chains with Chinese players. In 2024, however, EV sales fell by 9.3 per cent to 66,732 units, missing the Electric Vehicle Association of Thailand's goal of 80,000.[178] Thailand has the highest household debt levels in Southeast Asia, leading to more stringent bank loan approvals for many consumers.

By comparison, both Vietnam and Indonesia are embracing the influx of Chinese EV technology by facilitating strategic collaborations and investments. Indonesia, with its aim to become a leading EV battery producer by 2027, offers subsidies and tax incentives to foster EV industry growth. These efforts are mirrored in Vietnam, where the government is actively seeking to enhance its position in the automotive supply chain through similar incentives. In Vietnam, however, local incumbent Vinfast is outselling Chinese competitors like BYD as their entry-level model costs around $11,700 while BYD's models range from $25,800 to $56,667.[179] Out of the approximately 91,500 EVs sold in the country last year, over 87,000 were VinFast models. As wealthier individuals are generally more capable of adapting to new technologies, the transition to electric and hybrid vehicles is more feasible for them, which is evident from recent survey data. Nearly 70 per cent of approximately 1,100 Vietnamese residents from more affluent cities surveyed by KPMG in mid-2024 expressed their willingness to transition to either an electric or hybrid vehicle.[180] This presents an opportunity for BYD and other Chinese automakers to solidify their foothold in the market.

[173] Joseph Sipalan and Amy Sood, 'In Southeast Asia's EV race, car makers gun it for the chequered flag—as consumer demand lags behind', *South China Morning Post*, 4 November 2023, at https://www.scmp.com/week-asia/economics/article/3240229/southeast-asias-ev-race-car-makers-gun-it-chequered-flag-consumer-demand-lags-behind.
[174] Kenya Akama, 'China automakers doubled Thai market share in 2023 on EVs', *Nikkei Asia*, 2 February 2024, at https://asia.nikkei.com/Business/Automobiles/China-automakers-doubled-Thai-market-share-in-2023-on-EVs.
[175] Selina Ho and Chin-Hao Huang, 'Unpacking the drivers of Southeast Asia's policy towards China', *China Power*, 8 July 2024, at https://chinapower.csis.org/analysis/southeast-asias-policy-china/.
[176] 'Asean Statistical Highlights 2024', *Asean Stats*, accessed 3 March 2025, at https://www.aseanstats.org/wp-content/uploads/2024/10/ASH-2024-v1-1.pdf.
[177] Jeen Loo and Jason Liew, 'Southeast Asia's growing role in EVs: mitigating your supply chain risks', *StoneTurn*, 22 August 2024, at https://stoneturn.com/insight/southeast-asias-growing-role-in-evs-mitigating-your-supply-chain-risks/.
[178] Pathom Sangwongwanich 'Thailand car sales plunge to 15-year low as banks curb loans', *Bloomberg*, 28 January 2025, https://www.bloomberg.com/news/articles/2025-01-28/thailand-s-detroit-of-asia-nickname-hit-as-car-sales-fall-26-on-loan-squeeze.
[179] 'China EV makers including BYD face harsh reality in Southeast Asia Market', *Bloomberg*, 2 March 2025, at https://www.bloomberg.com/news/articles/2025-03-02/china-s-ev-makers-are-facing-a-reality-check-in-southeast-asia.
[180] 'The EV 2024 report: Vietnam's market', *KPMG*, 2024, at https://kpmg.com/vn/en/home/insights/2024/07/the-ev-landscape-in-vietnam.html.

Malaysia serves as an example of increasing car sales and attempting to establish localized production. Chinese automakers such as BYD and Geely have forged partnerships with local firms like Sime Darby Motors and Proton, reflecting the region's emphasis on building localized production capabilities and market-specific strategies. In mid-December 2024, Proton launched its first EV model with a starting price of just $24,400.[181] At the launch event, Malaysia's Prime Minister Anwar Ibrahim expressed his hope that the model could eventually be manufactured in Malaysia, with greater involvement from local companies in the production of EVs. Such collaborations are essential as they provide Southeast Asia with the capital required for infrastructure development—a crucial ingredient for full EV adoption. Southeast Asian governments, recognizing the necessity of these strategic alliances, are embracing policy frameworks that encourage infrastructure investments.

Challenges facing Southeast Asia

Despite Southeast Asia's readiness for this economic transformation, challenges persist, including those familiar in other world regions. With approximately 5 million cars compared to about 250 million motorcycles,[182] the market landscape in Southeast Asia is significantly more complex than that of China. Major barriers to EV adoption in the region include the lack of charging infrastructure, inadequate service and repair facilities, and concern about EV resale values. On the technology side, imports and investments from China can help alleviate some of these barriers, but bring challenges of their own.

One of the top barriers to EV adoption is range anxiety, which stems from limited charging infrastructure, particularly in the vast rural and remote areas of the region. This concern, coupled with the psychological barrier of running out of power, underscores the urgent need for investment in expanding charging stations and enhancing consumer education about EV capabilities.

Additionally, inadequate service and repair facilities present a critical issue for EV owners, as the scarcity of trained mechanics and service centres affects the reliability and ease of EV maintenance.[183] Affordability of EVs is another concern, as they remain a pricey alternative for many consumers despite government incentives aimed at closing the cost gap with traditional vehicles. Limited model availability further restricts consumer choice, although an increase in local EV production is expected to alleviate this over time.

The issue of resale value also dampens demand for used EVs, compounded by uncertainties about the lifespan of batteries and market dynamics. Rising insurance costs, driven by insufficient historical data, pose yet another financial hurdle for potential buyers, although growing local production and repair know-how might help reduce premiums. In this context, initiatives like BYD's acquisition of an online insurance unit to establish its own auto insurance provider could serve as a model.[184] By entering the auto insurance sector, BYD and similar automakers may enhance market stability and offer tailored insurance solutions for new-energy vehicles.

Moreover, the broader supply chain poses risks related to environmental and ethical concerns. For instance, the dependency on nickel in EV batteries brings attention to the challenges in Indonesia, including environmental degradation, land disputes, and labour issues. As Southeast Asia's role in the global EV supply chain expands, manufacturers must conduct thorough due diligence to mitigate these risks. This includes addressing the reputational and financial implications of environmental and labour-related issues, which might require compliance with emerging regulations like the European Corporate Sustainability Due Diligence Directive.

China's increasing collaboration with Southeast Asia in EV production and joint ventures highlights a strategic push to alleviate the overcapacity in China's EV market while tapping into the region's emerging demand. However, this expansion is not without its challenges, as Beijing remains in control of critical technological advancements and restrictions, particularly in areas like battery production.[185] While Southeast Asian nations are capitalizing on the short-term economic gains of hosting Chinese EV manufacturers, these partnerships come with long-term risks. The Chinese government, unwilling to allow full technology transfers, keeps a tight grip on critical intellectual property, meaning that Southeast Asia may not achieve full technological autonomy.

[181] Joseph Sipalan, 'Proton's new EV challenges BYD and Tesla in Malaysia's growing market', *South China Morning Post*, 16 December 2024, at https://www.scmp.com/week-asia/economics/article/3291062/proton-launches-malaysias-first-home-grown-ev-undercutting-byd-nearly-us10000.
[182] 'Paving the way for ASEAN to adopt electric motorcycles', *Southeast Asia Development Solutions*, 10 January 2025, at https://seads.adb.org/articles/paving-way-asean-adopt-electric-motorcycles.
[183] 'Navigating key challenges in Southeast Asia's EV market', *Southeast Asia Market Research*, 6 May 2024 at https://www.southeastasiamarketresearch.com/insight/key-challenges-for-ev-in-sea.
[184] Denise Jia and Wu Yujian, 'BYD takes over online insurer as base for Auto Insurance Unit', *Nikkei Asia*, 10 May 2023, at https://asia.nikkei.com/Spotlight/Caixin/BYD-takes-over-online-insurer-as-base-for-auto-insurance-unit.
[185] 'China proposes further export curbs on battery, critical minerals tech', *Reuters*, 2 January 2025, at https://www.reuters.com/technology/china-proposes-further-export-curbs-battery-critical-minerals-tech-2025-01-02/.

Furthermore, with EVs as one of China's pillar industries, its dominance in the sector, despite domestic economic headwinds, has drawn global scrutiny. This growing geopolitical rivalry between the US and China could place Southeast Asia in a precarious position, as the region becomes a focal point in the broader US–China competition. The risks are clear: while Southeast Asia benefits from Chinese investments, it must balance these partnerships carefully to avoid becoming caught in geopolitical tensions that could stymie its long-term development goals.

The existing dependency on technology from China will need to be managed to avoid geopolitical overreliance. Moreover, the rapid pace of change demands that Southeast Asian countries continue to adapt their strategies to leverage local strengths while mitigating risks associated with external dependencies. To capitalize on burgeoning opportunities, other global EV players looking to enter the Southeast Asian market must tailor their approaches to meet local preferences and overcome region-specific obstacles. They should also anticipate infrastructure barriers and work collaboratively with local governments to enhance the technological and economic landscape.

As Southeast Asia navigates its burgeoning role in the global EV supply chain, the region stands at a critical juncture, poised to capitalize on significant economic and strategic opportunities. The collaboration with Chinese EV manufacturers underscores the potential for accelerated growth and infrastructure development, yet it also brings accompanying challenges, such as dependency on imported technology and the need for diplomacy amidst global geopolitical tensions. The EV transition may yield substantial domestic and regional economic benefits, but will need to be managed with foresight and strategic planning to ensure sustainable growth, technological empowerment, and economic diversification across the region.

A NEW EU APPROACH TO BATTERY TECHNOLOGY LOCALIZATION

Julia Poliscanova

Rechargeable lithium-ion batteries are essential for Europe's decarbonization goals—both for electrifying transport and for integrating higher proportions of wind and solar. Because the technology will sit at the core of the energy and industrial system, it is also imperative that its adoption provide economic and technology development opportunities at home, and not just a new import and technology dependency on China.

This is why the European Union (EU) has been attempting to build a local battery value chain, notably cell manufacturing, since the creation of the European Battery Alliance in 2017. Since then, more than 1,000 GWh of lithium-ion battery manufacturing capacity has been announced across the member states by 2030, more than enough to meet the local demand for electric vehicle (EV) and energy storage applications.[186]

But how close is the EU to realizing its ambition of battery resilience? Who is behind the announced battery manufacturing plans, and is the continent on track to gain the needed skills and expertise in energy transition's key technology?

Who is building Europe's battery factories?

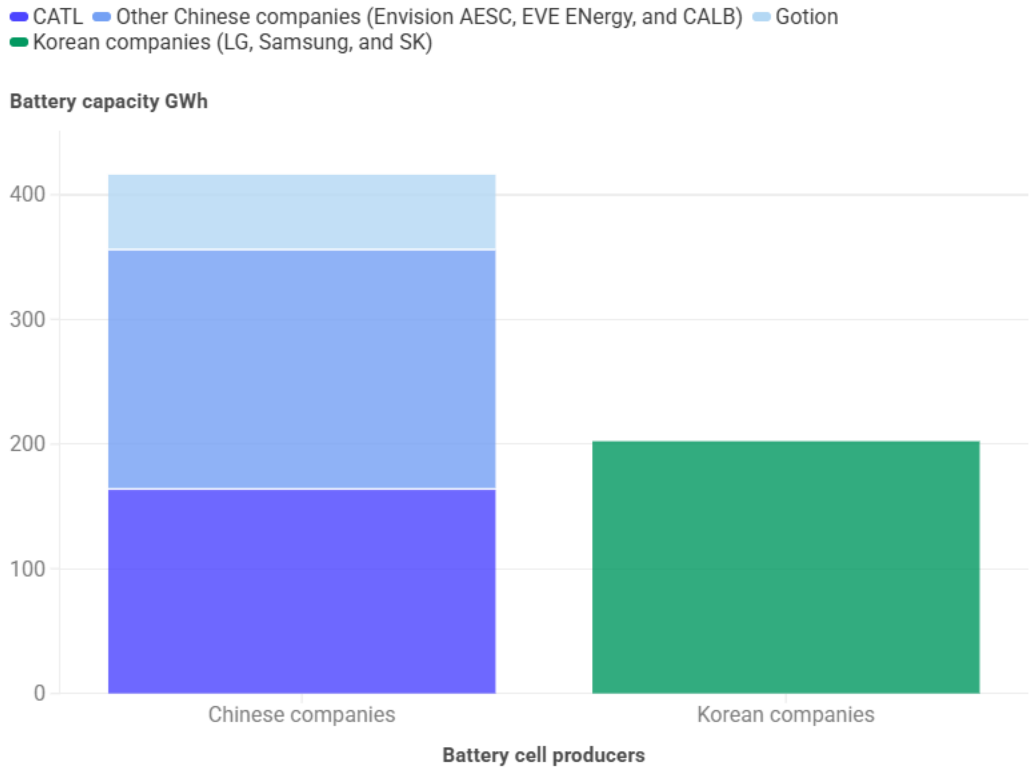
Half of all the 2030 EU battery cell gigafactory announcements are from European companies, including Northvolt, ACC, Verkor, and Volkswagen's PowerCo. While most are investing in the nickel-manganese-cobalt battery chemistries more popular in the West until now, some newer players, including Morrow and ElevenEs, are trying to scale the production of lower-cost lithium-iron-phosphate (LFP) chemistries. Until recently, only China has had significant LFP capacity, giving it an advantage in low-cost EV segments. LFP is now entering other markets as automakers push more affordable EV models for the mass market.

However, it is these European players, largely start-ups, that are at higher risk of delays and scaling difficulties. The now well-known story of Northvolt's struggles is not a unique case: at least 100 GWh of the planned EU-led capacity was cancelled or delayed in autumn 2024 alone, including Northvolt and ACC.[187] Ramping production from pilot lines to multiple-GWh production lines while keeping high yields and quality is taking time. This underlines the difficulty in acquiring local battery expertise and production, which is unlikely to happen as quickly as many policymakers and investors want.

[186] 'An industrial blueprint for batteries in Europe', *T&E*, 12 May 2024, at https://www.transportenvironment.org/articles/an-industrial-blueprint-for-batteries-in-europe.
[187] 'Trade defence: where's next for the EU's EV and battery trade policy', *T&E*, 4 October 2024, at https://www.transportenvironment.org/articles/batteries-the-next-frontier-in-trade-defence.

When it comes to Asian players, Chinese and South Korean shares of Europe's announced 2030 battery gigafactory plans are 27 per cent and 13 per cent, respectively. South Korean companies alone—LGES, Samsung, SDI, and SK On—already produced over 80 per cent of the EU's lithium-ion batteries in 2024. Going forward, the majority of the more likely battery investments in the EU are by Asian companies: around 440 GWh of new capacity has been announced by Chinese companies (mainly CATL and Gotion), and around 200 GWh by South Korean players (Figure 1). The reliance on Chinese players is even more pronounced on the battery component side (anodes and cathodes), underlining the growing role of Chinese companies in the EU's battery ecosystem.

Figure 1: Projected Asian battery production capacities in the EU by 2030



Source: T&E

Is Europe learning from Asian investment?

Chinese companies are leaders in battery technology, innovation, operational expertise, and global volumes of battery cell capacity installed. However, a key question for Europe is whether these battery manufacturing investments and partnerships in Europe bring genuine technology transfer and contribute to the development of local knowledge.

To assess this issue in detail, the nonprofit European Federation for Transport and Environment commissioned a number of independent experts (coordinated by Carbone 4).[188] The analysis looked at two current China–EU EV battery partnerships: the Volkswagen-Gotion partnership in Germany and the new Stellantis-CATL joint venture in Spain. While little information is available publicly, experts consulted through this study stressed that Chinese investments in Europe are largely about securing affordable battery supplies for EU automakers' operations, while comprehensive local knowledge or technology transfer provisions are missing.

The overall conclusion is that Chinese–EU battery partnerships today lack any meaningful knowledge sharing or local benefits. This is because no rules or framework requiring technology or skills transfer, or local content, currently exists in Europe. This means there is no political requirement to negotiate those as a condition to be on the European market.

This is in stark contrast to how the Chinese government treats foreign companies in China. Since the early 2000s, Chinese companies acquired traditional automotive expertise in part via the joint venture requirement, which mandated at least 51 per cent

[188] 'Assembly plant or battery powerhouse? analysis of foreign battery investments in EU', *T&E*, 18 February 2025, https://www.transportenvironment.org/articles/assembly-plant-or-battery-powerhouse-analysis-of-foreign-battery-investments-in-eu.

Chinese ownership for foreign car manufacturing.[189] Strict requirements on technology transfer and local research and development and performance requirements often benefiting local companies have long been in place in China. The government often imposes more favourable conditions on local companies, e.g. via lists or 'catalogues' which classify foreign investments as encouraged, restricted, or prohibited.[190]

The US Inflation Reduction Act similarly includes local content requirements, or in some cases free trade agreement qualifications, for certain subsidies or tax credits. This is followed by detailed implementation requirements, including the definition of 'foreign entities of concern' and rules limiting their direct, indirect, and effective control. The latter includes the control of production, supply chains, or maintenance and operation of critical equipment. Privately owned Chinese firms can still invest in the US, and EVs with Chinese content can be sold—albeit without a subsidy, making them more expensive. But the provisions of the Inflation Reduction Act have led to the CATL–Tesla and CATL–Ford LFP battery licensing deals.[191] These policies and the deals they catalysed can be seen as good practice the EU can build on.

The absence of technology transfer policy requirements will not necessarily undermine Europe's EV goals, as a plentiful supply of battery cells exists globally. In fact, the battery market is currently over-supplied.[192] However, the lack of comprehensive technology transfer, local content, or control provisions in the EU undermines the effort to gain a foothold in one of the energy transition's key technologies. This also poses geopolitical risks in case of trade tensions or political conflict.

What policy is necessary?

The lack of a comprehensive framework setting conditions on foreign direct investment, including batteries, is a key shortcoming. But the root cause for the EU's faltering battery plans (or clean tech more broadly for that matter) can be traced to a number of insufficient policies, including a lack of investment support for local clean energy companies, ineffective trade policy, and the absence of preferences for local content.

Europe lacks effective investment support for local clean energy technology companies and investments. Currently, the EU only provides initial research and development aid, as well as very limited CAPEX support via the Innovation Fund and loans from the European Investment Bank. Instead, the vast majority of scale-up and production support comes from national state aid. The EU's overly complex state aid requirements, which lack transparency, hinder successful green manufacturing ramp-up.[193] Approvals depend on slow, project-based political negotiation, often behind closed doors. The framework also lacks any made-in-EU requirements, failing to strengthen the EU's supply chain.

The EU also suffers from ineffective and weak trade policy. It seeks to adhere strictly to World Trade Organization rules, even as its trading partners take aggressive stances. While the EU made a progressive step putting in place anti-subsidy tariffs on Chinese EV imports, the import duty on lithium-ion battery cells entering the EU stands at a mere 1.3 per cent. This makes no sense for a region trying to localize manufacturing, especially in the face of significant over-capacity in China, making it easy to import cheaper batteries. Plainly put, weak trade policies reduce the incentive to scale local battery cell manufacturing in Europe. In comparison, the US is currently increasing its battery cell import duty from 25 per cent, and China has a 10 per cent import tariff for the EU.

Europe lacks policies to encourage local content or made-in-EU requirements. Again, this is in stark contrast to many regions attempting to build local manufacturing industries, including India, Brazil, and Turkey. Europe might be the last major economy left not having any local-value-add rules. This means that subsidies are often given to Asian companies without any conditions, thus creating jobs and supply chain opportunities in countries outside the EU rather than supporting struggling local start-ups.

If policy is not urgently changed, foreign battery and supply chain companies will find it easier to import knock-down kits, whereby key components are manufactured abroad, such as in China, and sent to Europe for final assembly. This would deprive Europe of knowhow, skills, and operational expertise in one of climate mitigation's core technologies.

[189] *Certain Measures on the Transfer of Technology*, European Commission, n.d., at https://policy.trade.ec.europa.eu/enforcement-and-protection/dispute-settlement/wto-dispute-settlement/wto-disputes-cases-involving-eu/wtds549-china-certain-measures-transfer-technology_en
[190] Ibid.
[191] 'Tesla to open new U.S. battery plant with equipment from CATL', *Bloomberg*, 31 January 2024, at https://www.bloomberg.com/news/articles/2024-01-31/tesla-to-open-new-us-battery-plant-with-equipment-from-catl?sref=M2YKkTZ6; 'Pentagon stings Chinese EV battery maker whose tech Ford plans to use at Marshall plant', *Detroit Free Press*, 7 January 2025, at https://eu.freep.com/story/money/cars/ford/2025/01/07/ford-chinese-ev-batteries-catl-marshall-megasite/77519832007/.
[192] David R. Baker, 'The world needs more batteries—but not this many', *Bloomberg*, 12 June 2024, at https://www.bloomberg.com/news/articles/2024-06-12/the-world-needs-more-batteries-for-electric-vehicles-but-not-this-many.
[193] 'EU Commission's State Aid plans fail to deliver for cleantech', *T&E*, 20 February 2025, https://www.transportenvironment.org/articles/eu-commissions-state-aid-plans-fail-to-deliver-for-cleantech.

But it does not have to be this way. Europe has power: its vast single market. This makes it attractive—and even imperative—for Asian companies to be present, especially as other global markets are becoming more closed.

What to do?

The EU should act quickly. The new European Commission and European Parliament, the upcoming Clean Industrial Deal, and a new trade and state aid reform agenda all offer a huge opportunity.

The EU should define conditions for access to its market, better support local players, and act with resolve. As a minimum, its toolbox should include both trade defence measures and changes to state aid to require local content.

Trade defence, such as tariffs, can limit the access of imported goods to its market and provide onshoring conditions to avoid those tariffs. As the first step, the EU should announce an anti-subsidy investigation into Chinese battery packs, cells, and components to level the playing field.

Conditionality should be added to state aid, EU funds, and public procurement to require more local content, local intellectual property transfer, local skills, and local supply chain investment. State aid should be urgently reformed, doing away with today's black-box thinking behind national subsidies and replacing it with ex-ante output-based criteria. All EU member states should be allowed to give up to €25/kWh for battery cells (including CAPEX and OPEX) without cumbersome approval processes or per-project haggling. In addition, local content rules should be included in broader EU funds, including the Innovation Fund, InvestEU, and the European Investment Bank programmes. Products manufactured in the EU or using EU-made materials should be equally prioritized in procurement bids ('Buy Europe').

Sustainability tools are needed to define what products can access the market, provided these are set simply (such as based on the emissions factors of the EU's grid vs foreign grids), without complex provisions that are open to misuse. One example is to agree to a grid-based CO_2 threshold for batteries, which would de facto limit batteries manufactured using energy from higher-carbon grids abroad. Another is to expand the French EV eco-bonus into a European 'eco-score' methodology for member states to use in their vehicle taxation. This would prioritize locally produced EVs made with clean steel, aluminium, and batteries.

Ultimately, the EU should set a comprehensive technology transfer framework, or foreign direct investment conditions. To be effective, this should include requirements for majority control by EU-based companies (preferably licensing deals), technology and intellectual property transfer, operational expertise and skills, and use of local suppliers.

Moving quickly and putting in place a framework to secure local battery expertise by leveraging its market power via tariffs, agreeing on a common >51 per cent joint venture framework vis-a-vis China, and reforming its subsidy framework can help Europe to turn its battery fortunes around and secure the local industry for its Green Deal goals.

CHINESE EV BATTERY MANUFACTURING IN HUNGARY: HELPING OR HINDERING THE GREEN TRANSITION?

Ágnes Szunomár

Policymakers are pursuing EV adoption as the automotive technology of the future due to its potential to reduce greenhouse gas emissions and dependence on fossil fuels. In line with the EU climate goals, including the 2035 zero-emission targets, the EU is investing hundreds of billions of euros into electromobility: pushing car manufacturers to produce EVs, boosting sales of these vehicles, and supporting the installation of a million charging stations.

Although the EU has been investigating whether Chinese EV companies have benefited from illegal state subsidies, it is hard to imagine that the region's ambitious EV strategy could be carried out without Chinese companies currently leading the EV industry—and, crucially, without the batteries produced by Chinese players. As a result, despite remaining tensions, Chinese EV companies are gaining a foothold in European markets to sell EVs and to supply European EV manufacturers with locally manufactured EV batteries. Hungary—host to most of the Chinese EV battery projects in Europe—exemplifies the emerging role of China, and presents an exceptional case study for how Chinese firms, technologies, and supply chains help and hinder Europe's green transition.

Automotive dependence

Despite its modest size, Hungary is the seventh-largest car producer in the EU and highly exposed to automotive foreign direct investment (FDI).[194] By participating in the manufacturing of internal combustion engine vehicles (ICEVs), Hungary has become a prominent player in the automotive supply chain, benefitting from its geographical proximity to European markets, manufacturing know-how, and relatively low wages. The main European—mainly German—original equipment manufacturers, such as Audi, BMW, and Mercedes, have been present for years in the country, where the automotive sector accounts for more than 26 per cent of manufacturing output, with 90 per cent of the assembled products being exported, according to Eurostat data. Vehicle manufacturing produces around 5–6 per cent of the Hungarian GDP, but together with automotive suppliers it accounts for 8–9 per cent.[195]

As the automotive industry slowly transitions from ICEVs to EVs, accompanied by a technological transformation, supply chains are also affected. Countries like Hungary, where labour-intensive, low-skill assembly of imported components constitutes the majority of local value added, are bound to face the adverse effects of this transition. Given that EVs are technologically distinct from ICEVs, and involve increasingly sophisticated labour-saving manufacturing processes, the inflow of new greenfield investments into Hungary's auto sector is likely to slow, weakening the country's potential for technology catch-up through supply chain integration.

To avoid being left behind economically and technologically, Central and Eastern European countries do not really have a choice when it comes to whether or not to get involved in EV manufacturing. However, they do have a certain degree of choice as to how they get involved and with whom they cooperate: besides relying on foreign investment, Czechia, Poland, and Slovakia aim to strengthen national development and their own capacities through domestic R&D spending, and are seeking to leverage the EV transition for industrial upgrading. In contrast, Hungary relies on external investors—mainly South Korean and Chinese—and labour-intensive, low-value-added activities.[196]

Increasing Chinese FDI, especially in EVs

Chinese companies have been investing in Hungary for more than two decades, particularly in the manufacturing sector. However, the overwhelming majority of Chinese investment projects in Hungary have been announced in the last three to four years, and the most important of these investments are in the EV battery space. According to annual stock figures, by 2022 China had become the eighth largest investor in Hungary, while based on annual flow data, in 2023 Hungary became the primary European

[194] 'Motor vehicle production volume in Europe in 2022, by country and vehicle type', *Statista*, 2023, at https://www.statista.com/statistics/585024/leading-car-automotive-manufacturer-europe-by-country/.
[195] Rhiannon Edwards, 'Hungary's automotive industry: moving on, moving up', *CIB Bank*, 21 March 2022, https://www.cib.hu/en/Maganszemelyek/our_world/growing-business/automotive.html.
[196] Ágnes Szunomár, *From Zero to Hero? Chinese Investment in Electric Vehicle Supply Chains in the Visegrád Four*, Prague, Czechia: Association for International Affairs, 25 January 2024, at https://chinaobservers.eu/from-zero-to-hero-chinese-investment-in-electric-vehicle-supply-chains-in-the-visegrad-four/.

destination for Chinese investments.[197] As a result, Hungary already ranks fourth globally in EV battery production, which will certainly continue to improve in the coming years with ongoing Chinese investments. Table 1 highlights major Chinese EV projects in Hungary.

Table 1: A selection of the largest Chinese EV-related FDI projects in Hungary

Chinese company	Year of investment	Location	Amount of FDI (million €)	Type of investment	Sectors
BYD (electric bus factory)	2017	Komárom (Komárom-Esztergom county)	35.5	Acquisition/ brownfield	Automotive EV
Semcorp	2020	Debrecen (Hajdú-Bihar county)	340	Greenfield	Automotive EV battery
Chervon China	2020	Miskolc (Borsod-Abaúj-Zemplén county)	50	Greenfield	Automotive Metal parts
Shenzhen Kedali	2020	Gödöllő (Pest county)	40	Greenfield	Automotive EV battery
Halms Hungary	2022	Debrecen (Hajdú-Bihar county)	43	Greenfield	Automotive EV battery
CATL	2022	Debrecen (Hajdú-Bihar county)	7,340	Greenfield	Automotive EV battery
Huayou Cobalt	2023	Ács (Komárom-Esztergom county)	1,400	Greenfield	Automotive EV battery
EVE Power	2023	Debrecen (Hajdú-Bihar county)	1,070	Greenfield	Automotive EV battery
Sunwoda Mobility Energy Technology	2023	Nyíregyháza (Szabolcs-Szatmár-Bereg county)	1,500	Greenfield	Automotive EV battery
BYD (EV factory)	2023	Szeged (Csongrád-Csanád county)	3,600	Greenfield	Automotive EV

According to company interviews, Chinese firms generally select Hungary to leverage their existing operations in Western Europe and to enhance their footprint in the broader European market. Companies seek a combination of European market access and cost efficiency—as demonstrated in the recent wave of Chinese investments in so-called export-platform-type FDI in Hungary. The export-oriented nature of the Chinese EV projects is evidenced by the planned capacity of the BYD automotive plant in the southern Hungarian city Szeged, which can produce 200,000 cars per year, while annual new car sales in Hungary only slightly exceed 100,000. German original equipment manufacturers active in Hungary are also partnering with Chinese EV battery plants there, all of which have export-to-sales ratios above 99 per cent.

Made-in-Hungary Chinese batteries: three arguments for and against

How do Chinese EV battery investments in Hungary reshape the green transition, and what are the socio-environmental consequences? Three arguments for and against these investments are worth highlighting.

First, although Chinese EV battery investments might reinforce Hungary's role in the EV value chain in the short and medium run, they have controversial effects in the long run. According to the Hungarian 2030 National Battery Industry Strategy (NBIS),[198] the battery value chain to be created in Hungary serves multiple purposes: it promotes the development of a competitive industry, increases economic value creation, creates new jobs, and makes Hungary a centre of the European battery value chain. With these initiatives and investments from South Korea and China in EV battery manufacturing, Hungary's position in global supply chains is undeniably expanding on both European and global scales.

Nonetheless, the NBIS runs in parallel to, but separate from, EU industrial policy. The EU's priority is to develop European batteries and create an indigenous European value chain, whereas the Hungarian strategy is based mainly on East Asian companies' investments. In the short and medium term, Chinese batteries will certainly be needed to meet European climate targets. In the longer term, however, as European innovation advances and local content policies grow more strict, Chinese-owned or -invested factories in Europe may become a liability.

[197] Andrea Éltető et al., *China's European Bridgehead? The Chinese Economic Presence in Hungary*, Budapest: Friedrich-Ebert-Stiftung, 2024, at https://library.fes.de/pdf-files/bueros/budapest/21458.pdf.
[198] *National Battery Sector Strategy*, Ministry for Innovation and Technology, 2022, https://kormany.hu/dokumentumtar/nemzeti-akkumulator-iparagi-strategia-2030.

In addition to this consideration, the main contradiction of Hungary becoming a European EV battery hub under the NBIS is that the country lacks raw materials such as lithium, graphite, and cobalt, as well as the water and energy resources necessary to produce EV batteries. The eastern part of the country, where most of the battery investments are planned, is affected by drought, and Hungary has relied heavily on energy imports for decades.

Second, Chinese batteries seem to be well positioned to support the EU's climate objectives, but they may threaten local environmental degradation by polluting water, air, and soil.[199] As regards water, hundreds of hectares of wetlands will be paved during factory construction; the factories' water-intensive operations will affect groundwater aquifers, jeopardizing the ecosystems reliant on them; and the factories' operations may also contaminate both ground and surface water. As for air pollution, current data show increased emissions of nitrogen oxides around existing South Korean-owned battery plants. Levels of nickel, cobalt, n-methyl-2-pyrrolidone (NMP), and other toxic substances were found to be three to five times the legal limit on several occasions.[200] Concerning soil pollution, contaminants from industry, heavy metals, and waste can cause irreversible damage to the environment.

Since the majority of Chinese battery factories have yet to commence operations in Hungary, there is limited tangible experience to analyse. However, several regulatory, transparency, and accountability issues exist with the South Korean EV battery companies currently operating in Hungary, indicating possible future complications with Chinese companies' environmentally harmful operations. The real problem lies not only in the potentially polluting actions of the EV battery companies, but in the weakness of accountability from the host country.

In Hungary, the Ministry of the Environment was abolished in 2010, and environmental authorities were subordinated to other government agencies, thereby rendering environmental protection entirely vulnerable to power politics.[201] EV battery plants are usually classified as 'priority investments of national economic importance', which enables them to be governed by expedited, streamlined procedures and reduced regulations.[202] When environmental problems arise, factories and authorities often withhold or delay information, investigations are insufficiently comprehensive, and factories are fined small sums under the existing light legislative framework.

Third, Chinese investments in the EV industry may create jobs in Hungary, but this also contributes to labour shortages and encourages the inflow of guest workers. In Hungary, the automotive sector employs over 150,000 people, providing 12 per cent of manufacturing jobs.[203] These jobs could be at risk as ICEV production is phased out. Accordingly, it is in the state's interest to safeguard these labour opportunities. Therefore, job creation and preservation is one of the main arguments in favour of battery factories, as the NBIS highlights. The NBIS even provides guidance on the location of future battery factories, highlighting the need to locate them in regions with an existing skilled labour force. This may explain the concentration of Chinese EV battery investments in eastern Hungary, where the unemployment rate has historically been higher.

On the flip side, however, unemployment in Hungary has nearly vanished in recent years: employment is now at its highest level since the democratic transition, and companies are currently seeking to fill approximately 100,000 positions.[204] Against this background, in 2023, the Hungarian prime minister predicted that Hungary would need 300,000 guest workers in the short term due to the establishment of battery plants. To this end, the Hungarian authorities have eased the work permit procedure for foreign workers, leading to a rise in the number of foreign (predominantly Asian) guest workers in Hungary. However, this process is not free of tensions: the first inflows of temporary guest workers from Southeast Asia has been a source of concern for the Hungarian public. The rather negative attitude of the Hungarian society is mainly due to the anti-immigration stance that the Hungarian government has taken since the 2015 European migration crisis.

[199] Andrea Éltető, *Why Is It Different? Specific Characteristics of the Hungarian Battery Industry: Legal Background and Environmental Impacts*, Working Paper No. 276, Budapest: HUN-REN Centre for Economic and Regional Studies, Institute of World Economics, August 2024, at https://vgi.krtk.hu/wp-content/uploads/2024/08/Elteto_WP_276.pdf.
[200] ENvironmental inFOrmation (ENFO) Database, accessed 7 March 2025, at https://web.okir.hu/hu/lair.
[201] Andrea Éltető, *Why Is It Different? Specific Characteristics of the Hungarian Battery Industry: Legal Background and Environmental Impacts*, Working Paper No. 276, Budapest: HUN-REN Centre for Economic and Regional Studies, Institute of World Economics, August 2024, at https://vgi.krtk.hu/wp-content/uploads/2024/08/Elteto_WP_276.pdf.
[202] Andrea Éltető, *Akkumulátorgyártás Magyarországon* [Battery manufacturing in Hungary], Working Paper No. 147, Budapest: HUN-REN Centre for Economic and Regional Studies, Institute of World Economics, March 2023, at https://vgi.krtk.hu/wp-content/uploads/2023/03/Elteto_MT_147.pdf.
[203] Rhiannon Edwards, 'Hungary's automotive industry: moving on, moving up', *CIB Bank*, 21 March 2022, https://www.cib.hu/en/Maganszemelyek/our_world/growing-business/automotive.html.
[204] John és Deák Szabó et al., 'The Hungarian automobile industry: towards an understanding of the transition to electromobility' in: *On the Way to Electromobility—A Green(er) but More Unequal Future?*, European Trade Union Institute, Brussels, 2020.

Conclusion

As this article has shown, Chinese EV-related investments in Hungary generate both opportunities and challenges at transnational, national, and local levels. While locally produced Chinese batteries may significantly contribute to the EU's climate goals and Hungary's economic growth through job creation and automotive transformation in the short run, they also risk imposing environmental burdens, social tensions, and even economic challenges in the longer run. Hungary, which relies heavily on the automotive industry, cannot afford to forgo the transition to EVs. However, the country's current strategy, which depends exclusively on foreign investments, appears to favour Chinese EV battery companies and European car manufacturers, rather than upgrading Hungary's supply chain.

Although the future of the fast-evolving EV industry remains unpredictable, a diversified development strategy grounded in indigenous R&D outcomes might significantly enhance the supply chain position of a small yet crucial participant in the European automotive sector. This is not only in Hungary's long-term interest but also crucial for a genuinely European transition to electromobility. Speaking about competitiveness during a press statement,[205] European Commission President von der Leyen stated that European automotive supply chains must become more resilient and adaptable, particularly concerning batteries. This is, however, challenging as European production is just about to scale up, while imported batteries remain cheaper. Neither making EVs more expensive nor creating new dependencies is a solution here, which is why, she added, direct support for EU battery manufacturers will be explored and European content requirements for battery cells and components will be phased in.

ROBOTAXI PROJECTS IN CHINA AND THEIR RELATIONSHIP TO AUTOMOTIVE SUPPLY CHAINS

Julian Schwabe

Autonomous driving, electric mobility, vehicle connectivity, and mobility services are four megatrends currently shaping the automotive industry.[206] In the realm of electromobility, the gradual replacement of the internal combustion engine with electric propulsion is highly disruptive because it renders comprehensive industrial development and production structures redundant over time, while new value chains, especially for the production of electric vehicle (EV) batteries, are emerging.[207] These are currently being dominated by Chinese firms, drastically changing competitive landscapes in the process.

Company ecosystems for developing autonomous driving technology are shaped by startups for software and hardware components. They follow different logics and show little overlap with EV battery production, but Chinese firms are at the forefront of development also in this domain. Both sectors, EVs and digital infrastructures, are consistently supported by state policy as part of government-designated strategic emerging industries. In contrast to EVs, the disruption which is entailed by autonomous driving technology concerns the definition of a car as the end-product. Will personal vehicles owned by private individuals continue to be the primary mode of transport within and between cities? Or will mobility services take over the client-facing function and define future mobility markets and products, reducing established automotive manufacturers to mere hardware suppliers?

The introduction of robotaxis for publicly available services in numerous cities brings all of the aforementioned megatrends together. They entail EVs, usually equipped with a variety of digital functions (e.g. for on-board entertainment), providing driverless mobility as a bookable service. While the sector is currently in its nascent stage, the potential of robotaxi services seems enormous: according to a McKinsey estimate, about 66 per cent of total passenger-kilometres driven in road transport in 2040 will be by autonomous vehicles, and 55 per cent will be autonomous as part of a mobility service.[208] With many projects and a vibrant pool of tech startups, China is at the forefront of developing autonomous driving technologies, and robotaxi projects in Chinese cities are an important incubator for this.

[205] 'Press statement by President von der Leyen on the Strategic Dialogue on the Future of the Automotive Industry', 5 March 2025, https://ec.europa.eu/commission/presscorner/detail/en/statement_25_656.
[206] David J. Teece, 'Tesla and the reshaping of the auto industry', *Management and Organization Review,* 14(3), 10 September 2018, at https://doi.org/10.1017/mor.2018.33.
[207] Julian Schwabe, 'Risk and counter-strategies: the impact of electric mobility on German automotive suppliers', *Geoforum,* 110, March 2020, at https://doi.org/10.1016/j.geoforum.2020.02.011.
[208] 'How China will help fuel the revolution in autonomous vehicles', *McKinsey*, 25 January 2019, at https://www.mckinsey.com/industries/automotive-and-assembly/our-insights/how-china-will-help-fuel-the-revolution-in-autonomous-vehicles.

Evolution of robotaxi projects in Chinese cities

Robotaxi services are available for public booking in many Chinese cities. Within the designated area, the self-driving vehicle brings one safely from A to B. The ride is slow, and sometimes available taxis can be few and far in between, however. Given the limited geographical area of operation, the service value of robotaxis is not yet competitive with regular taxi services. The main purpose of operating robotaxi services at this stage is not to provide competitive customer value, but to help develop and test self-driving technology.

Level 4 (L4) autonomous driving, or high automation where the vehicle can perform all driving tasks without human intervention under specific conditions, has gradually been introduced in Chinese cities over the past few years. The Autonomous Driving Road Testing Regulations, introduced in 2020, provided a legal framework for designating testing areas for L4 autonomous driving. This framework allowed companies to conduct road tests and gather data in controlled environments, setting the stage for more widespread adoption. To provide suitable framework conditions, Chinese cities have invested heavily in smart infrastructure (such as connected traffic lights, high-definition maps, and 5G networks) for autonomous vehicles to operate efficiently. This infrastructure is crucial for ensuring that vehicles can communicate with the city's systems and other road users.

The ambition of Chinese cities to be at the forefront in adopting self-driving vehicles is balanced against safety concerns. Most designated test areas for L4 autonomous driving are located on the outskirts of large cities (for example, Beijing Yizhuang and Shanghai Jiading), featuring wide and often empty streets, where traffic conditions are low in complexity and behaviours generally predictable. Cities have been cautious about opening more central areas with complex traffic conditions to L4 autonomous test driving due to concerns over behaviours that are more difficult to correctly evaluate. Urban centres with dense traffic, pedestrians, and unpredictable scenarios pose significant challenges for autonomous vehicles, which must navigate complex interactions with other road users. Despite technological advancements, local governments hesitate to grant unrestricted access in these areas until more data is collected from pilot projects in less congested zones.

Behind each robotaxi offering is a complex ecosystem of firms which intensively compete to be at the forefront of innovation. In its current phase, the sector is characterized by high uncertainty about future markets and high resource allocations for product development. Temporary, project-based inter-firm cooperations focused on optimizing technologies are the norm, and this also applies for existing robotaxi projects in China. The main companies populating the landscape are large mobility platforms—most prominently Didi Chuxing, Alibaba, and Apollo Go (as part of Baidu)—vehicle manufacturers, and AI startups like pony.ai, AutoX, and WeRide.ai (see Table 1).

Table 1: Robotaxi projects in China

Main operator (company type)	App	Initial robotaxi locations
Didi Chuxing (platform)	Didi	2020: Shanghai
GAC (automotive OEM)	Ontime	2022: Guangzhou
Pony.ai (AI-startup)	PonyPilot	2022: Beijing and Suzhou
Geely (automotive OEM)	Caocao	2022: Hangzhou
Alibaba (platform)	Amap	2020: Shanghai
SAIC (automotive OEM)	Xiangdao	2022: Shanghai, Suzhou, Shenzhen
Baidu (platform)	Apollo Go	2020: Beijing, Cangzhou, 2021: Shanghai, Guangzhou, Chongqing and others

Robotaxi operators offer their bookable services through mobility apps, some of which are well established and operated by large tech platform companies like Amap, Didi Chuxing, and Apollo Go, while automotive manufacturers GAC, Geely, and SAIC operate their own apps through which robotaxis can be booked (these are Xiangdao, Caocao, and Ontime). It is those companies who operate the robotaxi fleets, but they do this in project-based technology cooperations with AI startup firms who typically enter such cooperations through 'white box' agreements, in which the partners providing AI forward part of their technology to the main operator, who integrates the technology inputs into a functional autonomous driving system.

Existing robotaxi services are not intended to be profitable yet, but help develop autonomous driving technology. They generate large amounts of data on urban mobility patterns, user behaviour, and autonomous system performance and allow fleet operators to test, validate, and improve L4 autonomous driving in real-world urban environments. This helps refine their AI algorithms and autonomous systems, creating feedback loops for continuous improvement. However, the simplified driving environments on the outskirts of large Chinese cities somewhat limit technological advancement as the complexity of dense urban centres—with intricate traffic patterns, unexpected pedestrian behaviour, and diverse infrastructure challenges—aren't currently included in training L4 self-driving AI.

Startups as an integral part of autonomous driving development

China's autonomous driving startup landscape is highly competitive, driven by rapid technological advancements and significant investment. Companies like WeRide, Pony.ai, and AutoX are pioneering robotaxi services, while firms such as Inceptio Technology work on commercializing autonomous driving for trucks. Sensor technology providers like Hesai service the lidar market, supplying large industry players. Such startups help automakers like BYD integrate advanced driver-assistance systems into affordable vehicles.

The overall number of startups active in autonomous driving in China is estimated to be about 5,600.[209] They compete for venture capital and technical innovation. They move in delicate power relationships with larger, established vehicle manufacturers, tier 1 suppliers, and platform companies, which can develop a strategic interest to access their AI technologies and oftentimes act as venture-capital providers.

The vibrant startup ecosystem is embedded in structures of higher education. About 600,000 engineers graduate annually from Chinese universities,[210] providing a comprehensive talent pool from which such firms can recruit. In some cases, startups emerge as university spin-offs or are otherwise affiliated with university research where they can access research facilities, faculty expertise, and funding. The founders oftentimes come from established Chinese tech companies (especially from BAT: Baidu, Alibaba, and Tencent) and are supported by the world's second-largest venture capital market.[211]

As the case example of robotaxis illustrates, startups can play a vital role for established original equipment manufacturers (OEMs) but also large, tier 1 suppliers for developing new technologies and accessing innovation through dedicated partnerships. Among the thousands of existing startups, screening for viable technology partners can be a significant challenge, especially for international companies in China, who need to identify a startup's internal capabilities but also its standing with government authorities. An interview partner in this context highlighted more subtle details to be relevant in this context: for example, if a previously unknown startup captures a prominent speaking slot at a business conference, it can be interpreted as a sign of government goodwill.

Despite their geographical limitations, robotaxi projects are an important early adaptation for autonomous driving which contributes to the capability-building of auto manufacturers to offer EVs with self-driving function for private consumers. This happens through partnerships between vehicle OEMs and software firms, as both company types depend on the capabilities of the other, making strategic partnerships a viable option. An example is the 'God's Eye' system recently added in the BYD portfolio, which is powered by DeepSeek and supposedly offers up to L3 self-driving (meaning that a driver still needs to be present at all times) on highways and in cities. Baidu and Geely have been offering an EV with self-driving function under the brand Ji Yue. The models Ji Yue 01 and Ji Yue 07 are equipped with Baidu's Apollo suite, which claims to be capable of L4 self-driving capability. However, real-world deployment of full L4 capabilities depends on regulatory approvals and infrastructure readiness.

[209] Greg Gao, 'The number of China's autonomous driving tech companies exceed 5,600 as the auto industry accelerates the transition toward electrification and intelligence', *ijiwei*, accessed 3 March 2025, at https://jw.ijiwei.com/n/828677?utm.
[210] 'Engineering degree and education in China', *Best Education China*, accessed 3 March 2025, at http://www.besteduchina.com/engineering_education_in_china.html.
[211] Yan Xu, 'Harnessing venture capital in China', *Socio-Economic Review*, 22(1), January 2024, at https://doi.org/10.1093/ser/mwad037.

Internationalization and security concerns

International adaptation of autonomous driving technologies is hindered by geopolitical tensions and security concerns. China's regulatory environment presents significant challenges for international firms developing autonomous vehicles. Data security requirements (e.g. the cyber security law enacted in 2016 and the data security law of 2021) require that mapping and driving data remain within the country, limiting foreign companies' ability to leverage global datasets. High-precision mapping, essential for autonomous driving, is classified as a national security concern, restricting non-licensed firms from independently collecting such data. Instead, they must partner with state-approved local firms, adding bureaucratic hurdles and potential risks of intellectual property exposure. Testing and operational permits also favour domestic companies, with regulatory approvals often being more accessible to Chinese firms. Additionally, cybersecurity laws impose stringent requirements on data storage and transfer, making compliance complex for international firms. Likewise, the Unites States recently imposed a general ban on Chinese software technology in imported and domestically produced cars for security reasons. The ban is to take effect in 2027, but is likely to have immediate effect on value chains, as vehicle manufacturers active in the US market need to search for non-Chinese (and non-Russian) alternatives.

The EU has thus far refrained from moves that would explicitly exclude Chinese technology (even though the topic is intensively discussed, e.g. in the context of communication technologies from Huawei), and Chinese startups have been developing a modest presence in Europe as well. The adaptability of autonomous driving AI developed in China might be limited for Europe, because of the rather homogenous, low-complexity real-life environments under which Chinese self-driving AI is mostly trained, which differs significantly from real-life conditions in places outside of China.

At the same time, Chinese software firms do partner with European firms for autonomous driving projects in Europe, for example in the confined environment of shuttle services at the Zurich airport, where WeRide.ai serves as technology partner,[212] and pony.ai has recently opened an R&D centre in Luxembourg, from where new mobility partnerships are anticipated as well.[213] The presence of Chinese software startups at large industry fairs, like the International Automobile Exhibition in Munich, also exemplifies their ambition to enter overseas markets. A representative of a Chinese software startup reiterated their aim to seek partnerships with OEMs in Europe and highlighted that once the AI has achieved a certain readiness in China, the adaptations of the code for European environments are relatively minor (e.g. for signage and language), but data collection, high-precision mapping, and training of the AI in European environments is needed.

Future outlook

These dynamics show a picture of fragmented, project-oriented development of autonomous driving in which AI startups, established tech firms, and vehicle OEMs compete to pioneer L4 autonomous driving capabilities across diverse geographies. Robotaxi projects are an important early adaptation and incubator for technology development, and advanced self-driving functions for the private consumer market are gradually being introduced, even though it is still geographically constrained. Daimler's Drive Pilot system, offering L3 self-driving, is available in California, parts of Nevada, and, at limited speed, on German highways. L4 self-driving is not yet being offered in the private consumer market; and for the near future, the development can be expected to be geographically fragmented, as different technology partnerships develop place-specific L4 capabilities, the introduction of which depends on place-specific infrastructure development, regulation, and approval processes.

The expected impact of autonomous driving on energy systems and its contribution to decarbonizing the economy is still uncertain and depends on the adoption of self-driving technology, its predominant use case, and the degree to which it pushes electrification of road traffic. With in-built sensors, computers, and connectivity devices, a vehicle with L4 self-driving capability consumes more energy than a comparable conventional model.[214] However, if adopted widely for freight traffic or ride-hailing solutions, the potential to reduce energy consumption as a result of efficiency gains is significant. The estimated energy savings in automated truck fleets range between 4.5 and 15 per cent, while the energy-saving potential of autonomous vehicles for ride hailing could even amount to 74 per cent under optimal conditions.[215] Such dynamics point to the potential of developing mobility services as a business model, as mentioned at the beginning of this article, even though the actual willingness of commuters to replace a self-owned car by a robotaxi-powered ride-hailing service remains to be seen.

[212] 'WeRide erweitert europäische Präsenz mit autonomem Robobus-Shuttle-Einsatz am Flughafen Zürich', *Finanzen.net*, 10 January 2025, at https://www.finanzen.net/nachricht/aktien/weride-erweitert-europaeische-praesenz-mit-autonomem-robobus-shuttle-einsatz-am-flughafen-zuerich-14143338.

[213] Lei Kang, 'Chinese autonomous driving startup Pony AI opens European R&D center in Luxembourg', *CNEV Post*, 24 October 2024, at https://cnevpost.com/2024/10/24/pony-ai-opens-european-rd-center-luxembourg/.

[214] Jooyong Lee and Kara M. Kockelman, 'Energy implications of self-driving vehicles', *Environmental Science, Engineering, Economics*, January 2019, at https://www.caee.utexas.edu/prof/kockelman/public_html/TRB19EnergyAndEmissions.pdf.

[215] Shaopeng Zhong, et al., 'Energy and environmental impacts of shared autonomous vehicles under different pricing strategies', *Urban Sustain*, 3(8), 25 February 2023, at https://doi.org/10.1038/s42949-023-00092-2.

While electrification and self-driving functions can be adopted independently of each other, the current robotaxi fleets and vehicles with advanced autonomous driving capability for private consumer markets are almost all EVs. A full electrification of road traffic would significantly impact energy systems, as demand for gasoline would decrease proportionally to increasing electricity demand, requiring continuous investment in renewable power generation capacity, which will gradually replace infrastructures for crude oil extraction and processing. In China, for instance, full electrification of the transport system would drastically increase transport-related electricity demand, from 89 TWh in 2020 to a projected 1.185 TWh in 2050.[216] Development of and demand for autonomous driving and EV technologies may reinforce each other as part of a wider technological paradigm shift shaping the automotive industry which impacts energy systems, traffic management, business models, and even geopolitics.

BORROWING CHINA'S EV SUPPLY CHAIN PLAYBOOK

Anders Hove

In the past 12 months, China's lead in battery technology has continued to grow, pacing the country's rapid ascent to dominance in the electric vehicle (EV) space. Perhaps no single story can encapsulate this trend, but two recent news items provide a hint about what's to come. In October 2024, the world's largest battery manufacturer, Ningbo-based CATL, announced the launch of its new Freevoy battery, which combines lithium-ion batteries with sodium-ion batteries.[217] The Freevoy battery is designed for plug-in hybrid EVs, and CATL says it can enable long ranges and rapid charging—addressing two weaknesses of most currently available plug-in hybrids.

Meanwhile, CATL's main rival, Shenzhen-based BYD, announced it is already marketing its own sodium-ion batteries as an energy storage product, and believes the cost has already reached parity with its conventional lithium-iron phosphate (LFP) batteries, and poised to fall a further 70 per cent.[218]

When new disruptive technologies are entering the market, lagging companies and regions are often reluctant to jump in, citing 'technology risk', while others pin their hopes on startups and breakthrough technologies that offer a chance to disrupt current industry leaders. In contrast, these examples of sodium-ion technology show that disruptive technologies in the battery space are being scaled up and deployed in China first—often by the leading firms that might ordinarily be inclined to stick with established winners, such as the Chinese LFP batteries that have already revolutionized the global EV market. Chinese firms are not only at the technology frontier in EVs and batteries, but they are advancing their lead in emerging technologies—a success built on a combination of a thriving and ultra-competitive domestic market and years of accumulated battery manufacturing know-how.

How China passed the competition

China's EV success story traces back to the government's early decision to pursue the technology as a path for leapfrogging foreign automakers that had long dominated the country's vast and growing domestic car market. China's joint venture requirement for foreign automakers had helped localize manufacturing, but had seemingly done little to raise the quality of local players. With their simpler designs and components that draw from the electronics industry, EVs offered the potential—in President Xi Jinping's phrase in 2014—to transform China from a 'big car country' to a 'strong car country'.[219] Though one shouldn't exaggerate the consistency or far-sightedness of Chinese policy in this or other industrial fields, the sector has undoubtedly benefited from strong signals from the government that the industry would pursue EVs as a strategic emerging industry.

While the narrative on China often focuses on subsidies, government targets, or outright technology theft, it's important to recognize that China's EV and battery story has been led by a dynamic and innovative private sector. Outside China, Tesla remains one of the few successful startups to reach scale, suggesting that the car industry remains dominated by a few big players—often longstanding national champions. In contrast, China boasts dozens of major EV startups, many of which are now eagerly going abroad: Geely, Nio, Xpeng, Aito, and Li Auto, just to name a few. After launching its first EV in early 2024, electronics giant Xiaomi sold over 130,000 EVs in nine months, and plans over 300,000 sales for 2025.

[216] Hongyou Lu et al., *China's Electrification Pathways: Findings from the China Energy Outlook 2022*, Lawrence Berkeley National Laboratory, June 2022, at https://eta-publications.lbl.gov/sites/default/files/lbnl_china_electrification_report-final-rev2.pdf.
[217] Denis Bobylev, 'Avatr 07 and Neta L are first EVs with CATL's sodium-ion containing Freevoy battery', *Car News China*, 25 October 2024, at https://carnewschina.com/2024/10/25/avatr-07-and-neta-l-are-first-evs-with-catls-sodium-ion-containing-freevoy-battery/.
[218] Marija Maisch, 'New sodium-ion developments from CATL, BYD, Huawei', *PV Magazine*, 28 November 2024, at https://www.ess-news.com/2024/11/28/new-sodium-ion-developments-from-catl-byd-huawei/.
[219] '习近平:发展新能源汽车是迈向汽车强国的必由之路' [Xi Jinping: developing NEVs is essential for making China a strong car country], *Xinhua*, 24 May 2014, at http://www.xinhuanet.com/politics/2014-05/24/c_1110843312.htm.

Three factors help explain the rapid scale-up of private EV players that have emerged from obscurity in less than a decade. First and foremost, the vast size of China's domestic market has made it an immense prize for those companies nimble enough to be first to market with a quality product. Second, fierce domestic competition, not only from within China but even from within the same province or region of China, means companies have to move fast and not rest on a few early successes like Tesla with the Model 3 and Y. Third, vertically integrated or clustered supply chains enable rapid interactions with key suppliers, driving innovation and helping develop the local knowledge and skills base that can help scale up production. As a result of fierce competition and tight supply chain integration, Nio can turn out a new model in 36 months, while Geely's Zeekr sub-brand can produce a new model in 24 months—both far faster than the three to five years it might take counterparts abroad.[220]

To be sure, subsidies have been important—but many of those subsidies have gone to consumers within China, in the form of per-vehicle purchase subsidies that resemble those in the US and many European countries. Purchase subsidies have been scaled down, but tax breaks remain—again, encouraging consumers to buy, and thereby accelerating the adoption that makes China such an attractive market in the first place.

Nor should one overlook the importance of technology transfer to the Chinese story. After new-energy vehicles were named a strategic emerging industry in 2010,[221] various requirements were set to ensure domestic battery and EV manufacturers captured the resulting growth opportunities. Initially, the joint venture requirement was implemented to ensure EVs used batteries sourced 50 per cent from China. In 2017, requirements for recipients were set even higher, requiring domestic firms to demonstrate mastery of all core areas of EV production, ruling out the specialization of the Chinese supply chain in just one element of the making of EVs, and in effect localizing the industry from top to bottom.[222]

Innovation was also a focus of policy: the central government established technology lists and funded research centres, while local government directly encouraged foreign manufacturers to set up local research centres. The central government even set up a patent office near the headquarters of CATL, now the world's largest battery company, to accelerate the process of registering R&D results and moving them into the market. Leading players such as BYD, Nio, and Xpeng are constantly investing in R&D to stay ahead of the competition. BYD's R&D spending surpassed Tesla's in 2023—reaching $5 billion compared with Tesla's $4 billion, and up from under $1 billion in 2016. [223]

The results are clear: China now leads the world in EV and battery manufacturing, and the country is at or near the technology frontier in most aspects of the supply chain. China leads in power-train battery patents and is equal to the competition in battery patents.[224] CATL dominates the list of patent filings, followed by BYD, followed by Panasonic and Samsung.[225] China also leads in the field of EV-charging-related patents.[226] In the EV space, China has moved on from being a low-cost mass producer and now leads in a critical technology field.

Using the Chinese playbook

Other countries are clearly starting to study China's EV and battery development pathways—both advanced economies seeking to combat the threat China poses, and developing countries looking to adapt elements to their unique circumstances.

In the lead are global auto manufacturers, who are keen to license Chinese technologies—a major element of China's adoption of LFP technology over a decade ago. Whether Tesla licensing LFP technology from CATL, Ford licensing battery tech from Gotion, or Stellantis taking a stake in Leapmotor, international firms appear to recognize that working with Chinese leaders offers a better chance to catch up than going it alone.

[220] Selina Cheng and Yoko Kubota, 'How China is churning out EVs faster than everyone else', *Wall Street Journal*, 4 March 2024, at https://www.wsj.com/business/autos/how-china-is-churning-out-evs-faster-than-everyone-else-df316c71.

[221] '国务院关于加快培育和发展战略性新兴产业的决定' [China State Council decision on accelerating training and development of strategic emerging industries], China State Council, 18 October 2010, at https://www.gov.cn/zwgk/2010-10/18/content_1724848.htm.

[222] Yuki Kamikawa and Brummer, 'Cross-national and cross-sectoral dynamics of innovation policies: the case of lithium-ion battery technology for electric vehicles in the US and China', *Technological Forecasting & Social Change*, 201, 7 February 2024, at https://doi.org/10.1016/j.techfore.2023.123021; '新能源汽车生产企业及产品准入管理规定' [NEV production and products management measures], China Ministry of Industry and Information Technology, 1 July 2017, at https://www.gov.cn/xinwen/2016-08/16/5099738/files/830bc9a8f0ac42829b9bf47b1e8beb1d.pdf.

[223] Edward White and Wang Xueqiao, 'China's Xpeng to boost spending to survive "bloody sea" of EV competition', *Financial Times*, 19 February 2024, at https://www.ft.com/content/640bf423-0ae2-4992-8119-61cc6de29159.

[224] David Leggett, 'Signals: China leads on battery patent filings', *Just Auto*, 4 September 2023, at https://www.just-auto.com/news/signals-china-leads-on-battery-patent-filings/.

[225] 'Q4 2023 update: batteries related patent activity in the technology industry', *Verdict*, 30 January 2024, at https://www.verdict.co.uk/patent-activity-batteries-technology-industry/.

[226] Naoshije Shimizu, 'China takes commanding lead in EV charging patents', *Nikkei Asia*, 4 May 2023, at https://asia.nikkei.com/Business/Automobiles/China-takes-commanding-lead-in-EV-charging-patents.

As China assumes the mantle of free trade champion to protect itself and its carmakers from protectionist policies, and as the US rushes to impose tariffs and reverse spending under the Inflation Reduction Act, Europe is taking the lead in adapting successful industrial policies from the China EV playbook. Europe is considering requiring technology transfer and licensing as a condition for receiving subsidies, for example.[227] Domestic content requirements for critical minerals and batteries are on the books or under discussion. Often, such policies do not need to explicitly exclude Chinese players. To the contrary, Europe and other regions are now in the position of needing to work together with Chinese suppliers to ensure access to the latest technology and processes.

Chinese players, too, appear willing to play ball, not only setting up factories abroad and signing off on licensing arrangements, but actively building out supply chains abroad to reduce cost and develop a local technology ecosystem. Chinese companies are also taking the lesson on the benefits of vertical integration as they expand abroad. Battery manufacturing leader CATL recently announced plans to establish a US$1.5 billion fund to invest in local parts and suppliers in Europe.[228] CATL already makes batteries in Europe, but production and quality lag behind its facilities in China because local suppliers don't meet its needs or can't respond quickly enough to its requirements.

Chinese policymakers are clearly wary of the potential competition such deals entail. Not only has China gradually introduced various restrictions on minerals and critical materials, but a recent policy requires companies with LFP cathode technology to obtain approval for any investment or deal that could result in technology transfer. While the policy is not a prohibition, the signal to industry is clear—and more such restrictions are surely under consideration.

How to avoid falling further behind

Voices calling for protectionism or backward-looking policies are not limited to the tariffs imposed by US President Trump, which could snarl global trade and kneecap domestic production. Carmakers outside China have scaled back expansion plans, saying they need to wait for consumers to catch up—or for battery prices to come down. The collapse of Northvolt, cancellation of new battery plants in the US, and slower uptake of EVs in Europe and the US, have led companies up and down the supply chain to slow their expansion plans.

The industry risks entering a vicious cycle: lowered industry expectations lead policymakers to water down ambitious targets, while investors and entrepreneurs opt to wait for the next up-cycle before committing capital. Instead of integrating supply chains and signing deals to beat the competition, investors and companies announce they prefer to avoid taking technology risk, and say they'll wait for prices to fall sufficiently to make an EV product that appeals to price-conscious consumers. In this scenario, not only do Chinese EV makers extend their lead, but European and North American automakers fall further behind when it comes to acquiring the best and most cost-effective battery technologies.

Ultimately, Western carmakers and technology companies can't catch up by waiting for things to happen. Nor are efforts focused on technology breakthroughs with novel technologies—such as solid-state batteries or entirely new battery chemistries and formats—likely to succeed, given the Chinese advantage across the battery technology field and the fact that Chinese players are already active in almost all promising areas of battery development. While consistent policy support is needed, the policy debate about consumer EV subsidies, the handling of emissions fines, and target dates for eliminating internal combustion engine vehicles may be a distraction. Industrial policy could better follow the Chinese playbook by encouraging vertical integration and cluster formation to pool resources and promote domestic innovation. Cooperation with Chinese technology players will be necessary, but not sufficient, to reach the technology frontier. Developing skilled labour expertise and supply chain integration is key to manufacturing innovation—companies cannot simply outsource this to Chinese players if they want access to the latest tech or to have a chance at beating the China price. One thing seems apparent: industry and policy need to find ways of working better together—not lobbying to slow down, but to catch up.

[227] Alice Hancock et al., 'EU to demand technology transfers from Chinese companies', *Financial Times*, 19 November 2024, at https://www.ft.com/content/f4fd3ccb-ebc4-4aae-9832-25497df559c8.
[228] 'CATL prepares 1.5 billion fund to secure battery supply chain', *Mobility Portal Europe*, 15 July 2024, at https://mobilityportal.eu/catl-secure-battery-supply-chain/.

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